

---

# Regime-Aware Tactical Asset Allocation for Equities and Fixed Income: Independent and Coupled Jump-Model Labels

---

Team RLMM  
Machine Learning for Finance and Complex Systems

June 30, 2026

## Abstract

We study whether forecasted equity and bond return regimes improve monthly allocation across equities, bonds, and cash after transaction costs. The empirical design estimates realised regimes, forecasts next-month regimes, and maps forecast probabilities into portfolio weights. Unsupervised jump-model teachers estimate equity and bond return environments from realised return-based features. Student models are supervised only by these teacher labels and forecast next-month regimes from lagged macro-financial predictors; the COUPLED student also uses current-regime indicators and regime run length. The resulting forecast probabilities are converted into regime-conditional mean–variance allocations.

We compare two regime representations. The INDEPENDENT specification estimates separate two-state favourable and unfavourable labels for equities and bonds. The COUPLED specification estimates a three-state joint stock–bond label with calm, normal, and stressed regimes. Both representations distinguish the central portfolio mechanism: bonds support equity drawdowns in disinflationary episodes such as the dot-com bust and the global financial crisis, while inflation-driven sell-offs such as 1994 and 2022 weaken the bond hedge. In the student stage, regularised logistic regression and random forests produce informative next-month regime probabilities. In the allocation stage, selected dynamic strategies improve on the 60/40 benchmark along distinct margins, including lower drawdowns, higher Sharpe ratios, and higher terminal wealth, depending on risk aversion and turnover control.

## 1 Introduction

Balanced portfolios depend on how equity and bond returns interact across market environments. In disinflationary equity sell-offs, bond returns often rise and reduce portfolio losses. In inflation-driven sell-offs, equities and bonds can both lose value. A tactical allocation rule therefore needs information about the prevailing stock–bond environment when choosing next-month equity, bond, and cash exposures.

This paper studies monthly regime-aware allocation for a broad equity leg and a broad fixed-income leg. The equity leg represents the stock component of a balanced portfolio, and the bond leg represents the fixed-income component. The empirical design has three stages. First, unsupervised jump-model teachers estimate regime labels from equity and bond returns derived features. Second, student models forecast the next-month teacher labels from lagged macro-financial predictors; the COUPLED student also uses current-regime indicators and regime run length. Third, the allocation stage maps the forecast probabilities into regime-conditional return moments and monthly equity, bond, and cash weights.

We compare two regime-label structures. The INDEPENDENT approach estimates separate binary favourable-state labels for the equity and bond legs. The COUPLED approach estimates one joint stock–bond label with calm, normal, and stressed states. The independent

structure gives asset-level signals; the coupled structure gives a direct signal for the joint diversification environment.

The rest of the paper follows the same sequence. Section 2 describes the return series and predictors. Section 3 reports the realised regime labels. Section 4 reports next-month forecast performance. Section 5 evaluates the allocation results. The appendices report formulas, grids, diagnostics, and sensitivity checks.

## 2 Data

The empirical analysis uses monthly excess returns for the equity and bond legs; cash is the zero-excess-return asset in the allocation stage. The equity leg is the S&P 500. The bond leg is the Bloomberg aggregate bond-index. The one-month risk-free rate defines excess returns,

$$r_t^{\text{eq}} = \text{equity return}_t - rf_t, \quad r_t^{\text{bd}} = \text{bond return}_t - rf_t.$$

**Teacher features.** The INDEPENDENT equity teacher uses daily equity exponentially weighted returns and Sortino ratios over 21, 63, 126, and 252 trading days. The INDEPENDENT bond teacher uses monthly bond exponentially weighted returns, downside risk, Sharpe ratios, and volatility over one- to twelve-month horizons. The COUPLED teacher uses monthly equity and bond Sortino, drawdown, and trend features. Appendix A reports the formulas, horizons, and transformations.

**Student panel.** The student panels are monthly and lagged. The INDEPENDENT panel combines equity, Treasury, credit, commodity, volatility, macro, sentiment, and stock–bond correlation predictors, with causal winsorisation, rolling z-scores, and exponential smoothing. The COUPLED panel uses lagged macro-financial, Treasury, commodity, and regime-context predictors for the joint stock–bond label, and also includes current-regime indicators and regime run length. Appendices A and G report the predictor definitions and transformations.

## 3 Teacher: regime labelling with statistical jump models

The teacher stage defines the realised regimes that the student models later forecast. The jump-model teachers are unsupervised: they estimate regimes from return-based features. A regime label must be persistent and must separate different return outcomes. Persistence measures whether a state represents a sustained environment. State-conditional returns show whether the states distinguish favourable from adverse outcomes. Crisis-window diagnostics compare the labels with known stock–bond episodes.

### 3.1 The jump model

A jump model assigns each observation to a finite set of states while penalising frequent state changes. For a standardised feature vector  $x_t$ , state  $s_t$ , and state centres  $\{c_k\}$ , the state sequence solves

$$\min_{s_1, \dots, s_T, \{c_k\}} \sum_{t=1}^T d(x_t, c_{s_t}) + \lambda \sum_{t=2}^T \Lambda(s_{t-1}, s_t), \quad (1)$$

where  $d(\cdot, c_k)$  measures distance from state centre  $c_k$ ,  $\Lambda$  penalises switches, and  $\lambda$  controls persistence. Higher values of  $\lambda$  produce longer spells. Using an iterative approach, the model uses dynamic programming to find the state sequence from the current centers, then updates the centers based on that sequence. To avoid local minima, this loop is run from several different initializations.

### 3.2 Two regime parameterisations

While both regime specifications rely on the jump-model penalty, they define distinct state spaces. Both teachers are fit causally; thus, the label for month  $t$  is derived strictly from observations available at or before that point.

**Independent: two independent binary labels.** The INDEPENDENT teacher estimates separate two-state jump models for the equity and bond legs. For asset  $a \in \{\text{eq}, \text{bd}\}$ , feature vector  $x_i^a$ , and teacher member  $p$ , the state sequence solves

$$\min_{s_1^a, \dots, s_{T_a}^a, \{c_0^a, c_1^a\}} \sum_{i=1}^{T_a} d_p(x_i^a, c_{s_i^a}^a) + \lambda_p \sum_{i=2}^{T_a} \mathbf{1}\{s_i^a \neq s_{i-1}^a\}. \quad (2)$$

The equity teacher relies on daily observations, whereas the bond teacher uses monthly data. Both models employ an identical 12-year historical training window before starting and update at each month-end. During these updates, features are standardized using only the training sample available up to that specific date.

The teacher ensemble aggregates models across a grid of persistence penalties ( $\lambda_p$ ) using both  $L^2$  (squared Euclidean loss with mean centers) and  $L^1$  (absolute loss with medoid centers) specifications. Appendix B details the parameter grids and ensemble construction.

After each fit, hidden states are ranked by their realized in-sample Sharpe ratio, and the higher-Sharpe state ( $g^{a,p}$ ) is labeled “favorable.” Because the equity teacher runs on daily data while the bond teacher runs on monthly data, their monthly target labels are defined as:

$$S_m^{\text{eq},p} = \mathbf{1}\left\{\frac{1}{n_m} \sum_{i \in m} \mathbf{1}\{s_i^{\text{eq},p} = g^{\text{eq},p}\} \geq \frac{1}{2}\right\}, \quad S_m^{\text{bd},p} = \mathbf{1}\{s_m^{\text{bd},p} = g^{\text{bd},p}\}. \quad (3)$$

Thus, the monthly equity target is a majority vote of its daily states, whereas the bond target is simply that month’s state.

**Coupled Approach: Joint Regimes with Dynamic Coupling** The COUPLED approach models the stock–bond environment using three joint regimes defined by binary asset-class states:

$$\mathcal{S} = \{\text{calm} = (0, 0), \text{ normal} = (1, 0), \text{ stressed} = (1, 1)\}.$$

Here, the indicators denote the equity and bond states respectively (0 for benign, 1 for adverse). Calm indicates both assets are benign, normal indicates adverse equities alongside benign bonds, and stressed indicates both assets are adverse. The benign-equity/adverse-bond state (0, 1) is explicitly excluded.

For a joint state  $j = (j_E, j_B)$ , the state emission cost  $e_t(j)$  augments the asset-specific distances with a time-varying coupling penalty:

$$e_t(j) = d(z_t^{\text{eq}}, \mu_{j_E}^{\text{eq}}) + d(z_t^{\text{bd}}, \mu_{j_B}^{\text{bd}}) + \eta \left[ \kappa_t \mathbf{1}\{j \in D\} + (1 - \kappa_t) \mathbf{1}\{j \in A\} \right], \quad (4)$$

where  $A = \{\text{calm}, \text{stressed}\}$  and  $D = \{\text{normal}\}$  partition the state space into asset agreement and disagreement. The scalar  $\eta$  controls the coupling strength. This penalty acts as a soft bias within the emission cost, penalizing joint states that contradict the empirical stock–bond dependence signal while favoring those aligned with it.

The dynamic coupling parameter  $\kappa_t \in [0, 1]$  dictates the conditional probability of state disagreement. At each monthly iteration,  $\kappa_t$  is updated via an  $L^2$ -penalized logistic regression. The dependent variable is the asset-agreement indicator  $y_t = \mathbf{1}\{s_t^{\text{eq}} = s_t^{\text{bd}}\}$ , estimated

conditional on a vector of macroeconomic features  $x_t^{\text{mkt}}$ :

$$\kappa_t = \sigma(\beta_0 + \beta^\top x_t^{\text{mkt}}), \quad x_t^{\text{mkt}} = \{\rho_{12,t}^{\text{eq,bd}}, \text{trend}_t^{\text{eq}}, \text{trend}_t^{\text{bd}}, \text{dd}_t^{\text{eq}}, \text{dd}_t^{\text{bd}}\}. \quad (5)$$

The features comprise rolling asset-class correlation, trends, and peak-to-trough drawdowns. To guarantee causality, the regression is estimated on expanding historical samples using information up to month  $t$ . Finally, the sequence transition penalties scale independently across asset metrics:

$$\Lambda(i, j) = \lambda \mathbf{1}\{i_E \neq j_E\} + \lambda \mathbf{1}\{i_B \neq j_B\}.$$

**Relationship.** The two parameterizations construct structurally distinct regime objects. While the INDEPENDENT approach generates separate asset-level labels—allowing for all four permutations of favorable and unfavorable conditions—the COUPLED approach collapses this space into a single joint state. Specifically, calm maps to favorable equity and bonds, normal maps to unfavorable equity and favorable bonds, and stressed maps to unfavorable equity and bonds. Ultimately, the INDEPENDENT representation yields two separate asset-level signals, whereas the COUPLED alternative produces a unified signal for the joint stock–bond environment.

### 3.3 Ensembling and Voting

The final regime labels are aggregated across a grid of persistence penalties and specifications. For the INDEPENDENT approach, equity and bond paths are combined independently via a majority vote across all constituent models. For the COUPLED approach, the ensemble aggregates 22 joint-regime specifications formed by crossing distinct feature families with the  $L^1$  and  $L^2$  loss configurations. In both settings, the resulting vote shares quantify model consensus. Appendix B and Appendix D detail the explicit voting formulas, transition matrices, and model-level diagnostics.

### 3.4 Label results

The two teacher specifications produce persistent regimes with distinct return profiles. Figure 1 compares their label histories: the INDEPENDENT panel marks favourable-regime months for each leg, while the COUPLED panel shades the same stock–bond history by calm, normal, and stressed joint regimes. Appendix D reports coupled vote shares and the coupling-probability path.



**Figure 1:** Causal teacher-label representations of the stock–bond history. Both panels report labels formed using information available through the labelled month. Left (INDEPENDENT): equity and bond excess-return indexes with favourable-regime months marked for each leg. Right (COUPLED): cumulative excess-return indexes shaded by calm, normal, and stressed joint regimes.



**State economics.** Table 1 reports label-conditional return statistics. In the INDEPENDENT panel, favourable states have positive Sharpe ratios and unfavourable states have negative Sharpe ratios for both equity and bonds. In the COUPLED panel, calm is the strongest equity state, normal combines positive equity returns with bond support, and stressed records losses in both risky legs.

**Table 1:** Label-conditional economics. The table reports same-month annualised excess return, annualised volatility, Sharpe ratio, sample share, and average spell duration. The INDEPENDENT panel conditions on each leg’s binary label. The COUPLED panel conditions on the three joint regimes. Panel headers report the causal label sample used for each representation.

| Independent: two binary labels (1988–2024) |              |       |        |       |       |      | Coupled: three joint regimes (1988–2024) |          |       |        |       |       |      |
|--------------------------------------------|--------------|-------|--------|-------|-------|------|------------------------------------------|----------|-------|--------|-------|-------|------|
| Class                                      | Regime       | Share | Ret.   | Vol.  | Sh.   | Dur. | Class                                    | Regime   | Share | Ret.   | Vol.  | Sh.   | Dur. |
| Equity                                     | Favourable   | 76.3% | +13.8% | 12.1% | +1.14 | 14.3 | Equity                                   | Calm     | 13.8% | +21.0% | 8.1%  | +2.60 | 4.7  |
| Equity                                     | Unfavourable | 23.7% | −15.4% | 19.8% | −0.78 | 4.6  | Equity                                   | Normal   | 77.1% | +9.9%  | 14.8% | +0.67 | 12.2 |
| Bond                                       | Favourable   | 67.1% | +5.7%  | 3.2%  | +1.81 | 13.8 | Equity                                   | Stressed | 9.0%  | −40.9% | 13.7% | −2.99 | 2.2  |
| Bond                                       | Unfavourable | 32.9% | −4.1%  | 5.1%  | −0.81 | 6.8  | Bond                                     | Calm     | 13.8% | +0.8%  | 3.5%  | +0.22 | 4.7  |
|                                            |              |       |        |       |       |      | Bond                                     | Normal   | 77.1% | +3.9%  | 3.9%  | +1.00 | 12.2 |
|                                            |              |       |        |       |       |      | Bond                                     | Stressed | 9.0%  | −8.4%  | 5.0%  | −1.69 | 2.2  |

**Persistence.** Persistent labels make the forecast horizon economically usable: a one-month-ahead signal has allocation value only if regimes last long enough for portfolio weights to adjust without being dominated by turnover costs. Table 1 reports average spell durations, and Appendices B and D report the transition matrices and switch-count diagnostics.

**Crisis windows.** Table 2 compares the labels across historical stress windows. The comparison separates equity drawdowns with bond support from episodes in which equity and bond returns fall together.

**Table 2:** Crisis-window behaviour under both labellings. The INDEPENDENT columns report the bond label’s favourable share and Sharpe ratio. The COUPLED columns report the joint regime mix within each window and the equity and bond Sharpe ratios. Sharpe ratios use same-month annualised excess returns over each window.

| Window                  | INDEPENDENT bond |        | COUPLED regime share |        |         | COUPLED Sharpe |       |
|-------------------------|------------------|--------|----------------------|--------|---------|----------------|-------|
|                         | Fav.             | Sharpe | Calm                 | Normal | Stress. | Eq.            | Bd.   |
| 1990 recession          | 100.0%           | +1.33  | 0%                   | 100%   | 0%      | +0.05          | +1.41 |
| 1994 bond shock         | 8.3%             | −1.55  | 8%                   | 33%    | 58%     | −0.48          | −1.61 |
| Dot-com bust            | 87.5%            | +1.81  | 0%                   | 97%    | 3%      | −1.07          | +1.97 |
| Global financial crisis | 72.2%            | +0.74  | 0%                   | 94%    | 6%      | −1.58          | +0.74 |
| Euro sovereign 2011     | 100.0%           | +3.46  | 12%                  | 88%    | 0%      | −0.61          | +3.69 |
| Covid crash             | 100.0%           | +2.25  | 0%                   | 100%   | 0%      | −0.88          | +2.76 |
| Inflation 2022          | 0.0%             | −1.82  | 0%                   | 33%    | 67%     | −0.93          | −1.90 |

The crisis evidence links the regime labels to the allocation question. Disinflationary equity sell-offs favour bond exposure, while inflation-led sell-offs call for lower exposure to both risky legs.

**Feature separation.** Feature-separation diagnostics identify the inputs that define the teacher states. The INDEPENDENT equity labels load most on medium-horizon Sortino ratios and return trends; the bond labels load most on Sharpe ratios and downside-risk measures. In the COUPLED teacher, equity trend distinguishes calm months, and bond trend and bond

Sortino measures distinguish stressed months. Full rankings are reported in Appendices B and D.

#### 4 Student: forecasting the next-month regime

The student stage turns realised teacher labels into out-of-sample forecast probabilities. At the end of month  $t$ , the predictors  $X_t$  are observed and the target is the teacher label in month  $t+1$ . The INDEPENDENT student estimates two binary favourable-state probabilities, one for equity and one for bonds:

$$\hat{p}_{t+1|t}^a = \Pr(S_{t+1}^a = 1 \mid X_t), \quad a \in \{\text{eq, bd}\}. \quad (6)$$

The COUPLED student estimates the corresponding three-class probability vector over calm, normal, and stressed joint regimes. The student models use regularised logistic regression and random forests; Appendix E and Appendix G report the predictor designs, validation protocol, model grids, and full forecast diagnostics.

**Forecast quality.** Table 3 reports out-of-sample forecast performance. The INDEPENDENT forecasts are evaluated as two binary favourable-state predictions. The COUPLED forecast is evaluated as a three-class probability forecast over the joint stock–bond regimes. AUC values are therefore compared within panels, not as direct cross-panel levels: the INDEPENDENT tasks are binary per-leg forecasts, while the COUPLED task is a three-class joint-regime forecast. The allocation section tests whether these probability forecasts improve net portfolio outcomes.

**Table 3:** Out-of-sample student forecast quality. The INDEPENDENT rows report binary diagnostics for the equity and bond favourable-state forecasts. The full independent predictor-design results are in Table F.20. The COUPLED rows report diagnostics for the joint three-class forecast; AUC is the macro one-vs-rest average, and Brier and log loss are reported in their multiclass forms.

| Approach    | Task   | Model               | OOS sample | AUC   | Bal. acc. | Brier | Log loss |
|-------------|--------|---------------------|------------|-------|-----------|-------|----------|
| INDEPENDENT | Equity | Logistic regression | 1996–2024  | 0.929 | 0.846     | 0.088 | 0.296    |
| INDEPENDENT | Equity | Random forest       | 1996–2024  | 0.903 | 0.831     | 0.107 | 0.361    |
| INDEPENDENT | Bond   | Logistic regression | 1996–2024  | 0.892 | 0.831     | 0.128 | 0.462    |
| INDEPENDENT | Bond   | Random forest       | 1996–2024  | 0.873 | 0.800     | 0.147 | 0.471    |
| COUPLED     | Joint  | Logistic regression | 1996–2024  | 0.770 | 0.677     | 0.541 | 1.062    |
| COUPLED     | Joint  | Random forest       | 1996–2024  | 0.776 | 0.595     | 0.383 | 0.686    |

The INDEPENDENT results favour logistic regression on the headline binary metrics, while the COUPLED results show a trade-off between class accuracy and probabilistic scoring. Full predictor-design comparisons and probability-path diagnostics are reported in Appendices F and G.

**Predictor relevance.** Predictor-relevance diagnostics are reported in Appendices F and G. Equity-regime forecasts rely more on equity cross-section, industry, and style information. Bond-regime forecasts use a broader mix of equity, Treasury, and credit information. The COUPLED forecast draws on a more dispersed set of regime-context, commodity, Treasury, and macro-financial blocks, consistent with a joint stock–bond target.

## 5 Regime-aware allocation

The allocation stage assesses whether student forecast probabilities enhance a monthly equity-bond-cash portfolio after accounting for transaction costs. At the end of each month  $t$ , these forecasts are mapped into regime-conditional return moments, which then serve as inputs for a mean-variance optimization to determine target portfolio weights. We evaluate the realized returns net of one-way transaction costs, comparing the performance against a monthly rebalanced 60/40 and additional benchmarks.

For the INDEPENDENT allocator,  $\hat{p}_{t+1|t}^a$  is the end-of-month- $t$  probability that asset  $a \in \{\text{eq, bd}\}$  is favourable in month  $t + 1$ . State-specific means and variances are estimated from past returns assigned to each state and shrunk toward pre-out-of-sample unconditional moments with weight  $n_{s,t}^a/(n_{s,t}^a + 36)$ . The probability-mixture mean is

$$\hat{\mu}_t^a = (1 - \hat{p}_{t+1|t}^a)\hat{\mu}_{0,t}^a + \hat{p}_{t+1|t}^a\hat{\mu}_{1,t}^a. \quad (7)$$

The variance uses the corresponding second-moment mixture. The covariance matrix uses an expanding stock-bond correlation estimate shrunk toward the pre-prediction stock-bond correlation. Appendix H reports the full probability-to-moment and realised net-return formulas.

For the INDEPENDENT allocator, the monthly portfolio is long-only in equity, bonds, and cash and solves

$$w_t = \arg \max_{w \in \mathcal{W}} \left\{ w^\top \hat{\mu}_t - \gamma w^\top \hat{\Sigma}_t w - \tau c \|w - w_{t-1}\|_1 \right\}, \quad (8)$$

where  $\mathcal{W}$  is the long-only equity-bond-cash simplex,  $\gamma$  is risk aversion,  $c$  is the one-way transaction cost, and  $\tau$  scales turnover control. Appendix H reports the independent grid, selected-strategy settings, additional benchmarks, and parameter sweeps.

The COUPLED allocator uses the predicted probabilities of calm, normal, and stressed joint regimes in the same portfolio objective. Its conditional return moments are estimated from past returns assigned to the corresponding joint regimes, using either the most likely regime or a probability mixture, and its covariance matrix is estimated with an exponentially weighted stock-bond covariance with a 12-month half-life. Appendix I reports the coupled allocation details, benchmark rules, cost sensitivity, and parameter sweeps. The headline results for both approaches are evaluated against the 60/40 benchmark. The comparison is benchmark-relative within each approach; Appendices H and I report the approach-specific allocation grids, covariance estimators, and turnover-penalty conventions.

### 5.1 Allocation trade-off

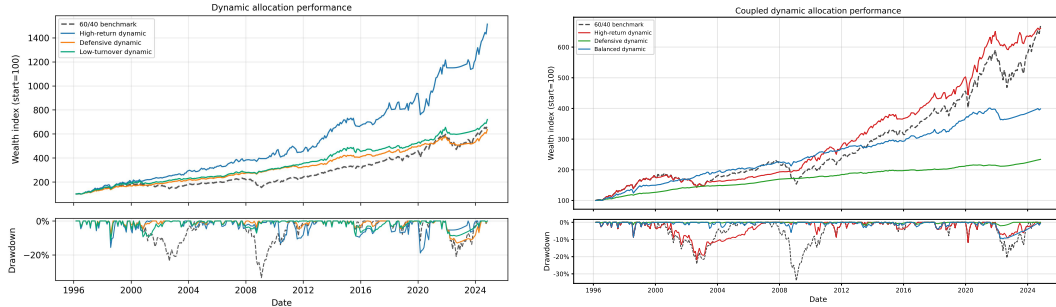
The allocation results show that regime forecasts have different portfolio roles across the selected dynamic strategies. Table 4 reports three representative roles from each allocation grid: a defensive allocation, an intermediate allocation, and a higher-return allocation. The selected rows summarise the main trade-off between drawdown control, trading intensity, and terminal wealth. Appendices H and I report the selection protocol, full allocation tables, cost sensitivity, and parameter sweeps. Additional independent benchmarks, including equity-only, bond-only, and volatility-targeted 60/40 allocations, are reported in Appendix H.5.

**Table 4:** Selected allocation results, out of sample. The table reports representative dynamic roles and 60/40 benchmarks. Returns are annualised excess returns net of 10 bps one-way transaction costs; “Final” indexes wealth to 100 at sample start. Both approaches use long-only equity–bond–cash weights and monthly rebalancing; the appendices report approach-specific moment estimates, grids, and turnover-penalty conventions.

| Approach    | Role         | Strategy                              | Ann. exc. | Vol.  | Sharpe | Max DD | Final  |
|-------------|--------------|---------------------------------------|-----------|-------|--------|--------|--------|
| INDEPENDENT | Defensive    | Random forest, smoothed predictors    | +4.4%     | 5.8%  | 0.76   | −13.0% | 636.6  |
| INDEPENDENT | Low-turnover | Logistic, z-score + smoothed z-score  | +4.8%     | 7.2%  | 0.67   | −11.5% | 720.0  |
| INDEPENDENT | High-return  | Logistic, smoothed predictors         | +7.6%     | 10.6% | 0.71   | −18.9% | 1513.6 |
| INDEPENDENT | Benchmark    | 60/40                                 | +4.5%     | 9.6%  | 0.47   | −33.4% | 669.8  |
| COUPLED     | Defensive    | Minimum variance, probability mixture | +0.80%    | 1.2%  | 0.64   | −2.0%  | 234.0  |
| COUPLED     | Balanced     | Mean–variance, probability mixture    | +2.74%    | 4.4%  | 0.63   | −9.6%  | 398.5  |
| COUPLED     | High-return  | Mean–variance, probability mixture    | +4.70%    | 7.6%  | 0.62   | −21.9% | 661.2  |
| COUPLED     | Benchmark    | 60/40                                 | +4.92%    | 9.6%  | 0.47   | −33.9% | 668.9  |

Table 4 shows that selected strategies improve on the 60/40 benchmark along different margins. Defensive allocations reduce drawdowns and raise Sharpe ratios; low-turnover and balanced allocations preserve part of this risk control with less aggressive rebalancing; high-return allocations accept larger drawdowns for greater participation in favourable regimes. The full metric set is reported in Tables I.41, H.24, and H.25.

Figure 2 reports the corresponding wealth and drawdown paths.



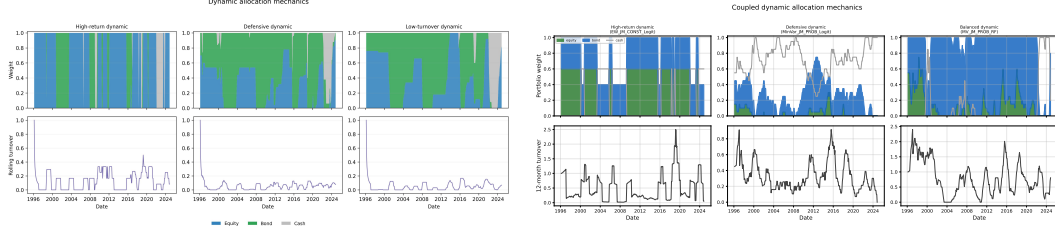
**Figure 2:** Dynamic allocation performance, out of sample. The panels report net wealth indexes and drawdowns for selected dynamic allocations and the 60/40 benchmark, with wealth indexed to 100 at the start of each evaluation sample.

## 5.2 Portfolio-weight behaviour

Figure 3 illustrates the empirical allocation mechanics and the resulting trade-offs between position smoothing and turnover. As visible in the structural profiles across weight plots, the optimization behavior splits along strategic objectives:

The high-return rules frequently trigger discrete, binary switches between asset classes, hitting corner solutions where the portfolio shifts entirely into a single dominant asset class. This behavior occurs because the optimization targets maximum conditional returns, causing aggressive portfolio rebalancing and sharp spikes in rolling turnover whenever a regime shift is predicted.

Conversely, the defensive and low-turnover configurations exhibit highly continuous, smooth weight trajectories. Rather than abruptly abandoning an asset class, these strategies scale exposures gradually in response to shifting forecast probabilities.



**Figure 3:** Portfolio weights and turnover, out of sample. The panels report equity, bond, and cash weights and turnover for the selected dynamic allocations.

### 5.3 Sensitivity

Appendices H and I report sensitivity to risk aversion, turnover penalties, transaction costs, probability smoothing, and moment shrinkage. The patterns are stable: higher risk aversion lowers risky exposure, smoothing and turnover penalties reduce trading, and higher transaction costs lower net performance.

## 6 Discussion and conclusion

This paper asked whether regimes estimated from stock and bond return behaviour can be forecast one month ahead and used in allocation. The answer is yes, but in a benchmark-relative sense. Regime forecasts create useful portfolio choices along different margins: drawdown control, Sharpe improvement, turnover reduction, and terminal wealth.

The first contribution is the construction of economically meaningful teacher labels. The INDEPENDENT labels separate favourable and unfavourable environments for each asset class: favourable equity and bond states have positive Sharpe ratios, while unfavourable states have negative Sharpe ratios. The COUPLED labels separate the stock–bond environment directly. Calm and normal regimes describe periods in which at least one risky leg remains useful, whereas stressed regimes identify months in which both equity and bond returns are adverse. This matters because the allocation problem is not only whether equities are attractive, but whether bonds still diversify equity risk.

The crisis-window evidence gives the clearest economic interpretation of the labels. During the dot-com bust, the global financial crisis, the euro-sovereign episode, and the Covid crash, bond exposure remained favourable and helped the balanced portfolio. During the 1994 bond shock and the 2022 inflation episode, bond exposure became part of the adverse environment. The regime models therefore recover the distinction that motivated the project: some equity sell-offs preserve the stock–bond hedge, while inflation-led sell-offs can damage both legs at the same time.

The second contribution is the separation between regime identification and regime forecasting. The teacher stage uses realised information to define the states. The student stage then asks whether those states are predictable from lagged macro-financial and regime-context variables. The out-of-sample diagnostics show that this step is feasible: the INDEPENDENT equity and bond labels are forecast with strong binary performance, and the COUPLED labels retain useful information despite the harder three-class joint-regime task. This makes the exercise more demanding than an ex-post regime interpretation.

The allocation results show where the forecasts have economic value. Defensive rules use adverse-regime probabilities to reduce drawdowns relative to 60/40. Low-turnover and balanced rules retain part of this risk control while limiting rebalancing. High-return rules use favourable forecasts to keep larger risky exposure and improve terminal wealth. The evidence therefore supports regime forecasts as allocation inputs, not as stand-alone market-timing signals.

## A Data Sources and Feature Construction

**Table A.5:** Teacher features by approach. The INDEPENDENT teacher uses daily equity features and monthly bond features. The COUPLED teacher uses a monthly joint stock–bond construction. Horizons are reported in trading days for daily equity features and in months for monthly features.

| Approach               | Feature                                                                | Definition                                                                                                                                         | Horizons                      |
|------------------------|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| INDEPENDENT (equity)   | Exponentially weighted return; Sortino ratio                           | Daily equity excess-return features based on recent mean performance and downside-risk-adjusted performance                                        | 21, 63, 126, 252 trading days |
| INDEPENDENT (bond)     | Exponentially weighted return; downside risk; Sharpe ratio; volatility | Monthly bond excess-return features based on recent mean performance, negative-return magnitude, rolling risk-adjusted performance, and volatility | 1, 3, 6, 12 months            |
| COUPLED (equity, bond) | Sortino ratio                                                          | Recent average excess return over downside risk (EWMA of squared negative excess returns)                                                          | 1, 2, 4, 12 m                 |
| COUPLED (equity, bond) | Drawdown level / term structure                                        | Log downside deviation; short minus long log downside deviation                                                                                    | 6 m; 2 vs 12 m                |
| COUPLED (equity, bond) | MA crossover / trend                                                   | Short minus long average excess return, scaled by downside risk; EWMA-standardised trend                                                           | 3 vs 12 m; 12, 24 m           |

### A.1 Time series

| Time series                                                                                                                       | Frequency used | Source    |
|-----------------------------------------------------------------------------------------------------------------------------------|----------------|-----------|
| S&P 500 Index level                                                                                                               | Daily          | Bloomberg |
| Bloomberg US Aggregate Bond Index level                                                                                           | Monthly        | Bloomberg |
| Fama–French market excess return, size, value, profitability, investment, momentum, and risk-free return                          | Monthly        | WRDS FF   |
| Six Fama–French size and book-to-market portfolio returns: small low, small medium, small high, big low, big medium, and big high | Monthly        | WRDS FF   |
| CRSP common-stock returns, prices, shares outstanding, SIC classifications, share classifications, and exchange identifiers       | Monthly        | WRDS CRSP |
| CRSP Treasury security returns, total amount outstanding, and maturity dates                                                      | Monthly        | WRDS Tsy  |
| CRSP fixed-term Treasury nominal yields at 1 month, 3 months, 1 year, 2 years, 5 years, 7 years, 10 years, 20 years, and 30 years | Monthly        | WRDS Tsy  |
| Equal-weight commodity portfolio excess return from <i>Commodities for the Long Run</i>                                           | Monthly        | AQR       |
| National Financial Conditions Index                                                                                               | Monthly        | FRED      |
| Adjusted National Financial Conditions Index                                                                                      | Monthly        | FRED      |

| Time series                                         | Frequency used | Source    |
|-----------------------------------------------------|----------------|-----------|
| Investment-grade corporate bond fund adjusted close | Monthly        | Yahoo     |
| High-yield corporate bond fund adjusted close       | Monthly        | Yahoo     |
| One-year Treasury yield                             | Monthly        | FRED      |
| Moody's Baa and Aaa corporate bond yields           | Monthly        | FRED      |
| Sticky-price core consumer inflation                | Monthly        | FRED      |
| CBOE VIX Index                                      | Monthly        | FRED      |
| Chicago Fed National Activity Index                 | Monthly        | FRED      |
| University of Michigan Consumer Sentiment Index     | Monthly        | FRED      |
| CRSP value-weighted equity market return            | Monthly        | WRDS CRSP |
| S&P 500 return                                      | Monthly        | WRDS CRSP |
| Aggregate bond return                               | Monthly        | Bloomberg |
| Ten-year Treasury bond return                       | Monthly        | WRDS Tsy  |

## A.2 Teacher-feature definitions

**Common return definitions.**

$$\alpha_h = 1 - \exp\left(\frac{\log(0.5)}{h}\right), \quad (9)$$

$$\text{EWM}_{h,t}(r) = \alpha_h r_t + (1 - \alpha_h) \text{EWM}_{h,t-1}(r), \quad (10)$$

$$r_t^- = \min\{r_t, 0\}, \quad (11)$$

$$D_{h,t}(r) = [\text{EWM}_{h,t}((r_t^-)^2)]^{1/2}. \quad (12)$$

**Equity teacher features.** Daily equity excess return:  $r_t^E$ . Horizons:  $h \in \{21, 63, 126, 252\}$ .

$$\text{EWMRet}_{h,t}^E = \text{EWM}_{h,t}(r^E), \quad (13)$$

$$\text{Sortino}_{h,t}^E = \frac{\text{EWM}_{h,t}(r^E)}{D_{h,t}(r^E) + 10^{-12}}. \quad (14)$$

**Bond teacher features.** Monthly bond excess return:  $r_t^B$ .

$$\text{EWMRet}_{h,t}^B = \text{EWM}_{h,t}(r^B), \quad h \in \{3, 6, 12\}, \quad (15)$$

$$\text{Sharpe}_{h,t}^B = \frac{\text{mean}(r_{t-h+1}^B, \dots, r_t^B)}{\text{sd}(r_{t-h+1}^B, \dots, r_t^B) + 10^{-12}}, \quad h \in \{3, 6, 12\}, \quad (16)$$

$$\text{LogDownside}_{h,t}^B = \log(D_{h,t}(r^B) + 10^{-12}), \quad h \in \{1, 3, 6, 12\}, \quad (17)$$

$$\text{Vol}_{6,t}^B = \text{sd}(r_{t-5}^B, \dots, r_t^B). \quad (18)$$

### A.3 Student base-feature definitions

**Rolling scalar features.**

$$\text{Sum}_{h,t}(x) = \sum_{j=0}^{h-1} x_{t-j}, \quad (19)$$

$$\text{Mean}_{h,t}(x) = \frac{1}{h} \sum_{j=0}^{h-1} x_{t-j}, \quad (20)$$

$$\text{Vol}_{h,t}(x) = \text{sd}(x_{t-h+1}, \dots, x_t). \quad (21)$$

**Return-block features.** For  $x_{1,t}, \dots, x_{K,t}$ ,

$$\text{Disp}_t = \text{sd}(x_{1,t}, \dots, x_{K,t}), \quad (22)$$

$$\text{BMW}_t = \max_k x_{k,t} - \min_k x_{k,t}. \quad (23)$$

**Correlation-matrix features.** For the 12-month rolling correlation matrix  $C_t$ ,

$$\bar{\rho}_t = \text{mean}_{i < j} (C_{ij,t}), \quad (24)$$

$$|\bar{\rho}|_t = \text{mean}_{i < j} |C_{ij,t}|, \quad (25)$$

$$\text{PC1}_t = \frac{\lambda_{\max}(C_t)}{\sum_j \lambda_j(C_t)}, \quad (26)$$

$$\text{Rotation}_t = \|C_t - C_{t-1}\|_F. \quad (27)$$

**Fisher correlation changes.**

$$\rho_t^c = \min\{\max\{\rho_t, -0.999\}, 0.999\}, \quad (28)$$

$$F_t = \frac{1}{2} \log\left(\frac{1 + \rho_t^c}{1 - \rho_t^c}\right), \quad (29)$$

$$\Delta F_t = F_t - F_{t-1}. \quad (30)$$

**Non-commutativity features.**

$$\Delta \Sigma_t = \Sigma_t - \Sigma_{t-1}, \quad (31)$$

$$\Lambda_t^\Sigma = \log\left(1 + 100 \frac{\|\Sigma_t \Delta \Sigma_t - \Delta \Sigma_t \Sigma_t\|_F}{\|\Sigma_t\|_F \|\Delta \Sigma_t\|_F}\right). \quad (32)$$

The same formula defines  $\Lambda_t^C$  after replacing  $\Sigma_t$  by  $C_t$ . Here  $\|\cdot\|_F$  denotes the Frobenius norm.

**Differences.**

$$\Delta x_t = x_t - x_{t-1}, \quad (33)$$

$$\Delta \log x_t = \log(x_t) - \log(x_{t-1}). \quad (34)$$



#### A.4 Student-feature transformations

**Causal winsorisation.**

$$L_t = Q_{0.01}(x_{t-120}, \dots, x_{t-1}), \quad (35)$$

$$U_t = Q_{0.99}(x_{t-120}, \dots, x_{t-1}), \quad (36)$$

$$\tilde{x}_t = \min\{\max(x_t, L_t), U_t\}. \quad (37)$$

Window: 120 months. Minimum prior observations: 60.

**Causal z-score.**

$$z_t = \frac{\tilde{x}_t - \text{mean}(\tilde{x}_{t-60}, \dots, \tilde{x}_{t-1})}{\text{sd}(\tilde{x}_{t-60}, \dots, \tilde{x}_{t-1}) + 10^{-12}}. \quad (38)$$

Window: 60 months. Minimum prior observations: 36.

**Smoothed z-score.**

$$z_t^s = \text{EWM}_{12,t}(z). \quad (39)$$

Half-life: 12 months.

**Student feature vector.** For base features  $j = 1, \dots, 134$ ,

$$Z_t = (z_{1,t}, z_{1,t}^s, \dots, z_{134,t}, z_{134,t}^s). \quad (40)$$

#### A.5 Coupled approach: return definitions and features

For half-life  $h$ , let  $\text{EWM}_h(\cdot)$  denote the exponentially weighted moving average with that half-life. For an excess-return series  $x_t$ , define downside deviation and the Sortino ratio as

$$\text{dd}_{h,t}(x) = \left[ \text{EWM}_h(\min(x_t, 0)^2) \right]^{1/2}, \quad (41)$$

$$\text{Sortino}_{h,t}(x) = \frac{\text{EWM}_h(x)}{\text{dd}_{h,t}(x) + \varepsilon}, \quad h \in \{1, 2, 4, 12\}. \quad (42)$$

With  $m_h = \text{EWM}_h(x)$ , the coupled teacher uses four additional return-based features:

$$\text{dd\_lvl}_t = \log(\text{dd}_{6,t}(x) + \varepsilon), \quad \text{dd\_diff}_t = \log(\text{dd}_{2,t} + \varepsilon) - \log(\text{dd}_{12,t} + \varepsilon), \quad (43)$$

$$\text{ma\_cross}_t = \frac{m_3 - m_{12}}{\text{dd}_{6,t} + \varepsilon}, \quad \text{trend}_t = \text{EWM}_6 \left( \frac{x_t - \text{EWM}_{12}(x)}{\text{EWM-sd}_{24}(x_t - \text{EWM}_{12}(x)) + \varepsilon} \right). \quad (44)$$

The market-derived block for the coupling logistic (5) uses the trailing 12-month equity–bond correlation  $\rho_{12,t}^{\text{eq,bd}}$ , together with each leg’s trend<sub>*t*</sub> and dd\_lvl<sub>*t*</sub>.

**Table A.7:** Coupled teacher feature families. Each family is estimated under both the  $L^1$  and  $L^2$  specifications.

| Family               | Features                                                |
|----------------------|---------------------------------------------------------|
| <i>all</i>           | Sortino <sub>1,2,4,12</sub> , dd_lvl, dd_diff, ma_cross |
| <i>sortino</i>       | Sortino <sub>1,2,4,12</sub>                             |
| <i>sortino_short</i> | Sortino <sub>1,2,4</sub>                                |
| <i>dd</i>            | dd_lvl, dd_diff                                         |
| <i>trend</i>         | ma_cross                                                |
| <i>risk</i>          | dd_lvl, Sortino <sub>4</sub>                            |
| <i>sortino_trend</i> | Sortino <sub>2,4</sub> , ma_cross                       |
| <i>dd_trend</i>      | dd_lvl, ma_cross                                        |
| <i>sortino_dd</i>    | Sortino <sub>2,4,12</sub> , dd_lvl, dd_diff             |
| <i>core</i>          | Sortino <sub>4</sub> , dd_lvl, ma_cross                 |
| <i>mom</i>           | Sortino <sub>2</sub> , ma_cross                         |

Crossing the eleven feature families with the  $L^1$  and  $L^2$  specifications gives 22 coupled teacher members. At each month, features are z-scored against the trailing 120-month window after winsorising that window at the 1%/99% quantiles. The current observation is clipped to the same bounds before scaling, so the transformation is strictly causal. Figures D.15 and D.16 plot the resulting standardised inputs.

## B Teacher Label Estimation Details

### B.1 Teacher panels and refitting schedule

**Table B.8:** Teacher panels

| Label               | Teacher features                         | Frequency | Initial sample |
|---------------------|------------------------------------------|-----------|----------------|
| Equity regime label | Daily equity return and Sortino features | Daily     | 12 years       |
| Bond regime label   | Monthly bond return and risk features    | Monthly   | 12 years       |

$$x_{t,j}^c = \min\{\max(x_{t,j}, \mu_j - 3\sigma_j), \mu_j + 3\sigma_j\}, \quad (45)$$

$$z_{t,j} = \frac{x_{t,j}^c - \mu_j}{\sigma_j}. \quad (46)$$

Daily equity models are refitted at month ends. Monthly bond models are refitted each month. Each refit uses only observations available at the refit date. In each refit,  $\mu_j$  and  $\sigma_j$  are computed from that training sample.

### B.2 Jump-model objective

For standardised teacher features  $z_t$ , the two-state objective is

$$\min_{s_1, \dots, s_T, c_0, c_1} \sum_{t=1}^T d(z_t, c_{s_t}) + \lambda \sum_{t=2}^T \mathbf{1}\{s_t \neq s_{t-1}\}, \quad s_t \in \{0, 1\}. \quad (47)$$

### B.3 $L^2$ jump model

$$\mathcal{T}_k = \{t : s_t = k\}, \quad k \in \{0, 1\}, \quad (48)$$

$$c_k = \frac{1}{|\mathcal{T}_k|} \sum_{t \in \mathcal{T}_k} z_t, \quad (49)$$

$$d_2(z_t, c_{s_t}) = \frac{1}{2} \|z_t - c_{s_t}\|_2^2. \quad (50)$$

### B.4 $L^1$ medoid jump model

$$\tilde{z}_k = \text{median}\{z_t : t \in \mathcal{T}_k\}, \quad (51)$$

$$m_k = \arg \min_{z_i : i \in \mathcal{T}_k} \|z_i - \tilde{z}_k\|_1, \quad (52)$$

$$d_1(z_t, m_{s_t}) = \|z_t - m_{s_t}\|_1 = \sum_{j=1}^p |z_{t,j} - m_{s_t,j}|. \quad (53)$$

### B.5 Penalty grids and optimisation

$$\lambda^E \in \{2, 4, 8, 16, 32, 64, 128\}, \quad (54)$$

$$\lambda^B \in \{0, 10^{-5}, 10^{-4}, 0.001, 0.00390625, 0.0078125, 0.015625, 0.03125, 0.0625, 0.125, 0.25, 0.5, 1, 2, 4\}. \quad (55)$$

Each fit uses five initialisations and at most 100 coordinate-descent iterations. Conditional on the state centres, the state path is updated by dynamic programming.

### B.6 State ordering

For fitted raw state  $k$ , define

$$\hat{\mu}_k = \text{mean}\{r_t : s_t = k\}, \quad (56)$$

$$\hat{\sigma}_k = \text{sd}\{r_t : s_t = k\}, \quad (57)$$

$$\widehat{\text{SR}}_k = \frac{\hat{\mu}_k}{\hat{\sigma}_k}. \quad (58)$$

The favourable state is

$$g = \arg \max_{k \in \{0, 1\}} \widehat{\text{SR}}_k. \quad (59)$$

If the Sharpe-ratio comparison is tied,

$$g = \arg \max_{k \in \{0, 1\}} \hat{\mu}_k. \quad (60)$$

### B.7 Monthly labels

For daily equity paths, the monthly favourable-state share is

$$p_m^E = \frac{1}{N_m} \sum_{t \in m} \mathbf{1}\{s_t = g\}, \quad (61)$$

and the monthly equity label is

$$S_m^E = \mathbf{1}\{p_m^E \geq 0.5\}. \quad (62)$$

For monthly bond paths,

$$S_m^B = \mathbf{1}\{s_m = g\}. \quad (63)$$

For asset  $a \in \{E, B\}$ , majority-vote labels are

$$\bar{S}_m^a = \frac{1}{K_a} \sum_{k=1}^{K_a} S_{m,k}^a, \quad (64)$$

$$S_m^{a,MV} = \mathbf{1}\{\bar{S}_m^a \geq 0.5\}. \quad (65)$$

The one-month-ahead student target is

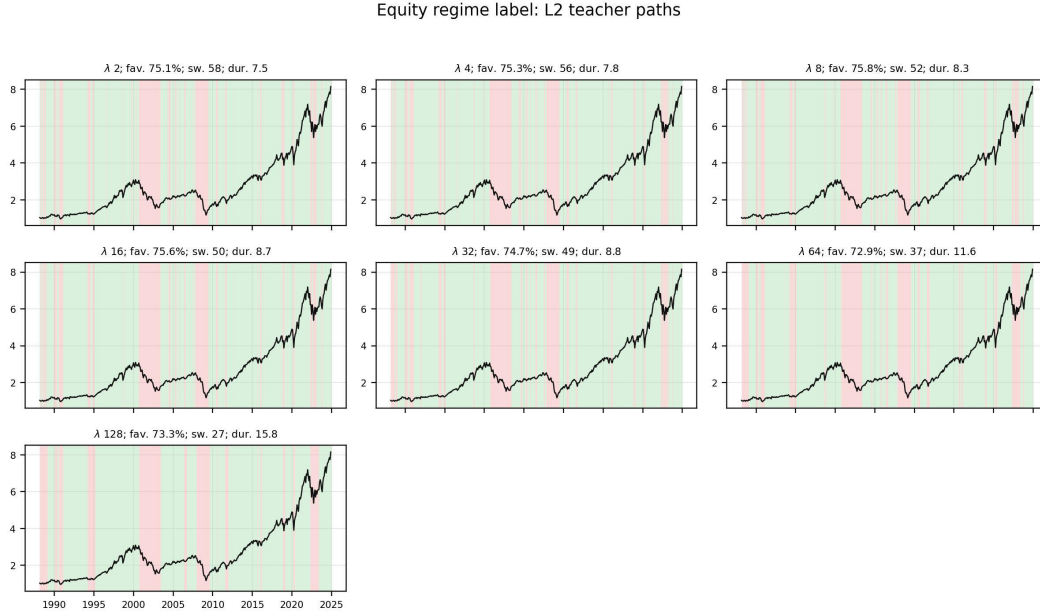
$$Y_{m+1}^a = S_{m+1}^{a,MV}. \quad (66)$$

## C Teacher Label and Feature Diagnostics

This appendix reports diagnostics for the equity and bond teacher labels. The diagnostics cover individual  $L^2$  and  $L^1$ -medoid teacher paths, jump-penalty sensitivity, majority-vote construction, transition behaviour, crisis-window behaviour, and teacher-feature state separation. These diagnostics document whether the labels are persistent, economically separated, and sufficiently stable to serve as supervised targets in the student stage. All return statistics use the monthly excess return of the labelled asset. The reported independent prediction and allocation results use the equity return–Sortino teacher label and the canonical bond teacher label; volatility/downside equity labels were candidate diagnostics and are not used in the reported allocation results.

### C.1 Individual teacher paths

Figures C.4–C.7 show the fitted teacher paths across jump-penalty values. Each panel plots the labelled asset’s excess-return index and shades months assigned to the favourable or unfavourable regime. The panel title reports the jump penalty, favourable-state share, number of switches, and average spell duration.



**Figure C.4:** Equity regime label:  $L^2$  teacher paths.

Equity regime label:  $L^1$ -medoid teacher paths

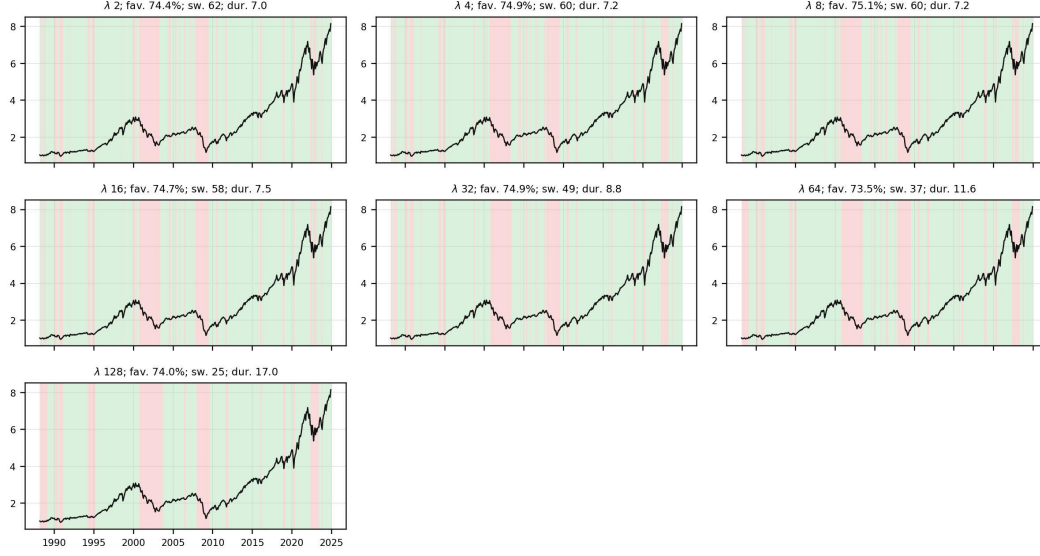


Figure C.5: Equity regime label:  $L^1$ -medoid teacher paths.

Bond regime label:  $L^2$  teacher paths

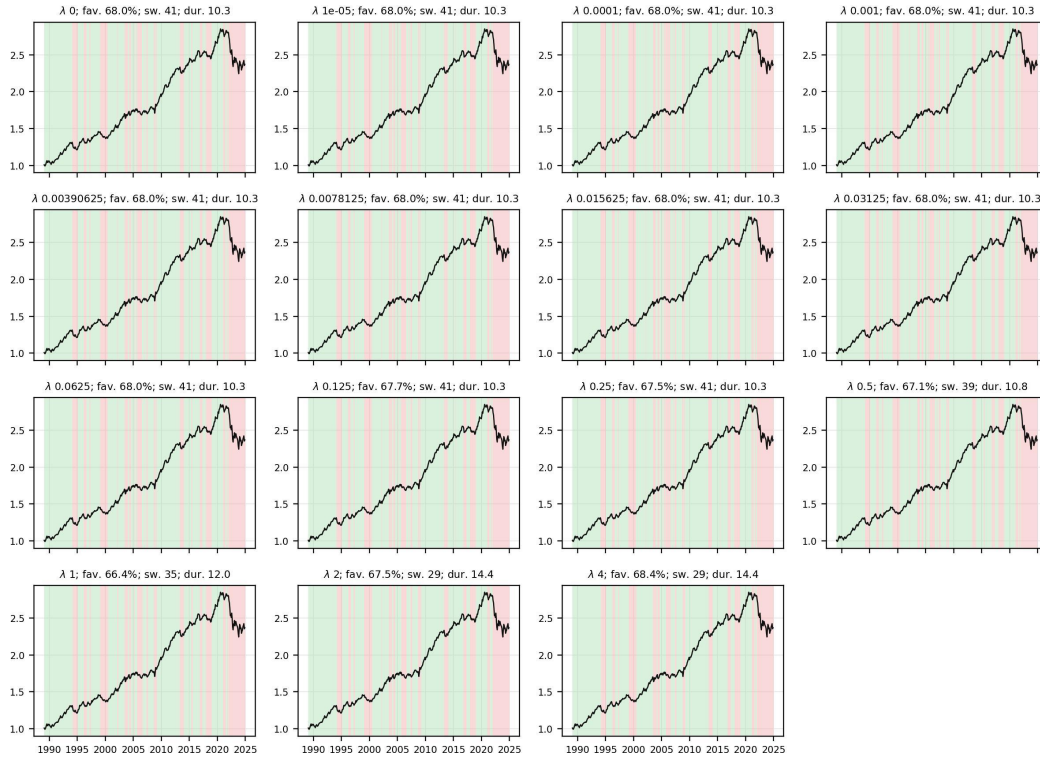


Figure C.6: Bond regime label:  $L^2$  teacher paths.

Bond regime label:  $L^1$ -medoid teacher paths

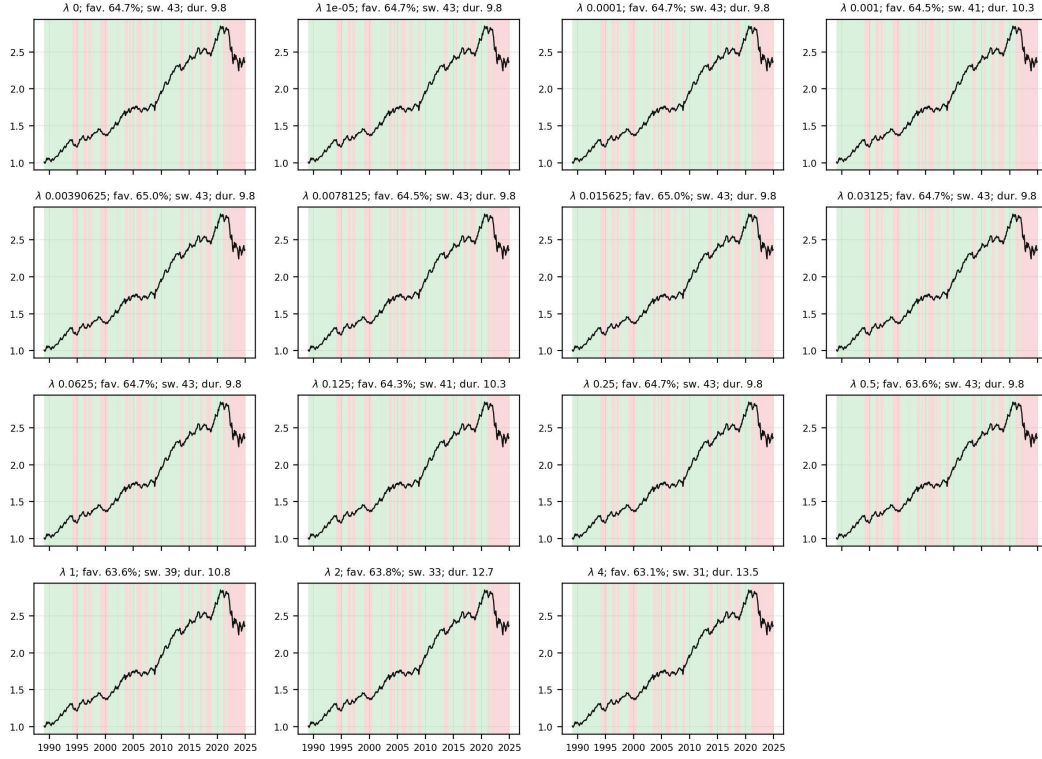


Figure C.7: Bond regime label:  $L^1$ -medoid teacher paths.

## C.2 Jump-penalty diagnostics

Figures C.8 and C.9 summarise how label properties vary across the jump-penalty grid. The panels compare the  $L^2$  and  $L^1$ -medoid teacher paths using favourable-state share, switch count, state durations, return separation, Sharpe separation, unfavourable-state volatility, and median feature-separation ratio.

Equity regime label: jump-penalty diagnostics

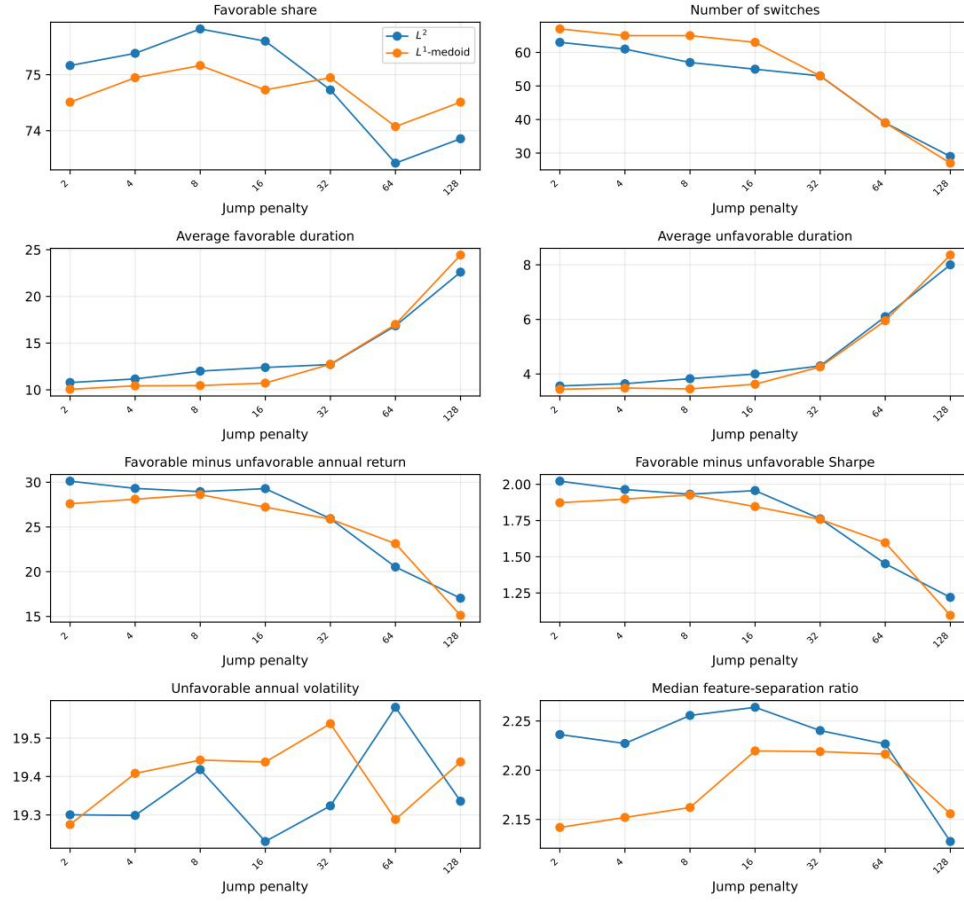


Figure C.8: Equity regime label: jump-penalty diagnostics.

Bond regime label: jump-penalty diagnostics

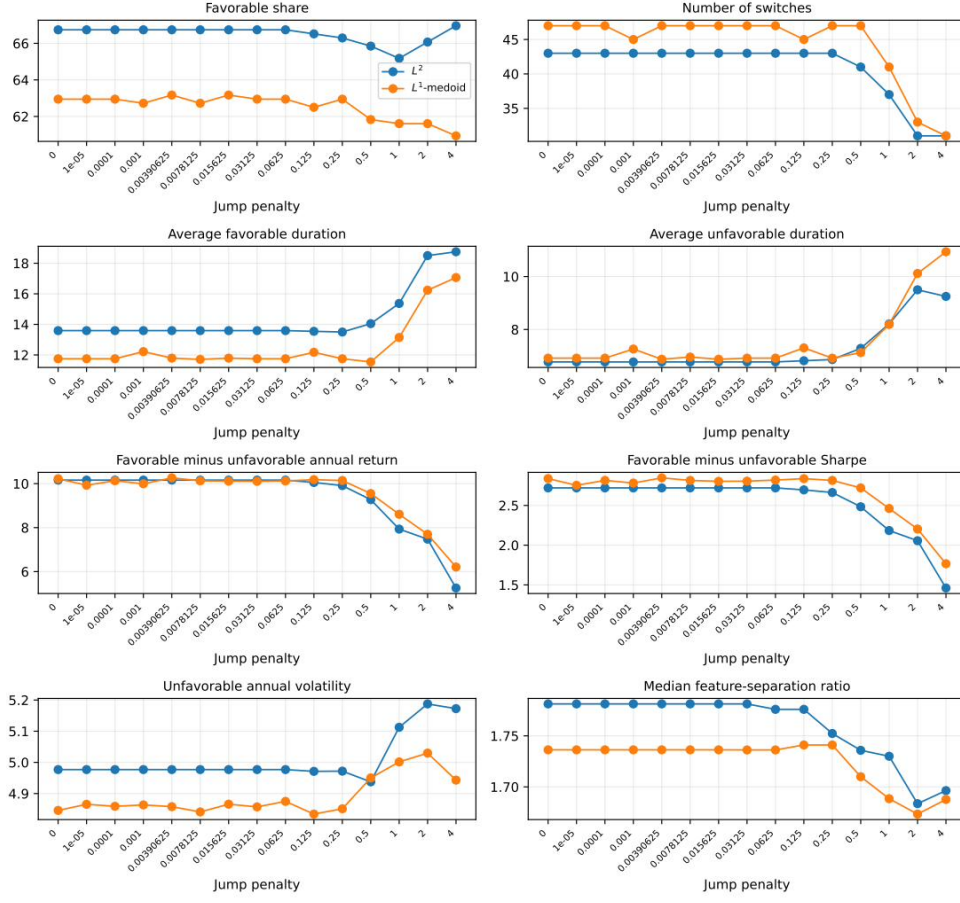


Figure C.9: Bond regime label: jump-penalty diagnostics.

### C.3 Majority-vote construction

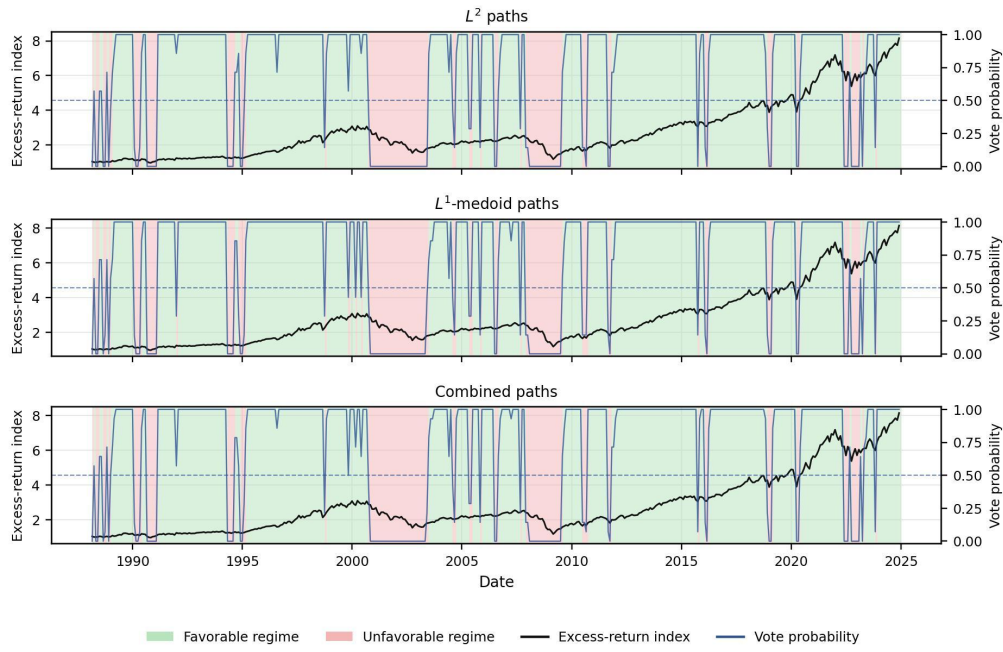
For asset  $a$ , the member vote share and final majority label are

$$\pi_m^a = \frac{1}{|\mathcal{P}_a|} \sum_{p \in \mathcal{P}_a} S_m^{a,p}, \quad S_m^a = \mathbf{1}\{\pi_m^a \geq 1/2\}. \quad (67)$$

Figures C.10 and C.11 show how the final teacher labels are formed from the  $L^2$ ,  $L^1$ -medoid, and combined teacher paths. Each panel plots the labelled asset's excess-return index, shades the majority-vote regime, and overlays the vote probability. The dashed horizontal line marks the one-half threshold used to form the majority label.

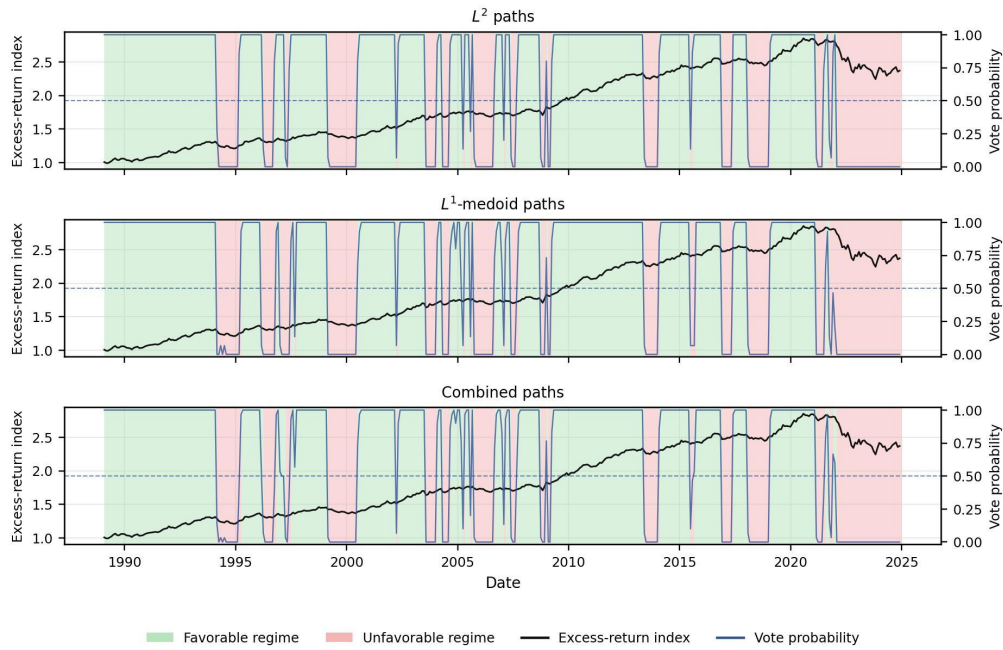


Equity regime label: majority-vote construction



**Figure C.10:** Equity regime label: majority-vote construction.

Bond regime label: majority-vote construction



**Figure C.11:** Bond regime label: majority-vote construction.

## C.4 Transitions, persistence, and crisis windows

Table C.9 reports transition probabilities and expected state durations for the final combined teacher labels.

**Table C.9:** Teacher regime label transition summary

| Label  | $P(U \rightarrow U)$ | $P(U \rightarrow F)$ | $P(F \rightarrow U)$ | $P(F \rightarrow F)$ | $E[D_U]$ | $E[D_F]$ |
|--------|----------------------|----------------------|----------------------|----------------------|----------|----------|
| Equity | 75.9%                | 24.1%                | 7.5%                 | 92.5%                | 4.2      | 13.3     |
| Bond   | 85.8%                | 14.2%                | 7.3%                 | 92.7%                | 7.1      | 13.8     |

Notes:  $F$  denotes the favourable regime and  $U$  denotes the unfavourable regime. The transition probabilities are estimated from the final monthly combined majority-vote label for each asset. Expected duration is computed as  $1/(1 - p_{ss})$  months for state  $s$ .

Table C.10 reports path-level persistence and label-conditional Sharpe statistics by asset, teacher model, and jump-penalty value.

**Table C.10:** Teacher path persistence summary

| Label  | Model         | $\lambda$  | Fav. share | Sw. | Avg. dur. | Fav. dur. | Unfav. dur. | Fav. Sharpe | Unfav. Sharpe |
|--------|---------------|------------|------------|-----|-----------|-----------|-------------|-------------|---------------|
| Equity | $L^1$ -medoid | 2          | 74.4%      | 62  | 7.0       | 10.3      | 3.6         | 1.1         | -0.68         |
| Equity | $L^1$ -medoid | 4          | 74.9%      | 60  | 7.2       | 10.7      | 3.7         | 1.1         | -0.7          |
| Equity | $L^1$ -medoid | 8          | 75.1%      | 60  | 7.2       | 10.7      | 3.7         | 1.1         | -0.72         |
| Equity | $L^1$ -medoid | 16         | 74.7%      | 58  | 7.5       | 11.0      | 3.9         | 1.1         | -0.66         |
| Equity | $L^1$ -medoid | 32         | 74.9%      | 49  | 8.8       | 13.2      | 4.4         | 1.1         | -0.61         |
| Equity | $L^1$ -medoid | 64         | 73.5%      | 37  | 11.6      | 17.1      | 6.2         | 1.1         | -0.55         |
| Equity | $L^1$ -medoid | 128        | 74.0%      | 25  | 17.0      | 25.2      | 8.8         | 0.86        | -0.24         |
| Equity | $L^2$         | 2          | 75.1%      | 58  | 7.5       | 11.1      | 3.8         | 1.2         | -0.79         |
| Equity | $L^2$         | 4          | 75.3%      | 56  | 7.8       | 11.5      | 3.9         | 1.1         | -0.76         |
| Equity | $L^2$         | 8          | 75.8%      | 52  | 8.3       | 12.4      | 4.1         | 1.1         | -0.75         |
| Equity | $L^2$         | 16         | 75.6%      | 50  | 8.7       | 12.8      | 4.3         | 1.1         | -0.76         |
| Equity | $L^2$         | 32         | 74.7%      | 49  | 8.8       | 13.2      | 4.5         | 1.1         | -0.61         |
| Equity | $L^2$         | 64         | 72.9%      | 37  | 11.6      | 16.9      | 6.3         | 1           | -0.44         |
| Equity | $L^2$         | 128        | 73.3%      | 27  | 15.8      | 23.1      | 8.4         | 0.92        | -0.31         |
| Bond   | $L^1$ -medoid | 0          | 64.8%      | 43  | 9.8       | 12.7      | 6.9         | 2           | -0.9          |
| Bond   | $L^1$ -medoid | 1e-05      | 64.8%      | 43  | 9.8       | 12.7      | 6.9         | 1.9         | -0.85         |
| Bond   | $L^1$ -medoid | 0.0001     | 64.8%      | 43  | 9.8       | 12.7      | 6.9         | 2           | -0.88         |
| Bond   | $L^1$ -medoid | 0.001      | 64.6%      | 41  | 10.3      | 13.3      | 7.3         | 2           | -0.86         |
| Bond   | $L^1$ -medoid | 0.00390625 | 65.0%      | 43  | 9.8       | 12.8      | 6.9         | 2           | -0.91         |
| Bond   | $L^1$ -medoid | 0.0078125  | 64.6%      | 43  | 9.8       | 12.7      | 7.0         | 2           | -0.88         |
| Bond   | $L^1$ -medoid | 0.015625   | 65.0%      | 43  | 9.8       | 12.8      | 6.9         | 2           | -0.88         |
| Bond   | $L^1$ -medoid | 0.03125    | 64.8%      | 43  | 9.8       | 12.7      | 6.9         | 2           | -0.88         |
| Bond   | $L^1$ -medoid | 0.0625     | 64.8%      | 43  | 9.8       | 12.7      | 6.9         | 2           | -0.88         |
| Bond   | $L^1$ -medoid | 0.125      | 64.4%      | 41  | 10.3      | 13.2      | 7.3         | 2           | -0.88         |
| Bond   | $L^1$ -medoid | 0.25       | 64.8%      | 43  | 9.8       | 12.7      | 6.9         | 2           | -0.89         |
| Bond   | $L^1$ -medoid | 0.5        | 63.7%      | 43  | 9.8       | 12.5      | 7.1         | 2           | -0.76         |
| Bond   | $L^1$ -medoid | 1          | 63.7%      | 39  | 10.8      | 13.8      | 7.8         | 1.9         | -0.63         |
| Bond   | $L^1$ -medoid | 2          | 63.9%      | 33  | 12.7      | 16.2      | 9.2         | 1.7         | -0.53         |
| Bond   | $L^1$ -medoid | 4          | 63.2%      | 31  | 13.5      | 17.1      | 9.9         | 1.5         | -0.33         |
| Bond   | $L^2$         | 0          | 68.1%      | 41  | 10.3      | 14.0      | 6.6         | 1.8         | -0.95         |
| Bond   | $L^2$         | 1e-05      | 68.1%      | 41  | 10.3      | 14.0      | 6.6         | 1.8         | -0.95         |
| Bond   | $L^2$         | 0.0001     | 68.1%      | 41  | 10.3      | 14.0      | 6.6         | 1.8         | -0.95         |
| Bond   | $L^2$         | 0.001      | 68.1%      | 41  | 10.3      | 14.0      | 6.6         | 1.8         | -0.95         |
| Bond   | $L^2$         | 0.00390625 | 68.1%      | 41  | 10.3      | 14.0      | 6.6         | 1.8         | -0.95         |
| Bond   | $L^2$         | 0.0078125  | 68.1%      | 41  | 10.3      | 14.0      | 6.6         | 1.8         | -0.95         |
| Bond   | $L^2$         | 0.015625   | 68.1%      | 41  | 10.3      | 14.0      | 6.6         | 1.8         | -0.95         |
| Bond   | $L^2$         | 0.03125    | 68.1%      | 41  | 10.3      | 14.0      | 6.6         | 1.8         | -0.95         |
| Bond   | $L^2$         | 0.0625     | 68.1%      | 41  | 10.3      | 14.0      | 6.6         | 1.8         | -0.95         |
| Bond   | $L^2$         | 0.125      | 67.8%      | 41  | 10.3      | 14.0      | 6.6         | 1.8         | -0.93         |
| Bond   | $L^2$         | 0.25       | 67.6%      | 41  | 10.3      | 13.9      | 6.7         | 1.8         | -0.91         |
| Bond   | $L^2$         | 0.5        | 67.1%      | 39  | 10.8      | 14.5      | 7.1         | 1.7         | -0.81         |
| Bond   | $L^2$         | 1          | 66.4%      | 35  | 12.0      | 15.9      | 8.1         | 1.6         | -0.59         |
| Bond   | $L^2$         | 2          | 67.6%      | 29  | 14.4      | 19.5      | 9.3         | 1.6         | -0.54         |

| Label | Model | $\lambda$ | Fav. share | Sw. | Avg. dur. | Fav. dur. | Unfav. dur. | Fav. Sharpe | Unfav. Sharpe |
|-------|-------|-----------|------------|-----|-----------|-----------|-------------|-------------|---------------|
| Bond  | $L^2$ | 4         | 68.5%      | 29  | 14.4      | 19.7      | 9.1         | 1.2         | -0.25         |

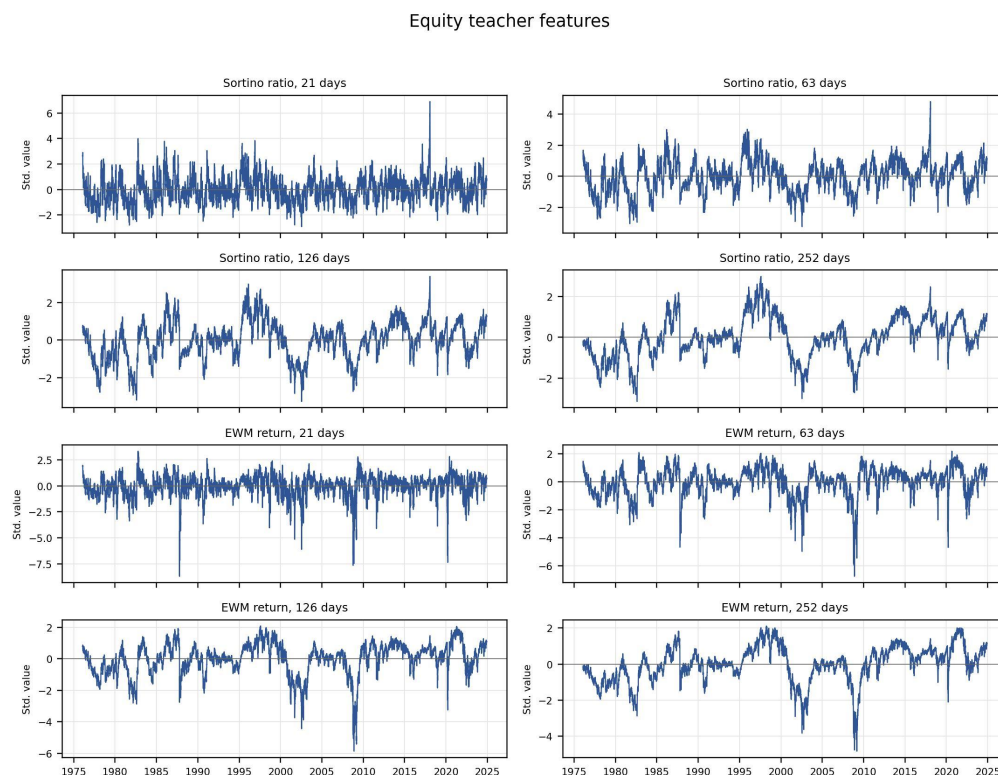
Table C.11 reports crisis-window behaviour for the final combined teacher labels. Favourable share is computed from the final monthly label within each window; return statistics use the labelled asset's monthly excess returns over the same window.

**Table C.11:** Teacher regime label crisis-window summary

| Label  | Window              | Months | Fav. share | Ann. ret. | Ann. vol. | Sharpe | Worst  | Max. DD |
|--------|---------------------|--------|------------|-----------|-----------|--------|--------|---------|
| Equity | 1990 recession      | 9      | 33.3%      | 0.8%      | 18.3%     | 0.045  | -10.0% | -16.3%  |
| Equity | 1994 bond shock     | 12     | 50.0%      | -4.9%     | 10.6%     | -0.46  | -4.8%  | -9.0%   |
| Equity | Dot-com bust        | 32     | 0.0%       | -19.9%    | 18.9%     | -1.1   | -11.1% | -46.9%  |
| Equity | GFC                 | 18     | 0.0%       | -29.6%    | 24.6%     | -1.2   | -17.0% | -50.7%  |
| Equity | Euro sovereign 2011 | 8      | 75.0%      | -10.6%    | 18.7%     | -0.57  | -7.2%  | -15.9%  |
| Equity | Covid crash         | 3      | 33.3%      | -33.8%    | 47.0%     | -0.72  | -12.6% | -12.6%  |
| Equity | Inflation 2022      | 12     | 41.7%      | -20.4%    | 22.9%     | -0.89  | -9.5%  | -21.3%  |
| Bond   | 1990 recession      | 9      | 100.0%     | 4.5%      | 3.4%      | 1.3    | -2.0%  | -2.0%   |
| Bond   | 1994 bond shock     | 12     | 8.3%       | -6.7%     | 4.3%      | -1.5   | -2.7%  | -7.8%   |
| Bond   | Dot-com bust        | 32     | 96.9%      | 6.5%      | 3.3%      | 1.9    | -1.8%  | -2.6%   |
| Bond   | GFC                 | 18     | 72.2%      | 3.3%      | 5.2%      | 0.64   | -2.4%  | -4.8%   |
| Bond   | Euro sovereign 2011 | 8      | 100.0%     | 8.8%      | 2.6%      | 3.5    | -0.3%  | -0.3%   |
| Bond   | Covid crash         | 3      | 100.0%     | 11.0%     | 4.9%      | 2.3    | -0.7%  | -0.7%   |
| Bond   | Inflation 2022      | 12     | 0.0%       | -14.9%    | 8.2%      | -1.8   | -4.5%  | -14.5%  |

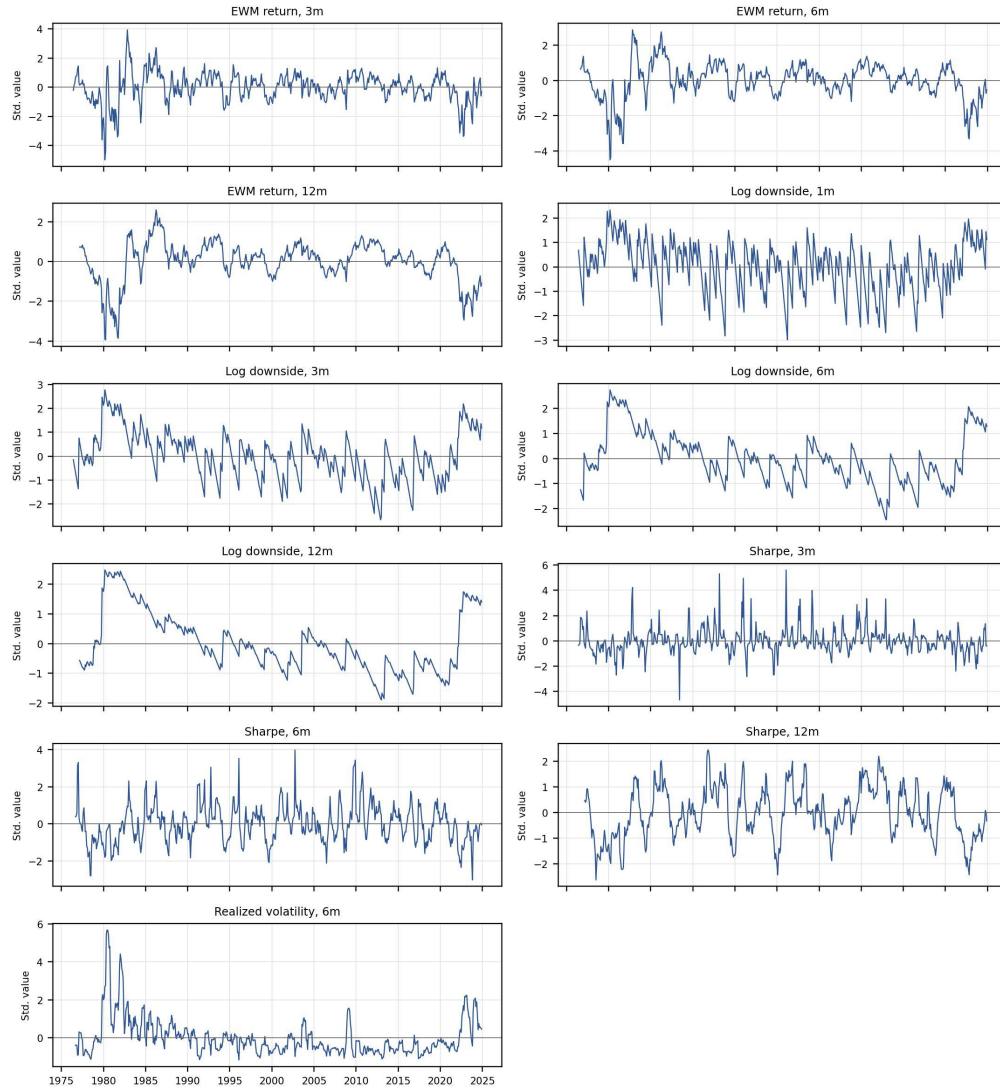
## C.5 Teacher features and state separation

Figures C.12 and C.13 show the standardised teacher features used to fit the equity and bond teacher paths. Standardisation is used only for visual comparability across features.



**Figure C.12:** Equity teacher features.

### Bond teacher features



**Figure C.13:** Bond teacher features.

Table C.12 reports the features with the strongest separation between favourable and unfavourable teacher states.

**Table C.12:** Teacher feature state-separation summary

| Label  | Model         | Feature                     | Median rank | Median sep. |
|--------|---------------|-----------------------------|-------------|-------------|
| Equity | $L^1$ -medoid | Sortino ratio, 63 days      | 1.0         | 2.53        |
| Equity | $L^1$ -medoid | Sortino ratio, 126 days     | 2.0         | 2.49        |
| Equity | $L^1$ -medoid | EWM return, 126 days        | 3.0         | 2.39        |
| Equity | $L^1$ -medoid | EWM return, 63 days         | 4.0         | 2.32        |
| Equity | $L^1$ -medoid | Sortino ratio, 252 days     | 5.0         | 2.08        |
| Equity | $L^1$ -medoid | EWM return, 252 days        | 6.0         | 1.99        |
| Equity | $L^1$ -medoid | Sortino ratio, 21 days      | 7.0         | 1.62        |
| Equity | $L^1$ -medoid | EWM return, 21 days         | 8.0         | 1.44        |
| Equity | $L^2$         | Sortino ratio, 63 days      | 2.0         | 2.59        |
| Equity | $L^2$         | Sortino ratio, 126 days     | 1.0         | 2.55        |
| Equity | $L^2$         | EWM return, 126 days        | 3.0         | 2.44        |
| Equity | $L^2$         | EWM return, 63 days         | 4.0         | 2.4         |
| Equity | $L^2$         | Sortino ratio, 252 days     | 5.0         | 2.13        |
| Equity | $L^2$         | EWM return, 252 days        | 6.0         | 2.03        |
| Equity | $L^2$         | Sortino ratio, 21 days      | 7.0         | 1.64        |
| Equity | $L^2$         | EWM return, 21 days         | 8.0         | 1.46        |
| Bond   | $L^1$ -medoid | Sharpe ratio, 12 months     | 1.0         | 2.25        |
| Bond   | $L^1$ -medoid | Sharpe ratio, 6 months      | 2.0         | 1.98        |
| Bond   | $L^1$ -medoid | Log downside risk, 3 months | 3.0         | 1.94        |
| Bond   | $L^1$ -medoid | Log downside risk, 1 month  | 4.0         | 1.93        |
| Bond   | $L^1$ -medoid | EWM return, 6 months        | 5.0         | 1.9         |
| Bond   | $L^1$ -medoid | EWM return, 3 months        | 6.0         | 1.74        |
| Bond   | $L^1$ -medoid | EWM return, 12 months       | 7.0         | 1.71        |
| Bond   | $L^1$ -medoid | Log downside risk, 6 months | 8.0         | 1.62        |
| Bond   | $L^2$         | Sharpe ratio, 12 months     | 1.0         | 2.14        |
| Bond   | $L^2$         | Sharpe ratio, 6 months      | 2.0         | 2.0         |
| Bond   | $L^2$         | Log downside risk, 3 months | 3.0         | 1.99        |
| Bond   | $L^2$         | Log downside risk, 1 month  | 4.0         | 1.99        |
| Bond   | $L^2$         | EWM return, 6 months        | 5.0         | 1.91        |
| Bond   | $L^2$         | EWM return, 3 months        | 6.0         | 1.78        |
| Bond   | $L^2$         | EWM return, 12 months       | 7.0         | 1.69        |
| Bond   | $L^2$         | Log downside risk, 6 months | 8.0         | 1.65        |

Notes: For each fitted teacher path, features are ranked by their separation between favourable and unfavourable state centroids. Median rank is the median of that within-path rank over all jump-penalty paths in the model family. Median separation is the median standardised centroid-separation ratio over those paths; larger values mean stronger separation between the two regimes.

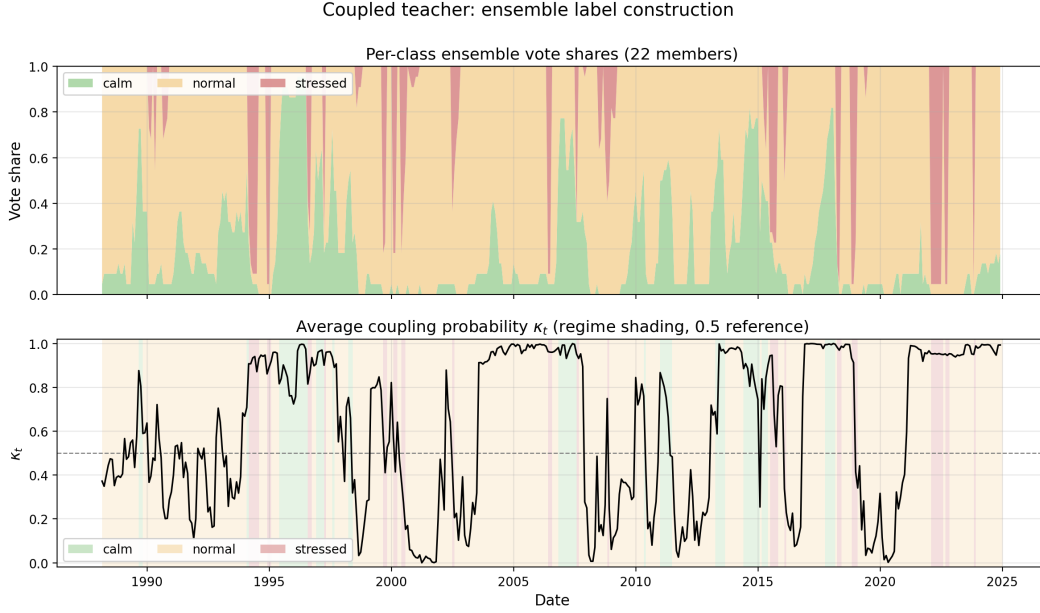
## D Coupled teacher: estimation details and diagnostics

### Objective, coupling logistic, and fitting

For standardised joint features  $(z_t^{\text{eq}}, z_t^{\text{bd}})$  the per-month cost is (4) and the path minimises the total of emission and jump costs by Viterbi. The  $L^2$  member uses Euclidean leg distances with mean centroids; the  $L^1$  member uses absolute leg distances with median centroids (two centroids per leg, indexed by the benign/adverse bit). The coupling probability  $\kappa_t$  in (5) is fitted by  $L^2$ -penalised logistic regression ( $C = 1$ ) with balanced class weights and clipped to  $[0, 1]$ . Within each monthly refit, the coupling regression is estimated on the expanding history available at the refit date. Each fit alternates: (i) re-estimate centroids and refit the coupling logistic given the provisional path; (ii) update the path by Viterbi given the centroids and  $\kappa_t$ . At most ten iterations, a single random restart plus a warm start from the previous month's path; the lowest-objective solution is kept. Per leg, the lower-mean state is set to adverse; a stressed-share cap then relabels to normal the stressed months whose equity moving-average crossover exceeds the 18% quantile of its trailing window, so only the weakest-trend months remain stressed. After a 132-month warm-up, each month is refit on the expanding history;  $(\lambda, \eta)$  is selected per month from  $\lambda \in \{2, 3, 4, 6\}$  and  $\eta \in \{1, 2, 4, 8\}$  to maximise the regime separation score subject to share, stressed-cap and minimum-run-length constraints (Table D.13).

**Table D.13:** Coupled teacher panel and causal refitting schedule.

| Setting                       | Value                                               |
|-------------------------------|-----------------------------------------------------|
| Legs                          | Equity (S&P 500 excess) and bond (aggregate excess) |
| Frequency                     | Monthly                                             |
| Warm-up (initial sample)      | 132 months (11 years)                               |
| Refit cadence                 | Every month                                         |
| Standardisation window        | 120 months, winsorised at 1%/99%                    |
| Random restarts per fit       | 1, plus a warm start from the previous month        |
| Coordinate-descent iterations | up to 10                                            |
| Jump-penalty grid $\lambda$   | $\{2, 3, 4, 6\}$                                    |
| Coupling-scale grid $\eta$    | $\{1, 2, 4, 8\}$                                    |
| Ensemble members              | 22 (11 feature families $\times \{L^1, L^2\}$ )     |
| Combination                   | Equal-weight vote (argmax of class counts)          |

**Figure D.14:** COUPLED ensemble label construction. The top panel reports per-class vote shares across the 22 coupled members. The bottom panel reports the average coupling probability  $\kappa_t$ , with regime shading and the 0.5 reference line.**Table D.14:** Coupled teacher: regime transition probabilities (%), row = from, column = to, with expected durations  $1/(1 - p_{ss})$  in months.

| From \ To | Calm | Normal | Stressed | $E[\text{dur}]$ |
|-----------|------|--------|----------|-----------------|
| Calm      | 78.7 | 14.8   | 6.6      | 4.7             |
| Normal    | 3.8  | 92.1   | 4.1      | 12.6            |
| Stressed  | 0.0  | 45.0   | 55.0     | 2.2             |

Table D.15 reports crisis-window behaviour for the final combined coupled labels. The stressed share is the fraction of months assigned to the stressed joint regime within each



window; return statistics use each leg’s monthly excess returns over the same window. This is the coupled counterpart of the independent crisis summary in Table C.11.

**Table D.15:** Coupled teacher: crisis-window summary by leg. Stressed share is the fraction of window months in the stressed joint regime; return statistics use the labelled leg’s monthly excess return over the window.

| Leg    | Window              | Months | Stressed sh. | Ann. ret. | Ann. vol. | Sharpe | Worst  | Max. DD |
|--------|---------------------|--------|--------------|-----------|-----------|--------|--------|---------|
| Equity | 1990 recession      | 9      | 0%           | +0.8%     | 17.4%     | +0.05  | −10.1% | −16.4%  |
| Equity | 1994 bond shock     | 12     | 58%          | −4.9%     | 10.1%     | −0.48  | −4.8%  | −9.0%   |
| Equity | Dot-com bust        | 32     | 3%           | −20.0%    | 18.6%     | −1.07  | −11.1% | −46.9%  |
| Equity | GFC                 | 18     | 6%           | −36.4%    | 23.1%     | −1.58  | −17.0% | −51.3%  |
| Equity | Euro sovereign 2011 | 8      | 0%           | −10.6%    | 17.5%     | −0.61  | −7.2%  | −15.9%  |
| Equity | Covid crash         | 3      | 0%           | −33.9%    | 38.4%     | −0.88  | −12.6% | −12.6%  |
| Equity | Inflation 2022      | 12     | 67%          | −20.4%    | 21.9%     | −0.93  | −9.5%  | −21.3%  |
| Bond   | 1990 recession      | 9      | 0%           | +4.5%     | 3.2%      | +1.41  | −2.0%  | −2.0%   |
| Bond   | 1994 bond shock     | 12     | 58%          | −6.7%     | 4.2%      | −1.61  | −2.7%  | −7.8%   |
| Bond   | Dot-com bust        | 32     | 3%           | +6.5%     | 3.3%      | +1.97  | −1.8%  | −2.6%   |
| Bond   | GFC                 | 18     | 6%           | +3.8%     | 5.2%      | +0.74  | −2.4%  | −4.8%   |
| Bond   | Euro sovereign 2011 | 8      | 0%           | +8.8%     | 2.4%      | +3.69  | −0.3%  | −0.3%   |
| Bond   | Covid crash         | 3      | 0%           | +11.0%    | 4.0%      | +2.76  | −0.7%  | −0.7%   |
| Bond   | Inflation 2022      | 12     | 67%          | −14.9%    | 7.9%      | −1.90  | −4.5%  | −14.5%  |

### Label construction and state summaries

Table D.16 records how the joint regime is assembled from the two per-leg state bits, and Table D.17 reports the label-conditional economics for both legs across the three joint regimes (the body reports a condensed version in Table 1).

**Table D.16:** Coupled teacher: label construction. The equity and bond per-leg state bits are estimated jointly through the coupling term, and their combination defines the three regimes.

| Component        | Teacher features                                                                                                             | Frequency | Initial sample |
|------------------|------------------------------------------------------------------------------------------------------------------------------|-----------|----------------|
| Equity state bit | Equity Sortino, drawdown and trend features                                                                                  | Monthly   | 11 years       |
| Bond state bit   | Bond Sortino, drawdown and trend features                                                                                    | Monthly   | 11 years       |
| Joint regime     | Combination (eq bit, bd bit) mapped to calm / normal / stressed, with the coupling probability $\kappa_t$ tying the two bits | Monthly   | 11 years       |

**Table D.17:** Coupled teacher: three-regime summary with spell counts. Share is the fraction of labelled months in each joint regime; equity and bond statistics are same-month annualised excess returns conditional on the regime; spells is the number of distinct runs.

| Regime   | Share | Equity (same-month) |       |        | Bond (same-month) |      |        | Avg. dur. | Spells |
|----------|-------|---------------------|-------|--------|-------------------|------|--------|-----------|--------|
|          |       | Ret.                | Vol.  | Sharpe | Ret.              | Vol. | Sharpe |           |        |
| Calm     | 13.8% | +21.0%              | 8.1%  | +2.60  | +0.8%             | 3.5% | +0.22  | 4.7       | 13     |
| Normal   | 77.1% | +9.9%               | 14.8% | +0.67  | +3.9%             | 3.9% | +1.00  | 12.2      | 28     |
| Stressed | 9.0%  | −40.9%              | 13.7% | −2.99  | −8.4%             | 5.0% | −1.69  | 2.2       | 18     |

### Member-level and feature-level diagnostics

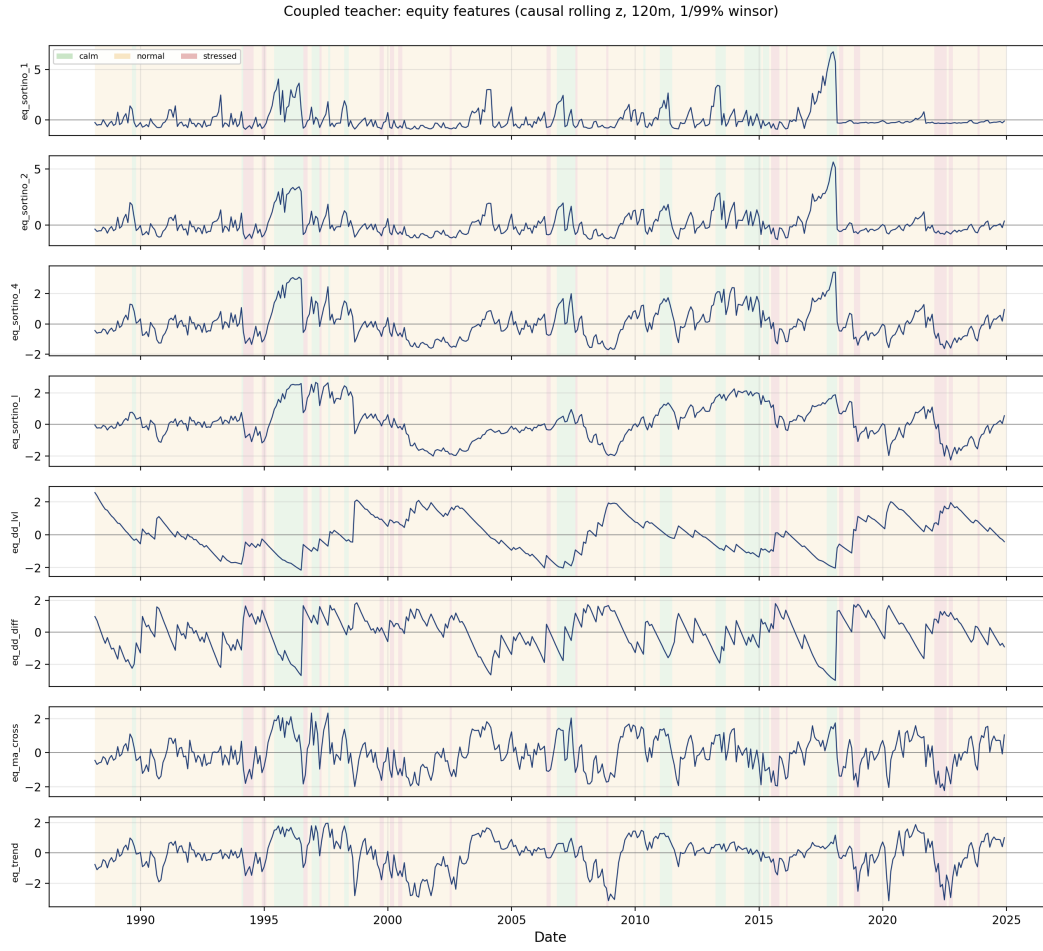
Table D.18 reports the twenty-two members. Most of the members select the smallest jump penalty  $\lambda = 2$ : the persistence in the final labels comes from the coupling term and the monthly refit, not from a large  $\lambda$ . Members separate well (most separation scores between 4 and 7), with Sortino- and momentum-based families separating best. Table D.19 ranks the per-leg features by how strongly their standardised mean separates across the three regimes.

**Table D.18:** Coupled teacher: member persistence and separation, one row per feature family and distance norm (22 members), sorted by the separation score.  $(\lambda, \eta)$  is the modal selected pair.

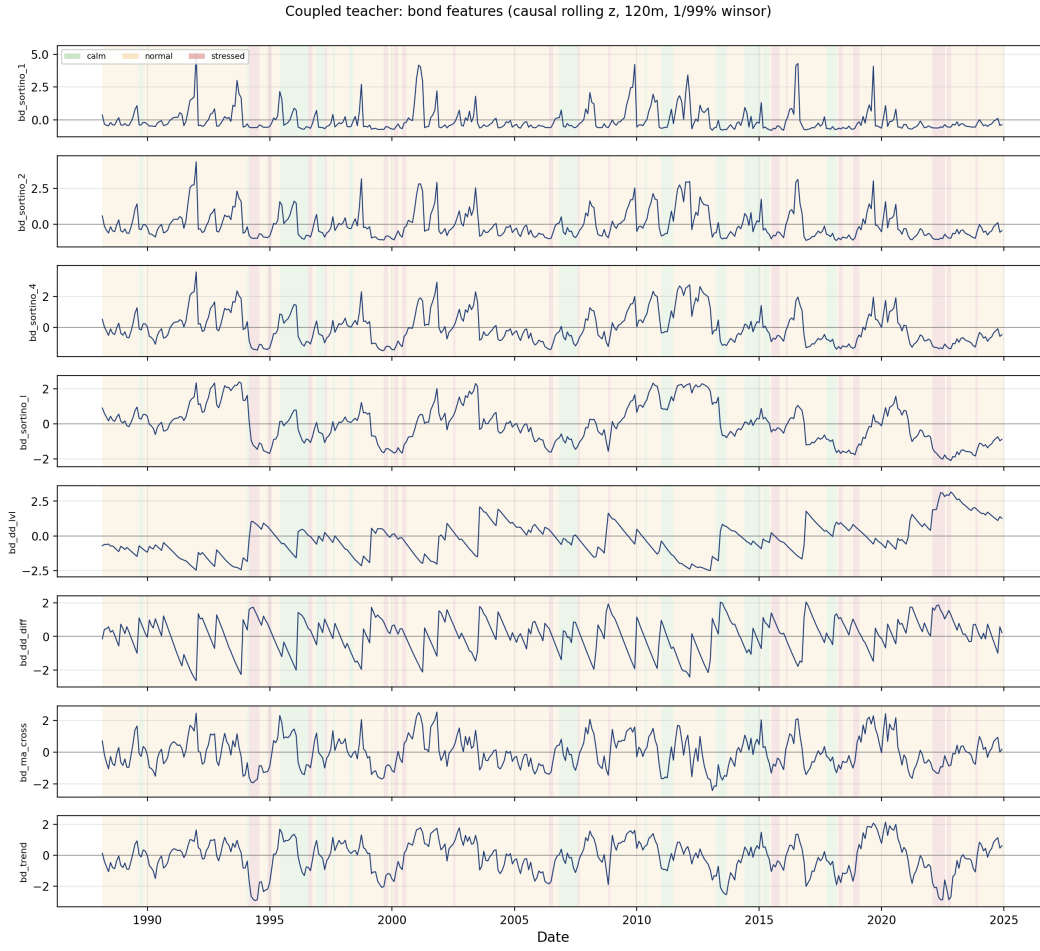
| Family        | Norm  | Calm | Normal | Stress. | Switches | Avg. dur. | Sep.  | $\lambda$ | $\eta$ |
|---------------|-------|------|--------|---------|----------|-----------|-------|-----------|--------|
| sortino       | $L^2$ | 0.03 | 0.84   | 0.13    | 59       | 7.4       | +7.07 | 2         | 8      |
| all           | $L^1$ | 0.19 | 0.70   | 0.10    | 86       | 5.1       | +6.24 | 2         | 1      |
| sortino_trend | $L^2$ | 0.06 | 0.83   | 0.11    | 55       | 7.9       | +5.96 | 2         | 4      |
| sortino       | $L^1$ | 0.19 | 0.70   | 0.11    | 77       | 5.7       | +5.89 | 2         | 2      |
| mom           | $L^2$ | 0.05 | 0.83   | 0.12    | 55       | 7.9       | +5.83 | 2         | 4      |
| core          | $L^1$ | 0.13 | 0.78   | 0.09    | 52       | 8.3       | +5.70 | 2         | 1      |
| sortino_dd    | $L^1$ | 0.18 | 0.72   | 0.10    | 62       | 7.0       | +5.69 | 2         | 1      |
| sortino_dd    | $L^2$ | 0.07 | 0.81   | 0.12    | 53       | 8.2       | +5.67 | 2         | 1      |
| sortino_short | $L^1$ | 0.19 | 0.71   | 0.09    | 71       | 6.1       | +5.60 | 2         | 2      |
| sortino_trend | $L^1$ | 0.28 | 0.61   | 0.11    | 77       | 5.7       | +5.29 | 2         | 2      |
| all           | $L^2$ | 0.22 | 0.71   | 0.07    | 51       | 8.5       | +5.10 | 2         | 2      |
| mom           | $L^1$ | 0.31 | 0.60   | 0.09    | 83       | 5.3       | +5.04 | 2         | 1      |
| risk          | $L^2$ | 0.21 | 0.71   | 0.07    | 40       | 10.8      | +4.60 | 2         | 2      |
| dd_trend      | $L^1$ | 0.11 | 0.81   | 0.08    | 46       | 9.4       | +4.59 | 2         | 8      |
| risk          | $L^1$ | 0.21 | 0.71   | 0.08    | 39       | 11.1      | +4.54 | 2         | 1      |
| dd_trend      | $L^2$ | 0.21 | 0.74   | 0.05    | 31       | 13.8      | +4.41 | 2         | 1      |
| core          | $L^2$ | 0.25 | 0.67   | 0.08    | 48       | 9.0       | +4.37 | 3         | 4      |
| trend         | $L^2$ | 0.27 | 0.66   | 0.07    | 53       | 8.2       | +4.33 | 2         | 1      |
| sortino_short | $L^2$ | 0.01 | 0.84   | 0.15    | 55       | 7.9       | +4.18 | 2         | 4      |
| dd            | $L^1$ | 0.21 | 0.72   | 0.08    | 41       | 10.5      | +4.07 | 2         | 4      |
| dd            | $L^2$ | 0.16 | 0.75   | 0.09    | 42       | 10.3      | +3.96 | 2         | 1      |
| trend         | $L^1$ | 0.66 | 0.33   | 0.01    | 63       | 6.9       | +2.15 | 2         | 8      |

**Table D.19:** Coupled teacher: feature state-separation. For each feature the standardised value is averaged within each regime; “Sep.” is the spread (max – min) of the three group means.

| Feature      | Sep. | Calm  | Normal | Stressed |
|--------------|------|-------|--------|----------|
| eq_sortino_4 | 2.62 | +1.72 | −0.13  | −0.90    |
| eq_sortino_2 | 2.54 | +1.72 | −0.12  | −0.82    |
| eq_ma_cross  | 2.30 | +0.89 | −0.04  | −1.41    |
| eq_sortino_1 | 2.17 | +1.56 | −0.10  | −0.61    |
| eq_dd_diff   | 2.08 | −1.05 | +0.01  | +1.03    |
| eq_trend     | 2.02 | +0.82 | −0.09  | −1.20    |
| eq_sortino_l | 1.99 | +1.60 | −0.07  | −0.39    |
| bd_trend     | 1.60 | −0.09 | +0.07  | −1.52    |
| bd_sortino_l | 1.43 | −0.06 | +0.14  | −1.30    |
| eq_dd_lvl    | 1.41 | −1.18 | +0.24  | +0.22    |
| bd_dd_lvl    | 1.29 | −0.45 | −0.31  | +0.84    |
| bd_sortino_4 | 1.25 | −0.13 | +0.12  | −1.13    |
| bd_ma_cross  | 1.05 | −0.11 | +0.10  | −0.95    |
| bd_dd_diff   | 1.02 | +0.05 | −0.04  | +0.98    |
| bd_sortino_2 | 0.95 | −0.17 | +0.08  | −0.87    |
| bd_sortino_1 | 0.62 | −0.23 | +0.05  | −0.58    |



**Figure D.15:** Coupled teacher: equity features after the causal rolling z-score (120-month window, 1%/99% winsorisation).

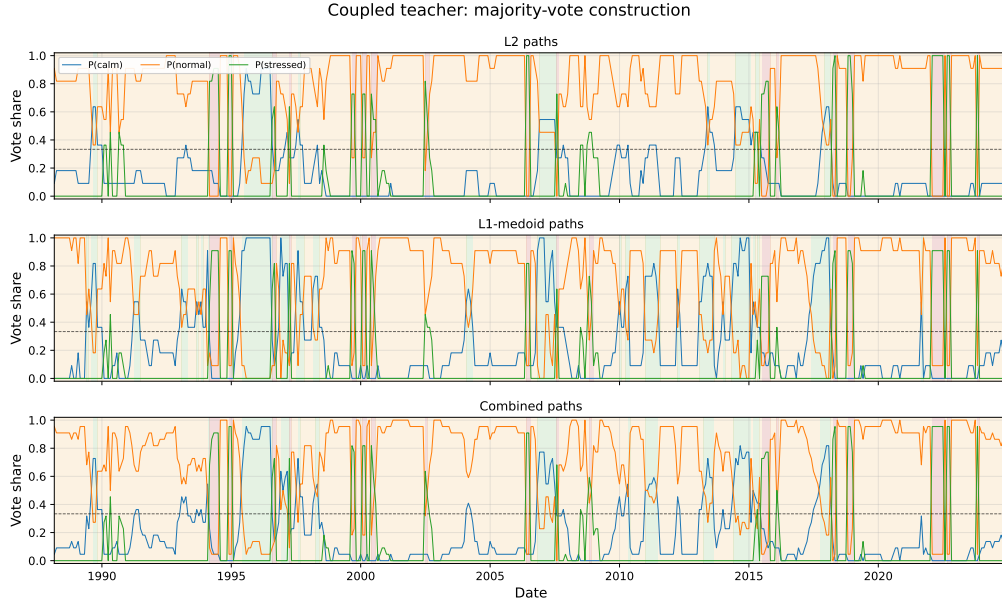


**Figure D.16:** Coupled teacher: bond features after the same causal rolling z-score. The flat left edge is the warm-up region before the standardisation window is populated.



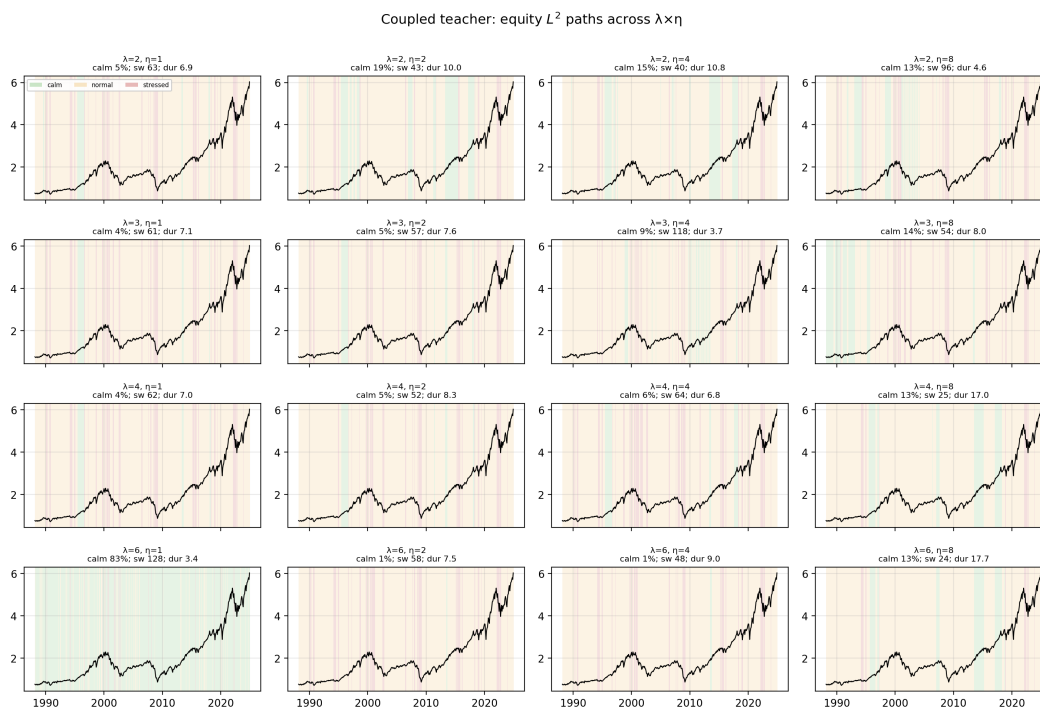
**Figure D.17:** Coupled teacher: per-member regime labels across time for the 22 ensemble members (rows), illustrating the agreement that produces the majority-vote label.

Figure D.18 shows how the final coupled label is formed from the candidate member paths under three scopes ( $L^2$  members,  $L^1$ -medoid members, and all 22 combined), the coupled counterpart of the per-leg independent majority-vote construction in Figures C.10–C.11.

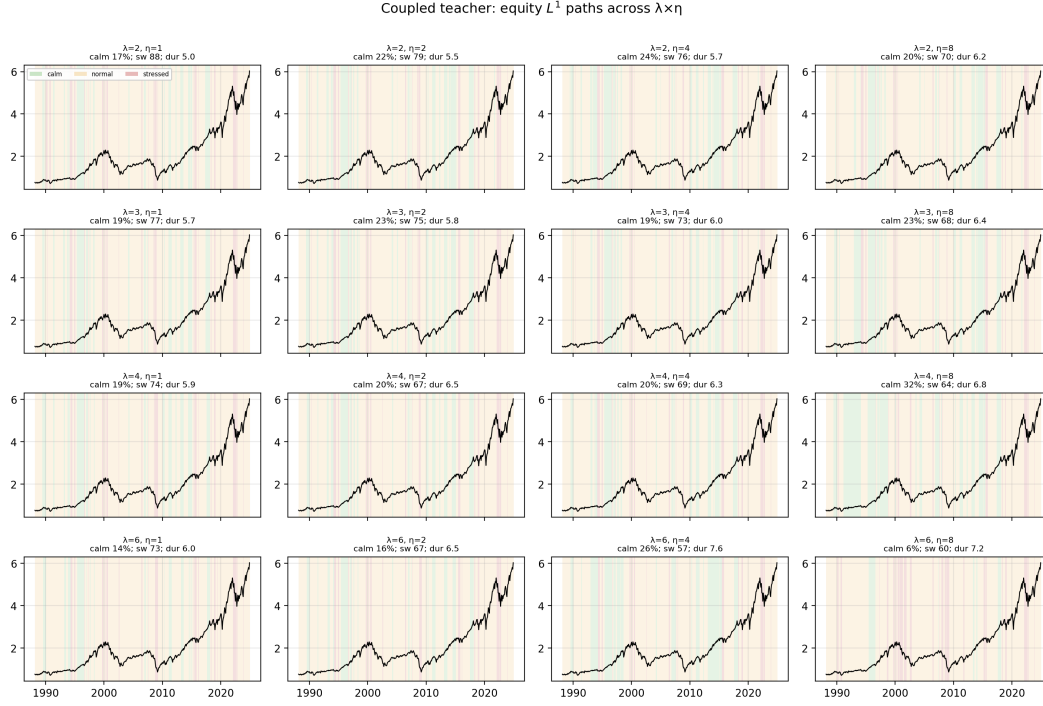


**Figure D.18:** Coupled teacher: majority-vote construction. Each panel restricts the ensemble to a scope ( $L^2$  members,  $L^1$ -medoid members, or all 22 combined), shades the equity excess-return index by the scope's majority joint regime, and overlays the majority-class vote share on the right axis. The dashed line marks an absolute majority (0.5); below it the winning class is a plurality among the three regimes.

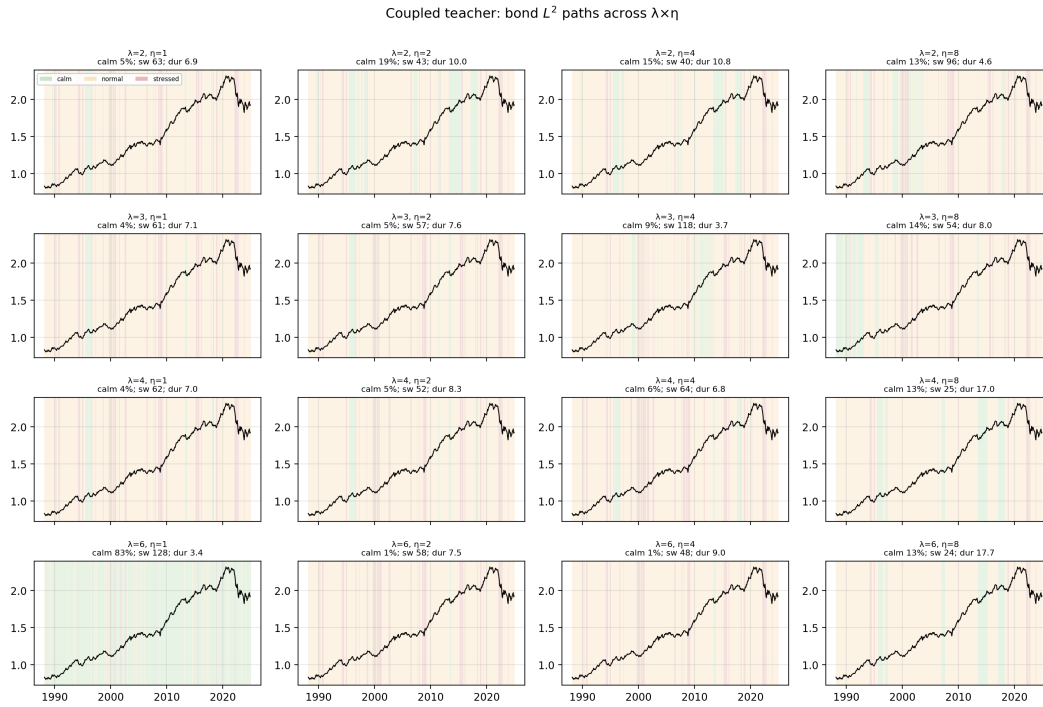
**Jump-penalty diagnostics.** To isolate the effect of persistence, the jump penalty is held fixed across a wider grid  $\lambda \in \{2, 3, 4, 6, 8, 12, 16\}$  at  $\eta = 2$  for the all-feature member under each norm (these paths form a sensitivity diagnostic, not the reported labels). A larger penalty produces longer, stickier regimes: switch counts fall and durations rise with  $\lambda$  while the calm share shrinks, and the separation score is highest at small penalties—consistent with the reported specification selecting  $\lambda = 2$  for most of the members (Figures D.19–D.25).



**Figure D.19:** Coupled teacher: equity  $L^2$  paths across the jump penalty (each panel fixes  $\lambda$ ,  $\eta = 2$ , and shades the equity excess-return index by the fitted regime).



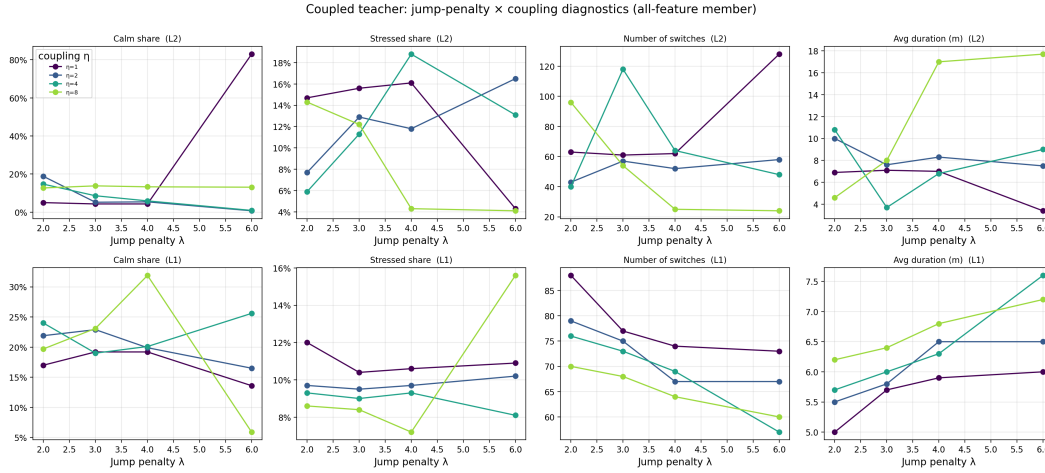
**Figure D.20:** Coupled teacher: equity  $L^1$  paths across the jump penalty.



**Figure D.21:** Coupled teacher: bond  $L^2$  paths across the jump penalty.



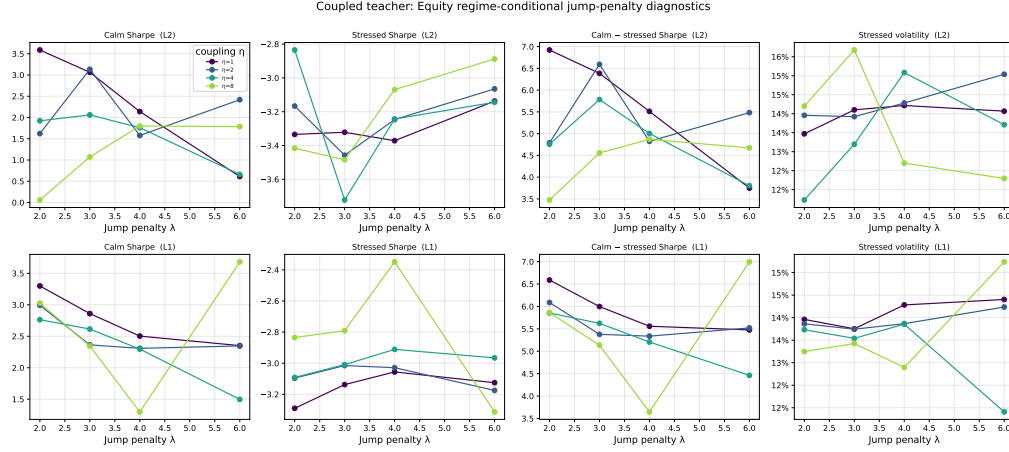
**Figure D.22:** Coupled teacher: bond  $L^1$  paths across the jump penalty.



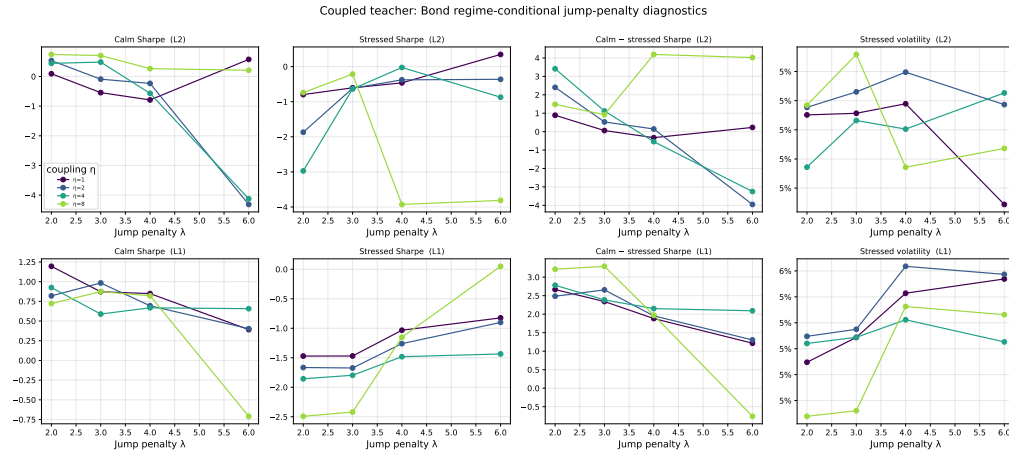
**Figure D.23:** Coupled teacher: jump-penalty diagnostics for the all-feature member—regime shares, switch count, average duration and separation score as functions of  $\lambda$ , for the  $L^1$  and  $L^2$  norms.

Figures D.24 and D.25 report the regime-conditional economics as a function of the jump penalty for each leg separately, the coupled counterpart of the per-leg independent jump-penalty diagnostics in Figures C.8 and C.9.





**Figure D.24:** Coupled teacher: equity regime-conditional jump-penalty diagnostics. For each norm ( $L^2$ ,  $L^1$ -medoid) and coupling value  $\eta$ , the panels report the equity calm-state Sharpe, stressed-state Sharpe, calm-minus-stressed Sharpe separation, and stressed-state volatility as functions of the jump penalty  $\lambda$ .



**Figure D.25:** Coupled teacher: bond regime-conditional jump-penalty diagnostics, defined as in Figure D.24 using the bond excess return.

## E Student Prediction Details

### E.1 Targets, features, and validation

The student models predict the one-month-ahead combined majority labels for the equity regime and the bond regime. The feature vector contains the z-score and smoothed z-score transformation of each monthly base predictor.

For each target and model family, prediction is performed with a rolling validation design. The initial training sample is 96 monthly observations, followed by a 36-month validation window. Candidate hyperparameters are selected on the validation data, and the selected setting is estimated on the full available sample before the forecast month. Validation ranking uses AUC, balanced accuracy, Brier score, net Sharpe ratio, drawdown improvement, volatility reduction, and turnover.

Predicted probabilities are averaged over seeds  $\{101, 111, 121\}$ .

## E.2 Model grids

The logistic student model uses median imputation, feature standardisation, balanced class weights, and  $L^2$  regularisation. The regularisation grid is

$$C \in \{0.03, 0.05, 0.10, 0.25, 0.50, 1.00\}. \quad (68)$$

The random-forest student model uses median imputation, 500 trees, bootstrap sampling, square-root feature subsampling, and balanced subsample class weights. The grid is

$$(d, \ell) \in \{(2, 10), (2, 20), (3, 10), (3, 20), (4, 20)\}, \quad (69)$$

where  $d$  is tree depth and  $\ell$  is the minimum leaf size.

## F Student regime prediction diagnostics

**Table F.20:** Out-of-sample student prediction summary

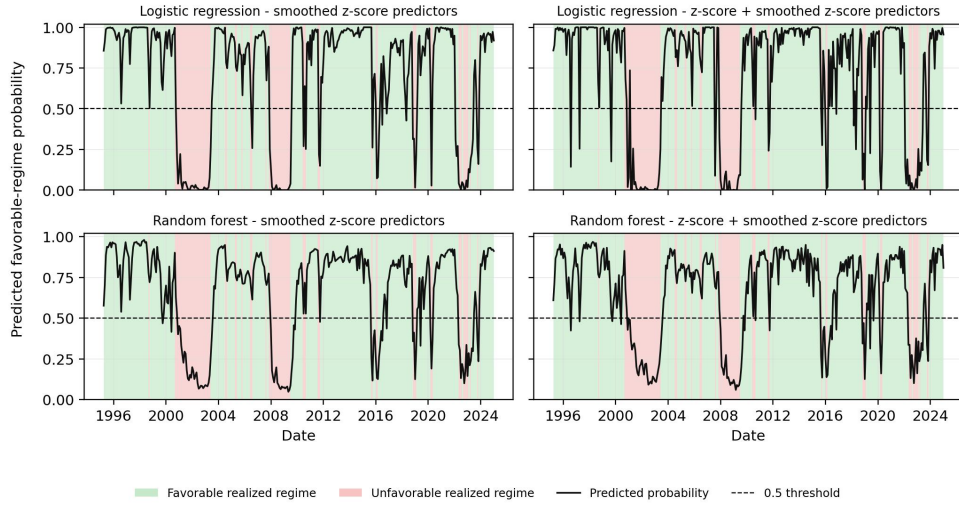
| Label         | Predictor set              | Model               | OOS sample      | Fav. share | Mean $\hat{p}$ | AUC   | Bal. acc. | Brier | Log loss |
|---------------|----------------------------|---------------------|-----------------|------------|----------------|-------|-----------|-------|----------|
| Bond regime   | z-score + smoothed z-score | Logistic regression | 1996-02–2024-12 | 62.2%      | 0.594          | 0.879 | 0.841     | 0.125 | 0.515    |
| Bond regime   | z-score + smoothed z-score | Random forest       | 1996-02–2024-12 | 62.2%      | 0.601          | 0.873 | 0.800     | 0.147 | 0.471    |
| Bond regime   | smoothed z-score           | Logistic regression | 1996-02–2024-12 | 62.2%      | 0.574          | 0.892 | 0.831     | 0.128 | 0.462    |
| Bond regime   | smoothed z-score           | Random forest       | 1996-02–2024-12 | 62.2%      | 0.590          | 0.849 | 0.784     | 0.153 | 0.484    |
| Equity regime | z-score + smoothed z-score | Logistic regression | 1995-03–2024-12 | 76.3%      | 0.707          | 0.929 | 0.846     | 0.088 | 0.296    |
| Equity regime | z-score + smoothed z-score | Random forest       | 1995-03–2024-12 | 76.3%      | 0.658          | 0.903 | 0.831     | 0.107 | 0.361    |
| Equity regime | smoothed z-score           | Logistic regression | 1995-03–2024-12 | 76.3%      | 0.700          | 0.918 | 0.862     | 0.086 | 0.307    |
| Equity regime | smoothed z-score           | Random forest       | 1995-03–2024-12 | 76.3%      | 0.660          | 0.902 | 0.837     | 0.105 | 0.355    |

*Note.* The table reports out-of-sample diagnostics from the selected rolling student models. AUC is the probability that a favourable-regime month receives a higher predicted probability than an unfavourable-regime month. Balanced accuracy is  $\frac{1}{2}\{TP/(TP + FN) + TN/(TN + FP)\}$ . The Brier score is  $n^{-1} \sum_i (\hat{p}_i - y_i)^2$ , and log loss is  $-n^{-1} \sum_i \{y_i \log(\hat{p}_i) + (1 - y_i) \log(1 - \hat{p}_i)\}$ , where  $y_i$  is the realised teacher label and  $\hat{p}_i$  is the predicted favourable-regime probability.

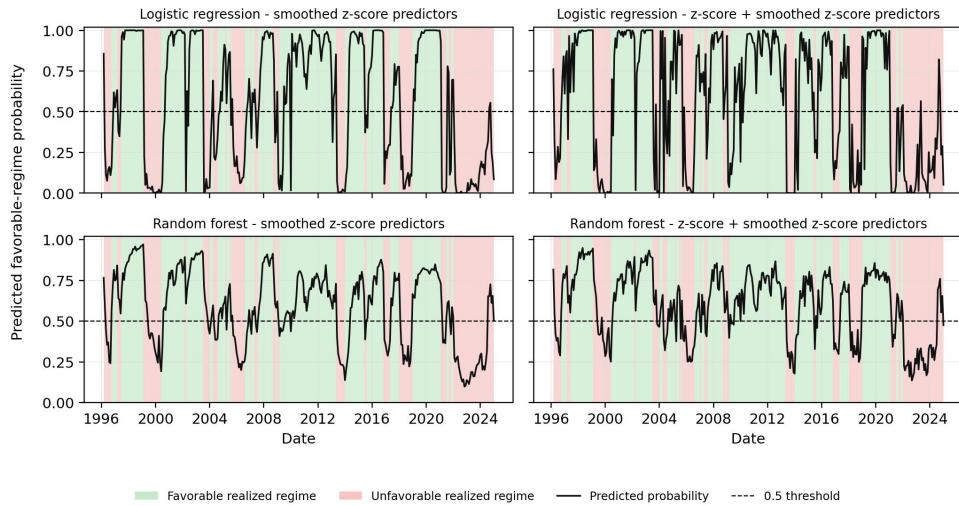
Equity-regime prediction is strongest for logistic regression. The logistic model with the combined z-score and smoothed z-score predictor set reaches an AUC of 0.929, balanced accuracy of 0.846, Brier score of 0.088, and log loss of 0.296. The smoothed-only logistic model gives the highest balanced accuracy for the equity label, 0.862, with a similar AUC of 0.918. Bond-regime prediction is strongest for logistic regression with the smoothed-only predictor set, which reaches an AUC of 0.892 and balanced accuracy of 0.831. Random forests also predict both labels, but their AUC and balanced-accuracy values are below the corresponding logistic-regression values in most specifications.

Figures F.26 and F.26 plot the out-of-sample probability paths. The shaded background marks the realised teacher regime, and the black line is the student-model probability assigned to the favourable regime.

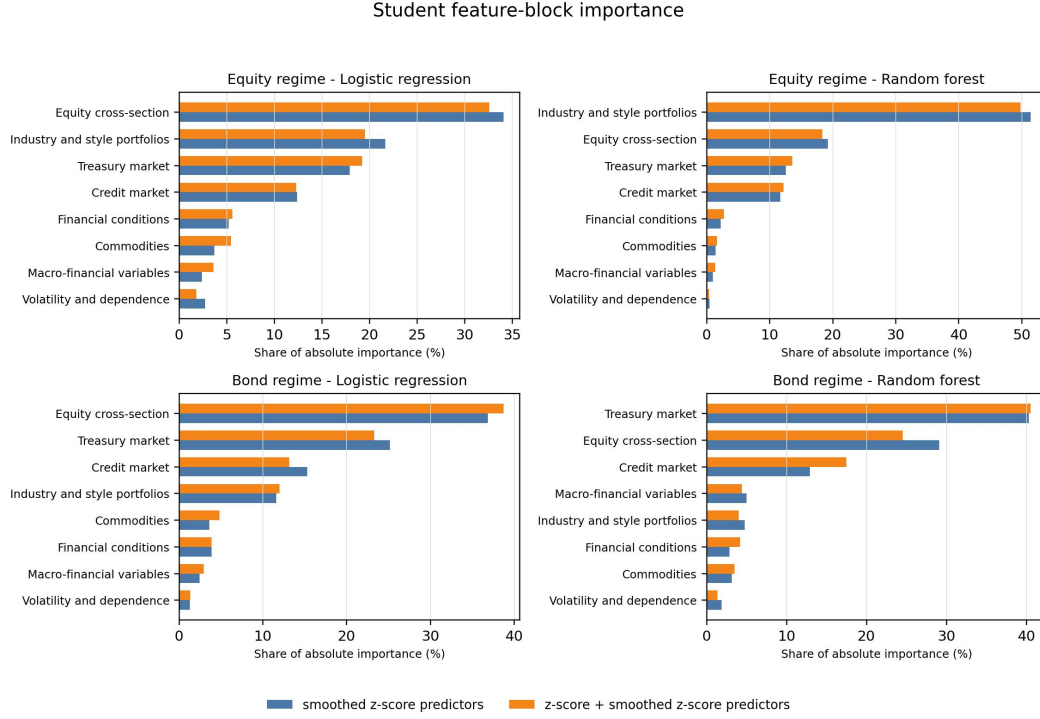
### Equity regime prediction: out-of-sample probabilities



### Bond regime prediction: out-of-sample probabilities



**Figure F.26:** INDEPENDENT student out-of-sample probability paths. The top panel reports equity favourable-regime probabilities, and the bottom panel reports bond favourable-regime probabilities. Shading marks the realised teacher label, the black line reports the predicted favourable-regime probability, and the dashed line marks the 0.5 threshold.



**Figure F.27:** INDEPENDENT student feature-block relevance. Logistic-regression relevance is measured by absolute coefficient magnitude. Random-forest relevance is measured by impurity-based feature importance. Values are aggregated within predictor blocks.

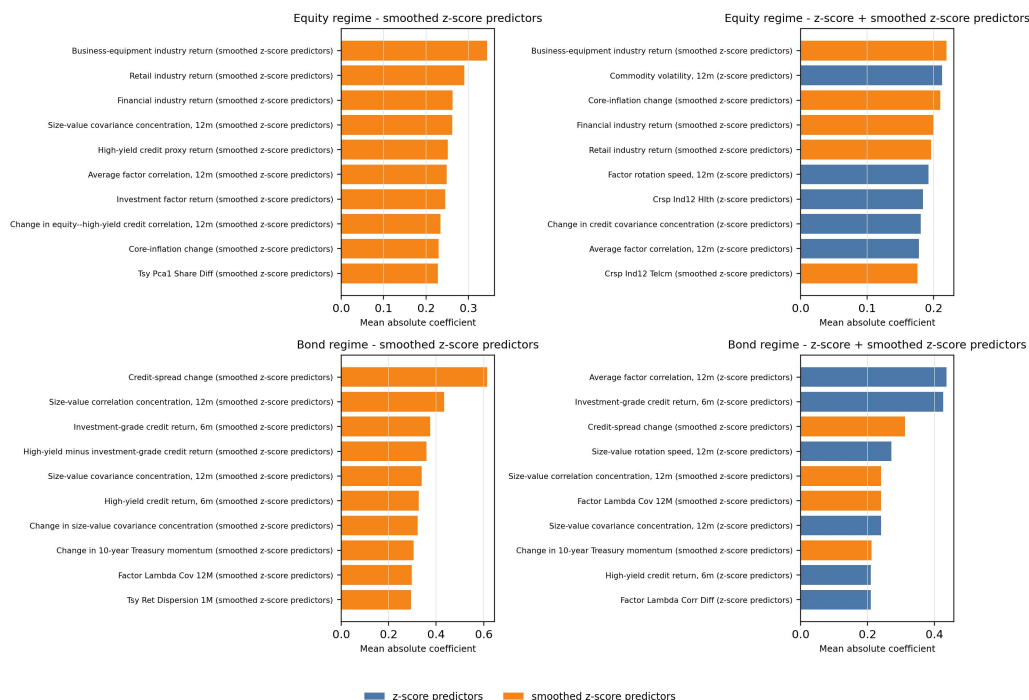
Table F.21 reports the most frequently selected hyperparameter setting over the rolling out-of-sample test months. The table reports which specification is most often selected within each target, predictor set, and model class.

**Table F.21:** Most frequent selected student-model specification

| Label         | Predictor set                         | Model               | Most frequent setting                          | Sel. share | Months |
|---------------|---------------------------------------|---------------------|------------------------------------------------|------------|--------|
| Bond regime   | z-score + smoothed z-score predictors | Logistic regression | $C = 0.03$ , class weight=balanced             | 56.2%      | 347    |
| Bond regime   | z-score + smoothed z-score predictors | Random forest       | trees=500, depth=3, leaf=10, max features=sqrt | 32.3%      | 347    |
| Bond regime   | smoothed z-score predictors           | Logistic regression | $C = 0.03$ , class weight=balanced             | 38.6%      | 347    |
| Bond regime   | smoothed z-score predictors           | Random forest       | trees=500, depth=2, leaf=10, max features=sqrt | 30.8%      | 347    |
| Equity regime | z-score + smoothed z-score predictors | Logistic regression | $C = 0.03$ , class weight=balanced             | 60.1%      | 358    |
| Equity regime | z-score + smoothed z-score predictors | Random forest       | trees=500, depth=3, leaf=10, max features=sqrt | 67.6%      | 358    |
| Equity regime | smoothed z-score predictors           | Logistic regression | $C = 0.03$ , class weight=balanced             | 56.4%      | 358    |
| Equity regime | smoothed z-score predictors           | Random forest       | trees=500, depth=3, leaf=10, max features=sqrt | 70.7%      | 358    |

*Note.* The table reports the most frequently selected hyperparameter setting across out-of-sample rolling test months. The selection share is the fraction of out-of-sample months in which that setting is selected for the corresponding regime label, model class, and predictor set.

### Individual predictor importance, logistic regression



**Figure F.28:** Student prediction individual predictor importance.

*Note.* Individual predictor importance uses model-level importances from the rolling prediction exercise. For logistic regression, importance is the absolute value of the fitted coefficient after preprocessing. For random forests, importance is impurity-based feature importance. Mean absolute importance is averaged across rolling selected-model fits and seeds. When both the z-score and smoothed z-score transformations of the same predictor are available within a predictor set, the plotted value is the larger mean absolute importance for that economic predictor. Each panel shows the eight largest economic predictors by the maximum mean absolute importance across the two predictor sets for the same label and model.

The individual-predictor diagnostics are consistent with the block-level figure. Equity-regime prediction relies heavily on industry returns, style-portfolio returns, equity-factor information, and cross-market credit correlation. Bond-regime prediction relies on credit-spread changes, investment-grade and high-yield credit returns, Treasury returns, and equity-factor dependence measures. These predictors explain why the equity cross-section, Treasury market, and credit market blocks receive high importance shares in the block-level results.

The feature-block definitions are: equity cross-section contains equity factors, size-value variables, and equity dispersion or dependence measures; Treasury market contains Treasury returns, yield-curve and rate changes, bond momentum, and bond-market dependence; credit market contains investment-grade and high-yield credit returns, credit spreads, dispersion, and credit dependence; commodities contain commodity returns, momentum, volatility, Sharpe ratios, and commodity dependence; financial conditions contain NFCI, adjusted NFCI, their changes, and their spread; macro-financial variables contain activity, inflation, and sentiment variables; volatility and dependence contains volatility and remaining correlation or concentration measures; and industry and style portfolios contain remaining industry and style-portfolio returns.

## G Coupled student: estimation details

**Coupled student.** The target is the next-month three-class regime. At each refit the five predictors most correlated with the next-month regime over the trailing window are kept, and the current-regime indicators and run-length are appended. Two classifiers are fitted with balanced class weights: an  $L_2$ -penalised multinomial logistic regression and a random forest. Estimation is walk-forward (132-month training, 60-month validation by balanced accuracy, 6-month test), giving the out-of-sample span 1996-02 to 2024-10.

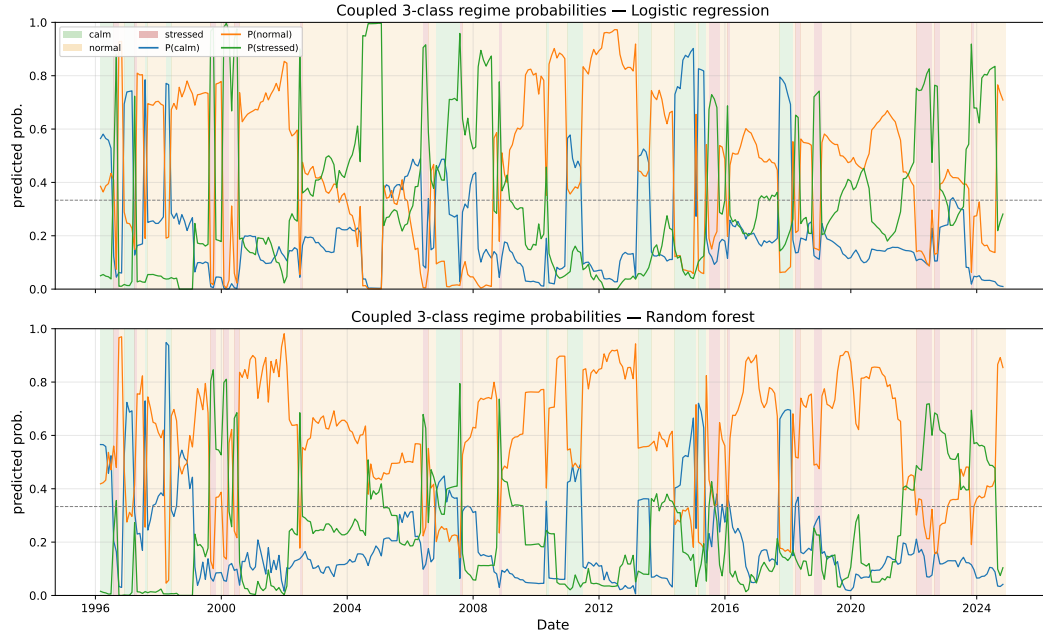
**Coupled student forecast diagnostics.** Table G.22 reports the out-of-sample probabilistic metrics for the two coupled students, and Table G.23 reports the most frequently selected hyper-parameter setting across the walk-forward windows. The predicted three-class probability paths are shown in Figure G.29. Figure G.30 reports block-level relevance, and Figure G.31 reports individual-predictor importance.

**Table G.22:** Coupled student: out-of-sample three-class forecast metrics (OOS 1996-02 to 2024-10). Balanced accuracy averages per-class recall; AUC is the macro one-vs-rest area; Brier and log loss are probabilistic scores (lower is better).

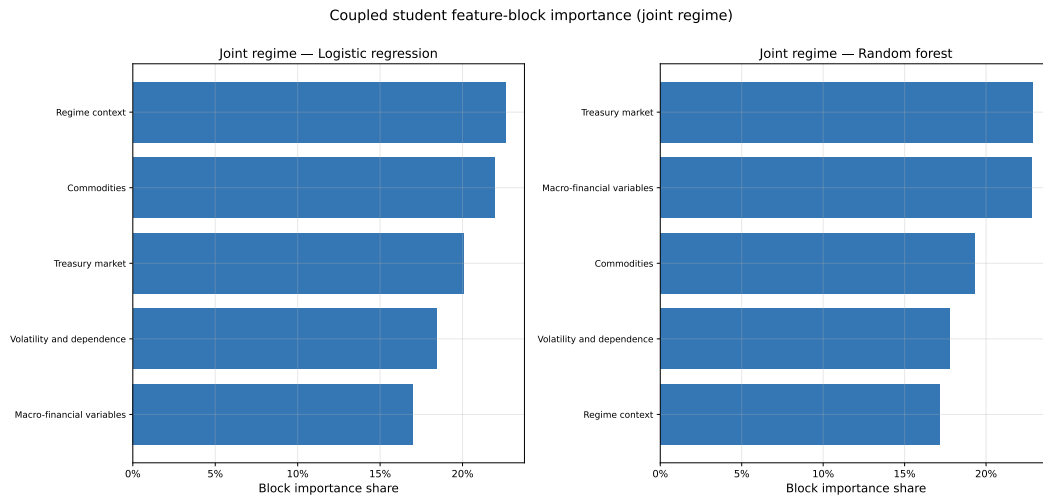
| Model               | Accuracy | Balanced acc. | AUC (macro OVR) | Brier | Log loss |
|---------------------|----------|---------------|-----------------|-------|----------|
| Logistic regression | 0.678    | 0.677         | 0.770           | 0.541 | 1.062    |
| Random forest       | 0.748    | 0.595         | 0.776           | 0.383 | 0.686    |

**Table G.23:** Coupled student: most frequently selected hyper-parameter setting across the walk-forward out-of-sample windows, with the share of windows in which it was chosen.

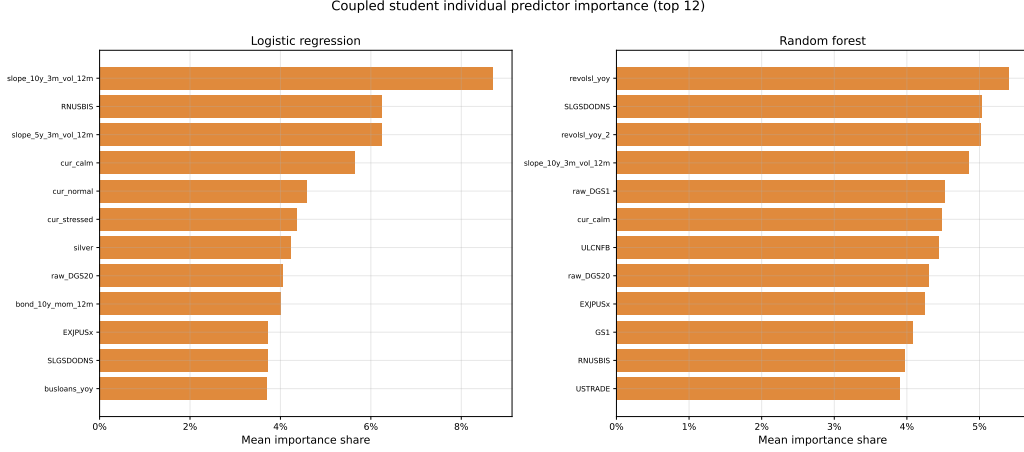
| Model               | Most frequent setting         | Selection share | OOS windows |
|---------------------|-------------------------------|-----------------|-------------|
| Logistic regression | $C = 0.03$                    | 81.0%           | 58          |
| Random forest       | $d = 4$ , leaf = 5, $n = 300$ | 72.4%           | 58          |



**Figure G.29:** COUPLED student out-of-sample probability paths for the three joint regimes (calm, normal, stressed), shown for the logistic and random-forest students. Shading marks the realised teacher regime, the black lines report the predicted regime probabilities, and the dashed line marks the 0.5 threshold.



**Figure G.30:** COUPLED student feature-block relevance for the three-class next-month regime forecast. Logistic-regression relevance is measured by absolute coefficient magnitude. Random-forest relevance is measured by impurity-based feature importance. Values are aggregated within predictor blocks.



**Figure G.31:** Coupled student: individual-predictor importance for the next-month regime forecast.

## H Allocation Details

### H.1 Probability smoothing and shrinkage

Let  $p_{i,m}$  be the predicted favourable-regime probability for asset  $i \in \{E, B\}$ . Predicted probabilities are used directly or smoothed with exponential half-life

$$h \in \{0, 3, 6\}. \quad (70)$$

For each asset and state, historical next-month excess-return moments are estimated from past observations and shrunk toward pre-prediction unconditional moments:

$$a_{i,s,m} = \frac{n_{i,s,m}}{n_{i,s,m} + 36}, \quad (71)$$

$$\tilde{\mu}_{i,s,m} = a_{i,s,m} \hat{\mu}_{i,s,m} + (1 - a_{i,s,m}) \mu_{i,0}, \quad (72)$$

$$\tilde{\sigma}_{i,s,m}^2 = a_{i,s,m} \hat{\sigma}_{i,s,m}^2 + (1 - a_{i,s,m}) \sigma_{i,0}^2. \quad (73)$$

### H.2 Probability-to-moment mapping

The probability-mixture mean and variance are

$$\mu_{i,m} = (1 - p_{i,m}) \tilde{\mu}_{i,0,m} + p_{i,m} \tilde{\mu}_{i,1,m}, \quad (74)$$

$$\sigma_{i,m}^2 = (1 - p_{i,m}) (\tilde{\sigma}_{i,0,m}^2 + \tilde{\mu}_{i,0,m}^2) + p_{i,m} (\tilde{\sigma}_{i,1,m}^2 + \tilde{\mu}_{i,1,m}^2) - \mu_{i,m}^2. \quad (75)$$

The equity–bond correlation is estimated from expanding historical returns and shrunk toward the pre-prediction correlation using the same shrinkage form. The covariance matrix is then formed from the conditional volatilities and the shrunk equity–bond correlation.

### H.3 Portfolio grid and net returns

The reported  $L^1$  allocation uses the cash-allowed long-only budget

$$w_E + w_B \leq 1,$$

with equity and bond weights each capped at one and selected on a 0.02 grid. The reported allocation grid varies

$$\gamma \in \{0.25, 0.5, 1, 2, 3, 5\},$$



$$c \in \{0.0005, 0.0010, 0.0025\},$$

$$h \in \{0, 3, 6\},$$

and

$$\lambda_{\text{trade}} \in \{0, 0.0005, 0.001, 0.0025, 0.005, 0.0075, 0.01, 0.015, 0.02, 0.03, 0.05\}.$$

Here  $h$  is the probability-smoothing half-life and  $\lambda_{\text{trade}}$  is the turnover penalty in the allocation objective. Monthly net portfolio return is

$$1 + r_{m+1}^p = (1 - c\|w_m - w_{m-1}\|_1) \left( 1 + w_{E,m}R_{E,m+1} + w_{B,m}R_{B,m+1} + w_{C,m}R_{m+1}^f \right). \quad (76)$$

#### H.4 Allocation diagnostics and parameter sensitivity

This appendix reports the allocation diagnostics behind the dynamic allocation results in Section 5.1. Table H.24 reports the main independent allocation summary using 10 bps one-way transaction costs. Table H.25 gives the baseline parameter settings for the three dynamic strategies. Figure H.32 reports the corresponding predicted probabilities, portfolio weights, turnover, wealth, and drawdown.

**Table H.24:** Allocation performance summary

| Role                                                                                                                                                                                                                                                                                                                | Model               | Predictor set              | Ann. ret. | Ann. vol. | Sharpe | Max DD | Ann. turn. | Final   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|----------------------------|-----------|-----------|--------|--------|------------|---------|
| 60/40 benchmark                                                                                                                                                                                                                                                                                                     | --                  | --                         | 4.5%      | 9.6%      | 0.47   | -33.4% | 19.7%      | 669.78  |
| High-return dynamic                                                                                                                                                                                                                                                                                                 | Logistic regression | smoothed z-score           | 7.6%      | 10.6%     | 0.71   | -18.9% | 152.1%     | 1513.59 |
| Defensive dynamic                                                                                                                                                                                                                                                                                                   | Random forest       | smoothed z-score           | 4.4%      | 5.8%      | 0.76   | -13.0% | 59.8%      | 636.60  |
| Low-turnover dynamic                                                                                                                                                                                                                                                                                                | Logistic regression | z-score + smoothed z-score | 4.8%      | 7.2%      | 0.67   | -11.5% | 51.4%      | 719.99  |
| <i>Note.</i> Dynamic strategy returns are net of transaction costs. The 60/40 benchmark is rebalanced monthly and reported net of the same one-way transaction-cost convention. The benchmark and dynamic strategies are evaluated using the same monthly return, drawdown, turnover, and final-wealth definitions. |                     |                            |           |           |        |        |            |         |

**Table H.25:** Selected dynamic allocation specifications

| Role                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Model         | Predictor set              | $\gamma$ | $\tau$ | Cost   | Sm. | Ann. ret. | Sharpe | Max DD | Turn.  | Final  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|----------------------------|----------|--------|--------|-----|-----------|--------|--------|--------|--------|
| High-return                                                                                                                                                                                                                                                                                                                                                                                                                                              | Logistic      | smoothed z-score           | 0.5      | 0      | 10 bps | 0   | 7.6%      | 0.71   | -18.9% | 152.1% | 1513.6 |
| Defensive                                                                                                                                                                                                                                                                                                                                                                                                                                                | Random forest | smoothed z-score           | 3        | 0.001  | 10 bps | 0   | 4.4%      | 0.76   | -13.0% | 59.8%  | 636.6  |
| Low-turnover                                                                                                                                                                                                                                                                                                                                                                                                                                             | Logistic      | z-score + smoothed z-score | 3        | 0.001  | 10 bps | 3   | 4.8%      | 0.67   | -11.5% | 51.4%  | 720.0  |
| <i>Note.</i> Parameter settings for the three dynamic strategies shown in the main allocation figures. “Role” abbreviates High-return/Defensive/Low-turnover dynamic; “Logistic” is logistic regression; predictor sets are smoothed z-score and z-score + smoothed z-score. $\gamma$ is risk aversion, $\tau$ the turnover penalty, “Sm.” the probability-smoothing half-life (months), “Turn.” annual turnover, “Final” terminal wealth (start = 100). |               |                            |          |        |        |     |           |        |        |        |        |

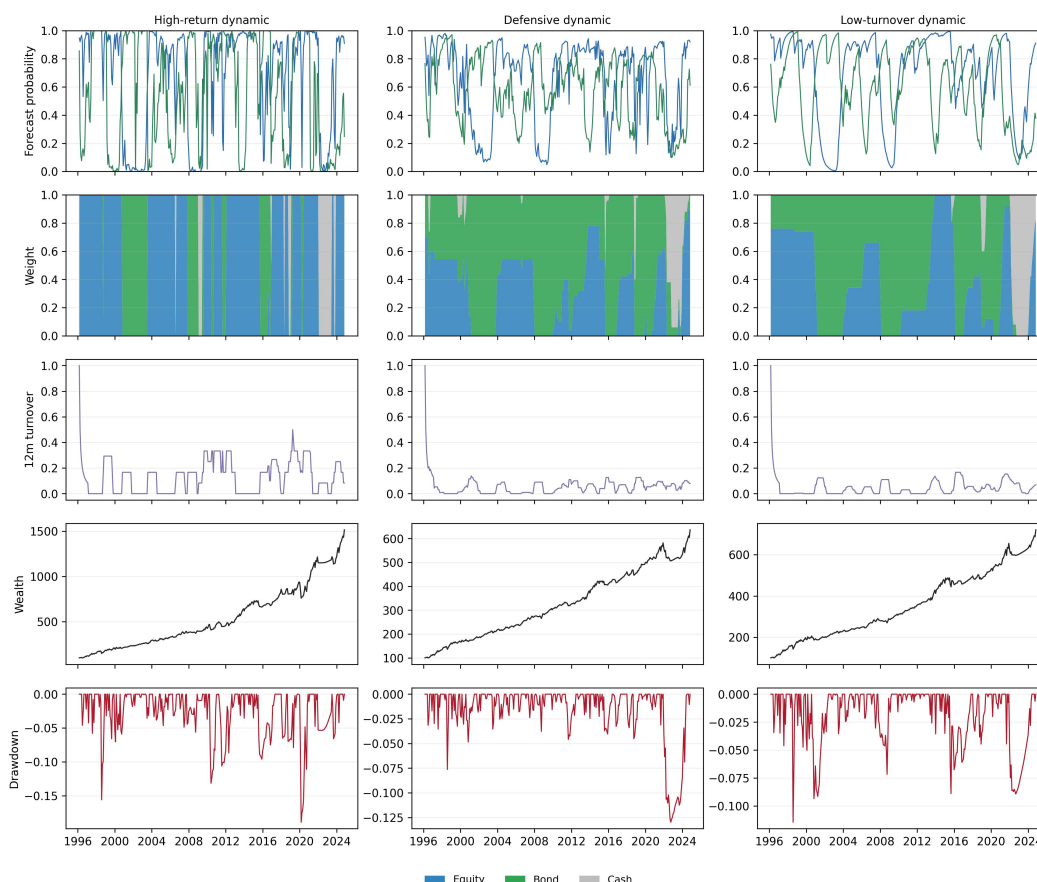
**Headline-row interpretation.** The three independent dynamic rows are representative points from the allocation grid. They are selected to illustrate distinct allocation trade-offs. The high-return row shows a logistic-regression specification with high terminal wealth and materially lower drawdown than the 60/40 benchmark. The defensive row shows a random-forest specification with lower volatility, lower drawdown, and a higher Sharpe ratio. The low-turnover row shows a logistic-regression specification with probability smoothing and annual turnover near 51%. The full parameter sweeps around these rows are reported below.

## H.5 Additional independent benchmarks

The additional benchmarks use the same monthly excess-return columns as the independent allocation sample ending in 2024-10. Cash is treated as zero excess return because the independent allocation stage uses equity and bond excess returns but no separate cash-return column. These benchmarks are included to compare the regime-aware rules with simple dynamic or single-asset alternatives that do not use regime forecasts.

| Benchmark                 | Ann. return | Ann. vol. | Sharpe | Max drawdown | Ann. turnover | Final wealth |
|---------------------------|-------------|-----------|--------|--------------|---------------|--------------|
| 60/40 benchmark           | 4.5%        | 9.6%      | 0.47   | -36.2%       | 19.7%         | 359.35       |
| Equity-only               | 6.9%        | 15.4%     | 0.44   | -62.2%       | 0.0%          | 505.52       |
| Bond-only                 | 2.1%        | 4.1%      | 0.51   | -21.3%       | 0.0%          | 177.25       |
| Volatility-targeted 60/40 | 3.6%        | 8.9%      | 0.41   | -35.8%       | 8.0%          | 229.92       |

**Table H.26:** Additional independent dynamic-allocation benchmarks. Returns are monthly excess returns over the independent allocation sample ending in 2024-10. Cash is treated as zero excess return. The volatility-targeted 60/40 rule uses a trailing 36-month realised volatility estimate, a 10% annual volatility target, and leverage capped at one.



**Figure H.32:** Allocation mechanics for selected dynamic strategies

Note: The panels report predicted favourable-regime probabilities, portfolio weights, rolling 12-month turnover, wealth, and drawdown for the three selected dynamic strategies.

The mechanics figure shows that the three dynamic strategies use the same regime forecasts

in different ways. The high-return strategy takes large equity positions and accepts higher turnover. The defensive strategy uses more bond exposure and reduces equity exposure when forecasts weaken. The low-turnover strategy changes allocations more slowly and uses probability smoothing to reduce trading.

### H.5.1 Parameter sensitivity around selected strategies

The sensitivity checks vary allocation parameters around the three selected dynamic strategies. In each one-parameter sweep, the reported parameter varies and the remaining allocation parameters are fixed at the strategy-specific baseline values in Table H.25. The transaction-cost and turnover-penalty table varies these two parameters jointly and keeps the remaining parameters fixed at the same baseline values.



**Figure H.33:** Risk-aversion sweep, high-return dynamic

Note: Risk aversion varies across panels. The model, predictor set, turnover penalty, transaction cost, and smoothing half-life are fixed at the strategy-specific baseline values in Table H.25.

**Table H.27:** Risk-aversion sweep, high-return dynamic

| Risk aversion | Ann. return | Ann. vol. | Sharpe | Max. drawdown | Monthly CVaR 1% | Final wealth | Ann. turnover |
|---------------|-------------|-----------|--------|---------------|-----------------|--------------|---------------|
| 0.25          | 7.6%        | 10.7%     | 0.71   | -20.0%        | -11.1%          | 1523.40      | 153.7%        |
| 0.5           | 7.6%        | 10.6%     | 0.71   | -18.9%        | -10.8%          | 1513.59      | 152.1%        |
| 1             | 7.4%        | 10.4%     | 0.72   | -15.6%        | -9.7%           | 1460.98      | 156.0%        |
| 2             | 6.4%        | 9.7%      | 0.66   | -15.6%        | -9.3%           | 1101.27      | 176.4%        |
| 3             | 5.4%        | 8.5%      | 0.64   | -13.5%        | -8.5%           | 850.63       | 177.4%        |
| 5             | 4.1%        | 5.8%      | 0.70   | -8.0%         | -5.6%           | 588.05       | 151.7%        |



**Figure H.34:** Risk-aversion sweep, defensive dynamic

Note: Risk aversion varies across panels. The model, predictor set, turnover penalty, transaction cost, and smoothing half-life are fixed at the strategy-specific baseline values in Table H.25.

**Table H.28:** Risk-aversion sweep, defensive dynamic

| Risk aversion | Ann. return | Ann. vol. | Sharpe | Max. drawdown | Monthly CVaR 1% | Final wealth | Ann. turnover |
|---------------|-------------|-----------|--------|---------------|-----------------|--------------|---------------|
| 0.25          | 6.1%        | 10.6%     | 0.58   | -18.2%        | -10.5%          | 1035.50      | 99.2%         |
| 0.5           | 5.9%        | 10.2%     | 0.58   | -17.0%        | -10.0%          | 976.60       | 100.3%        |
| 1             | 6.0%        | 9.7%      | 0.62   | -17.1%        | -9.9%           | 1001.81      | 91.1%         |
| 2             | 5.4%        | 7.6%      | 0.70   | -12.1%        | -7.5%           | 837.17       | 77.9%         |
| 3             | 4.4%        | 5.8%      | 0.76   | -13.0%        | -5.5%           | 636.60       | 59.8%         |
| 5             | 3.5%        | 4.4%      | 0.78   | -12.4%        | -4.2%           | 495.49       | 43.1%         |



**Figure H.35:** Risk-aversion sweep, low-turnover dynamic

Note: Risk aversion varies across panels. The model, predictor set, turnover penalty, transaction cost, and smoothing half-life are fixed at the strategy-specific baseline values in Table H.25.

**Table H.29:** Risk-aversion sweep, low-turnover dynamic

| Risk aversion | Ann. return | Ann. vol. | Sharpe | Max. drawdown | Monthly CVaR 1% | Final wealth | Ann. turnover |
|---------------|-------------|-----------|--------|---------------|-----------------|--------------|---------------|
| 0.25          | 5.8%        | 11.9%     | 0.49   | -23.9%        | -11.7%          | 936.18       | 66.1%         |
| 0.5           | 5.5%        | 11.1%     | 0.49   | -22.6%        | -11.1%          | 865.89       | 68.2%         |
| 1             | 5.5%        | 10.0%     | 0.55   | -20.4%        | -10.6%          | 862.11       | 67.2%         |
| 2             | 5.4%        | 8.9%      | 0.61   | -15.6%        | -9.0%           | 849.15       | 61.4%         |
| 3             | 4.8%        | 7.2%      | 0.67   | -11.5%        | -7.1%           | 719.99       | 51.4%         |
| 5             | 3.8%        | 5.0%      | 0.76   | -9.9%         | -4.5%           | 545.07       | 35.5%         |

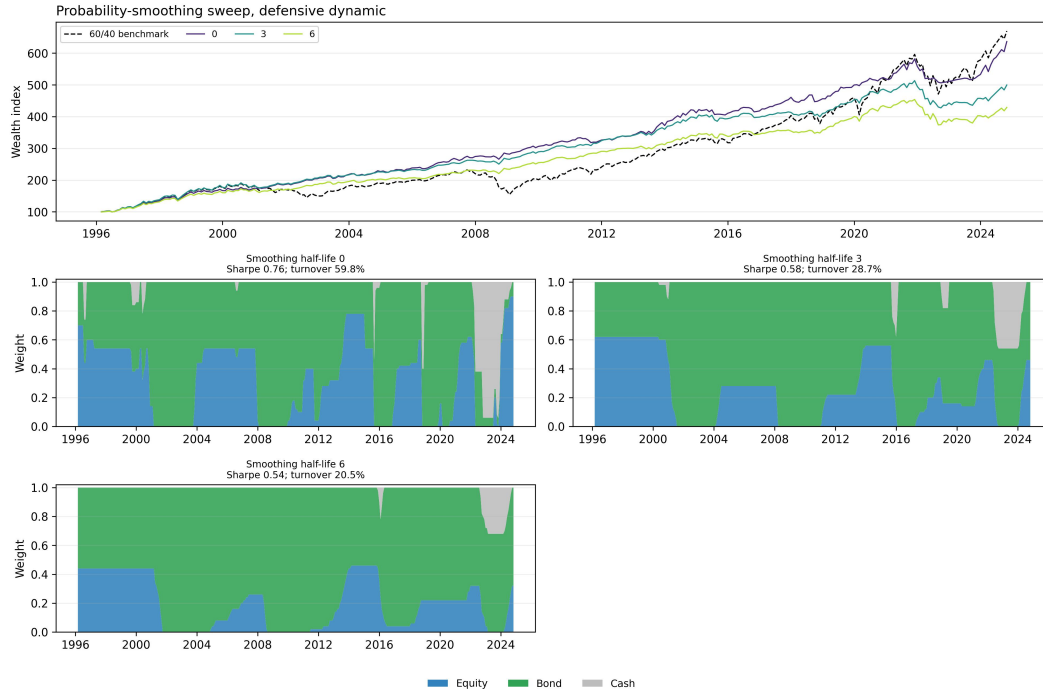


**Figure H.36:** Probability-smoothing sweep, high-return dynamic

Note: Probability smoothing varies across panels. The model, predictor set, risk aversion, turnover penalty, and transaction cost are fixed at the strategy-specific baseline values in Table H.25.

**Table H.30:** Probability-smoothing sweep, high-return dynamic

| Smoothing half-life | Ann. return | Ann. vol. | Sharpe | Max. drawdown | Monthly CVaR 1% | Final wealth | Ann. turnover |
|---------------------|-------------|-----------|--------|---------------|-----------------|--------------|---------------|
| 0                   | 7.6%        | 10.6%     | 0.71   | -18.9%        | -10.8%          | 1513.59      | 152.1%        |
| 3                   | 6.0%        | 11.3%     | 0.53   | -20.0%        | -11.5%          | 1007.50      | 82.5%         |
| 6                   | 6.3%        | 12.0%     | 0.52   | -23.4%        | -11.7%          | 1073.65      | 47.4%         |

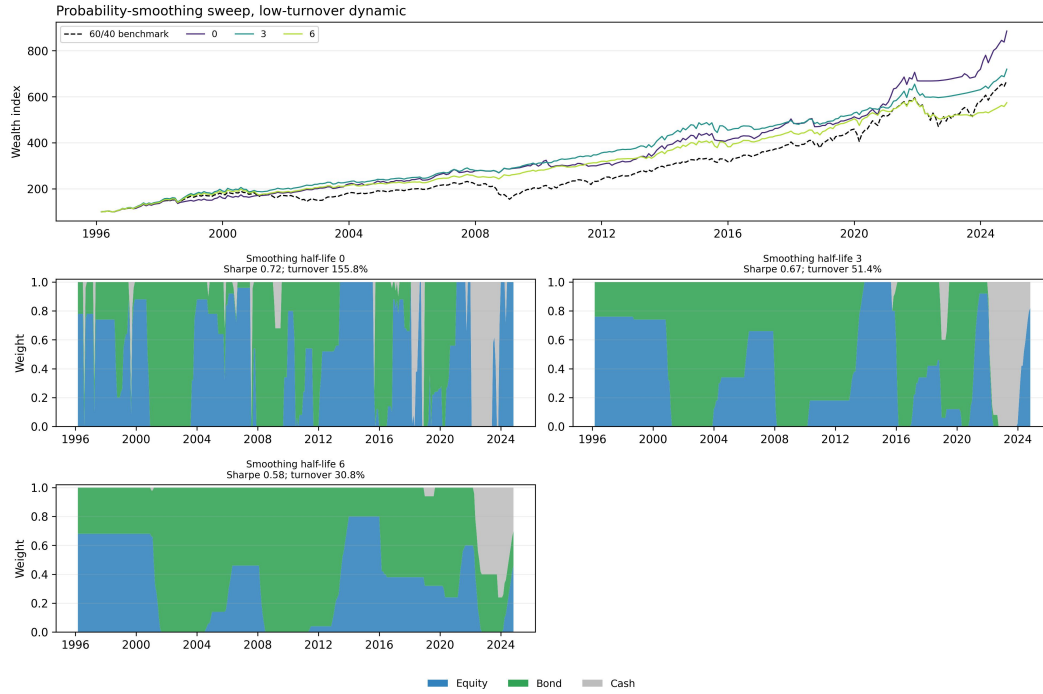


**Figure H.37:** Probability-smoothing sweep, defensive dynamic

Note: Probability smoothing varies across panels. The model, predictor set, risk aversion, turnover penalty, and transaction cost are fixed at the strategy-specific baseline values in Table H.25.

**Table H.31:** Probability-smoothing sweep, defensive dynamic

| Smoothing half-life | Ann. return | Ann. vol. | Sharpe | Max. drawdown | Monthly CVaR 1% | Final wealth | Ann. turnover |
|---------------------|-------------|-----------|--------|---------------|-----------------|--------------|---------------|
| 0                   | 4.4%        | 5.8%      | 0.76   | -13.0%        | -5.5%           | 636.60       | 59.8%         |
| 3                   | 3.5%        | 6.0%      | 0.58   | -16.7%        | -6.0%           | 499.69       | 28.7%         |
| 6                   | 2.9%        | 5.4%      | 0.54   | -17.6%        | -5.2%           | 429.04       | 20.5%         |



**Figure H.38:** Probability-smoothing sweep, low-turnover dynamic

Note: Probability smoothing varies across panels. The model, predictor set, risk aversion, turnover penalty, and transaction cost are fixed at the strategy-specific baseline values in Table H.25.

**Table H.32:** Probability-smoothing sweep, low-turnover dynamic

| Smoothing half-life | Ann. return | Ann. vol. | Sharpe | Max. drawdown | Monthly CVaR 1% | Final wealth | Ann. turnover |
|---------------------|-------------|-----------|--------|---------------|-----------------|--------------|---------------|
| 0                   | 5.6%        | 7.7%      | 0.72   | -11.1%        | -7.6%           | 885.67       | 155.8%        |
| 3                   | 4.8%        | 7.2%      | 0.67   | -11.5%        | -7.1%           | 719.99       | 51.4%         |
| 6                   | 4.0%        | 6.9%      | 0.58   | -15.2%        | -7.0%           | 574.31       | 30.8%         |



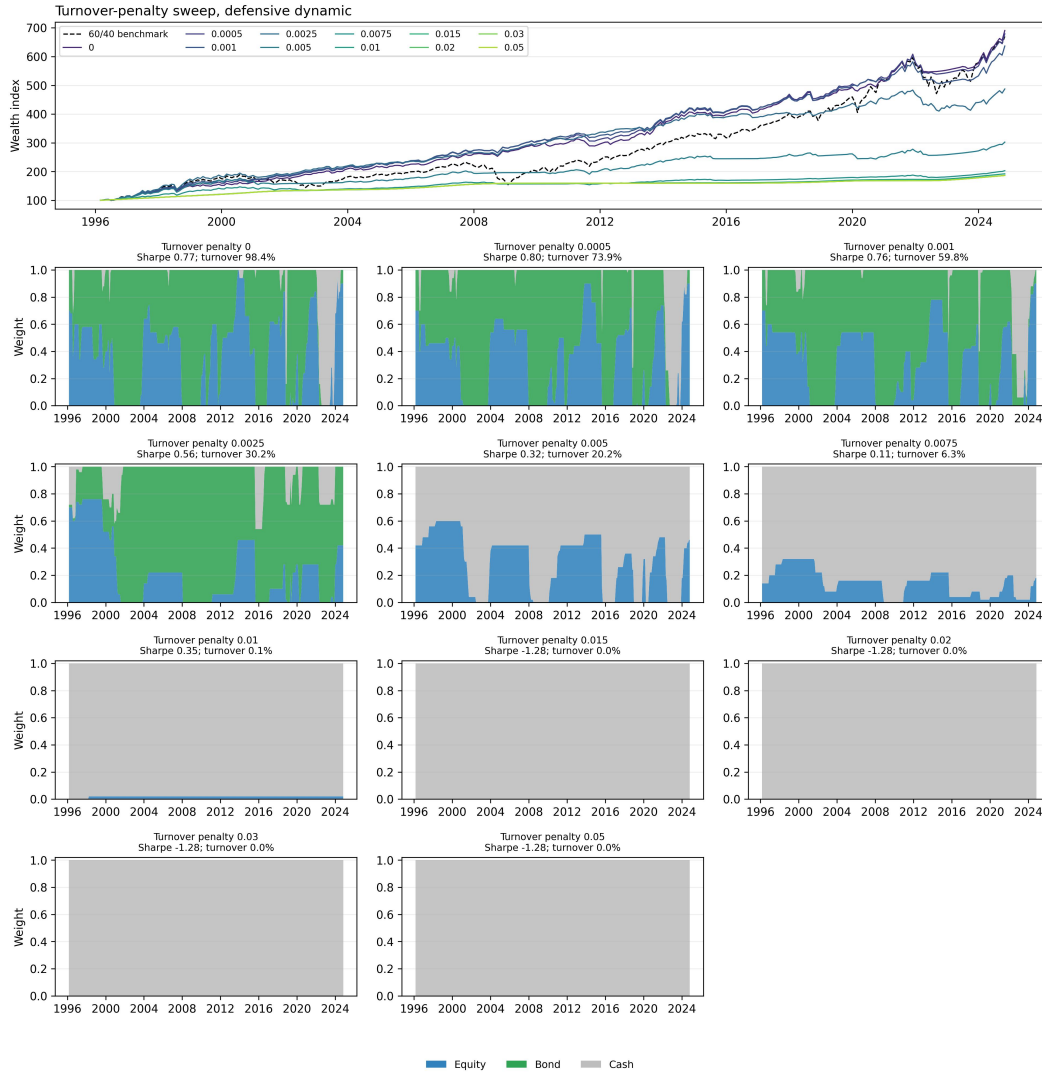


**Figure H.39:** Turnover-penalty sweep, high-return dynamic

Note: Turnover penalty varies across panels. The model, predictor set, risk aversion, transaction cost, and smoothing half-life are fixed at the strategy-specific baseline values in Table H.25.

**Table H.33:** Turnover-penalty sweep, high-return dynamic

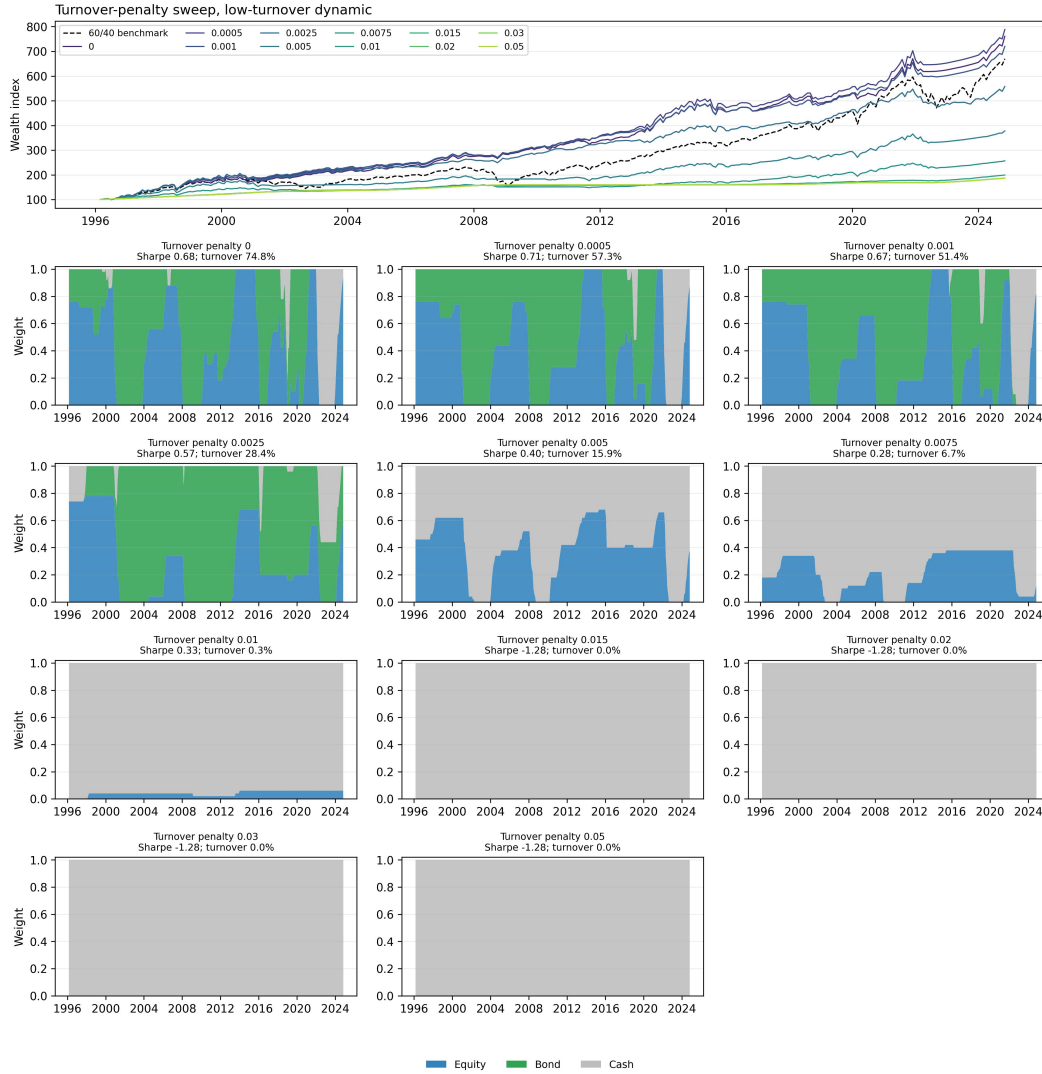
| Turnover penalty | Ann. return | Ann. vol. | Sharpe | Max. drawdown | Monthly CVaR 1% | Final wealth | Ann. turnover |
|------------------|-------------|-----------|--------|---------------|-----------------|--------------|---------------|
| 0                | 7.6%        | 10.6%     | 0.71   | -18.9%        | -10.8%          | 1513.59      | 152.1%        |
| 0.0005           | 7.6%        | 10.7%     | 0.71   | -20.0%        | -11.1%          | 1525.04      | 135.5%        |
| 0.001            | 7.3%        | 10.7%     | 0.69   | -20.0%        | -11.1%          | 1412.70      | 125.9%        |
| 0.0025           | 5.2%        | 10.2%     | 0.51   | -20.0%        | -11.2%          | 796.46       | 114.5%        |
| 0.005            | 5.8%        | 11.2%     | 0.51   | -21.1%        | -11.3%          | 936.68       | 59.1%         |
| 0.0075           | 4.2%        | 12.1%     | 0.35   | -33.5%        | -11.7%          | 608.48       | 48.1%         |
| 0.01             | 0.8%        | 5.5%      | 0.14   | -22.2%        | -5.9%           | 232.06       | 8.8%          |
| 0.015            | -0.0%       | 0.0%      | -1.28  | 0.0%          | -0.0%           | 186.68       | 0.0%          |
| 0.02             | -0.0%       | 0.0%      | -1.28  | 0.0%          | -0.0%           | 186.68       | 0.0%          |
| 0.03             | -0.0%       | 0.0%      | -1.28  | 0.0%          | -0.0%           | 186.68       | 0.0%          |
| 0.05             | -0.0%       | 0.0%      | -1.28  | 0.0%          | -0.0%           | 186.68       | 0.0%          |

**Figure H.40:** Turnover-penalty sweep, defensive dynamic

Note: Turnover penalty varies across panels. The model, predictor set, risk aversion, transaction cost, and smoothing half-life are fixed at the strategy-specific baseline values in Table H.25.

**Table H.34:** Turnover-penalty sweep, defensive dynamic

| Turnover penalty | Ann. return | Ann. vol. | Sharpe | Max. drawdown | Monthly CVaR 1% | Final wealth | Ann. turnover |
|------------------|-------------|-----------|--------|---------------|-----------------|--------------|---------------|
| 0                | 4.7%        | 6.1%      | 0.77   | -10.2%        | -5.9%           | 690.57       | 98.4%         |
| 0.0005           | 4.6%        | 5.8%      | 0.80   | -11.1%        | -5.2%           | 678.95       | 73.9%         |
| 0.001            | 4.4%        | 5.8%      | 0.76   | -13.0%        | -5.5%           | 636.60       | 59.8%         |
| 0.0025           | 3.4%        | 6.1%      | 0.56   | -15.1%        | -6.4%           | 487.25       | 30.2%         |
| 0.005            | 1.7%        | 5.2%      | 0.32   | -12.0%        | -5.9%           | 302.54       | 20.2%         |
| 0.0075           | 0.3%        | 2.7%      | 0.11   | -8.8%         | -3.3%           | 203.22       | 6.3%          |
| 0.01             | 0.1%        | 0.3%      | 0.35   | -0.5%         | -0.3%           | 192.37       | 0.1%          |
| 0.015            | -0.0%       | 0.0%      | -1.28  | 0.0%          | -0.0%           | 186.68       | 0.0%          |
| 0.02             | -0.0%       | 0.0%      | -1.28  | 0.0%          | -0.0%           | 186.68       | 0.0%          |
| 0.03             | -0.0%       | 0.0%      | -1.28  | 0.0%          | -0.0%           | 186.68       | 0.0%          |
| 0.05             | -0.0%       | 0.0%      | -1.28  | 0.0%          | -0.0%           | 186.68       | 0.0%          |

**Figure H.41:** Turnover-penalty sweep, low-turnover dynamic

Note: Turnover penalty varies across panels. The model, predictor set, risk aversion, transaction cost, and smoothing half-life are fixed at the strategy-specific baseline values in Table H.25.

**Table H.35:** Turnover-penalty sweep, low-turnover dynamic

| Turnover penalty | Ann. return | Ann. vol. | Sharpe | Max. drawdown | Monthly CVaR 1% | Final wealth | Ann. turnover |
|------------------|-------------|-----------|--------|---------------|-----------------|--------------|---------------|
| 0                | 5.0%        | 7.4%      | 0.68   | -10.8%        | -7.1%           | 760.69       | 74.8%         |
| 0.0005           | 5.1%        | 7.2%      | 0.71   | -11.5%        | -7.2%           | 788.23       | 57.3%         |
| 0.001            | 4.8%        | 7.2%      | 0.67   | -11.5%        | -7.1%           | 719.99       | 51.4%         |
| 0.0025           | 3.9%        | 6.8%      | 0.57   | -13.3%        | -6.9%           | 557.59       | 28.4%         |
| 0.005            | 2.5%        | 6.2%      | 0.40   | -13.8%        | -6.4%           | 377.90       | 15.9%         |
| 0.0075           | 1.1%        | 4.0%      | 0.28   | -8.2%         | -4.2%           | 257.09       | 6.7%          |
| 0.01             | 0.2%        | 0.7%      | 0.33   | -1.6%         | -0.7%           | 199.35       | 0.3%          |
| 0.015            | -0.0%       | 0.0%      | -1.28  | 0.0%          | -0.0%           | 186.68       | 0.0%          |
| 0.02             | -0.0%       | 0.0%      | -1.28  | 0.0%          | -0.0%           | 186.68       | 0.0%          |
| 0.03             | -0.0%       | 0.0%      | -1.28  | 0.0%          | -0.0%           | 186.68       | 0.0%          |
| 0.05             | -0.0%       | 0.0%      | -1.28  | 0.0%          | -0.0%           | 186.68       | 0.0%          |

**Table H.36:** Transaction-cost and turnover-penalty sensitivity

| Strategy             | Turnover penalty | 5 bps | 10 bps | 25 bps |
|----------------------|------------------|-------|--------|--------|
| High-return dynamic  | 0.0000           | 0.77  | 0.74   | 0.67   |
| High-return dynamic  | 0.0005           | 0.75  | 0.74   | 0.62   |
| High-return dynamic  | 0.0010           | 0.75  | 0.72   | 0.53   |
| High-return dynamic  | 0.0025           | 0.64  | 0.55   | 0.44   |
| High-return dynamic  | 0.0050           | 0.59  | 0.56   | 0.47   |
| High-return dynamic  | 0.0100           | 0.15  | 0.17   |        |
| Defensive dynamic    | 0.0000           | 0.75  | 0.78   | 0.69   |
| Defensive dynamic    | 0.0005           | 0.79  | 0.81   | 0.61   |
| Defensive dynamic    | 0.0010           | 0.81  | 0.77   | 0.57   |
| Defensive dynamic    | 0.0025           | 0.63  | 0.58   | 0.43   |
| Defensive dynamic    | 0.0050           | 0.38  | 0.35   | 0.20   |
| Defensive dynamic    | 0.0100           | 0.15  | 0.35   |        |
| Low-turnover dynamic | 0.0000           | 0.71  | 0.70   | 0.63   |
| Low-turnover dynamic | 0.0005           | 0.71  | 0.73   | 0.61   |
| Low-turnover dynamic | 0.0010           | 0.74  | 0.69   | 0.59   |
| Low-turnover dynamic | 0.0025           | 0.62  | 0.60   | 0.50   |
| Low-turnover dynamic | 0.0050           | 0.43  | 0.43   | 0.38   |
| Low-turnover dynamic | 0.0100           | 0.23  | 0.33   |        |

*Note.* Entries are Sharpe ratios. For each dynamic strategy, the model, predictor set, risk aversion, and smoothing half-life are fixed at the strategy-specific baseline values in Table H.25.

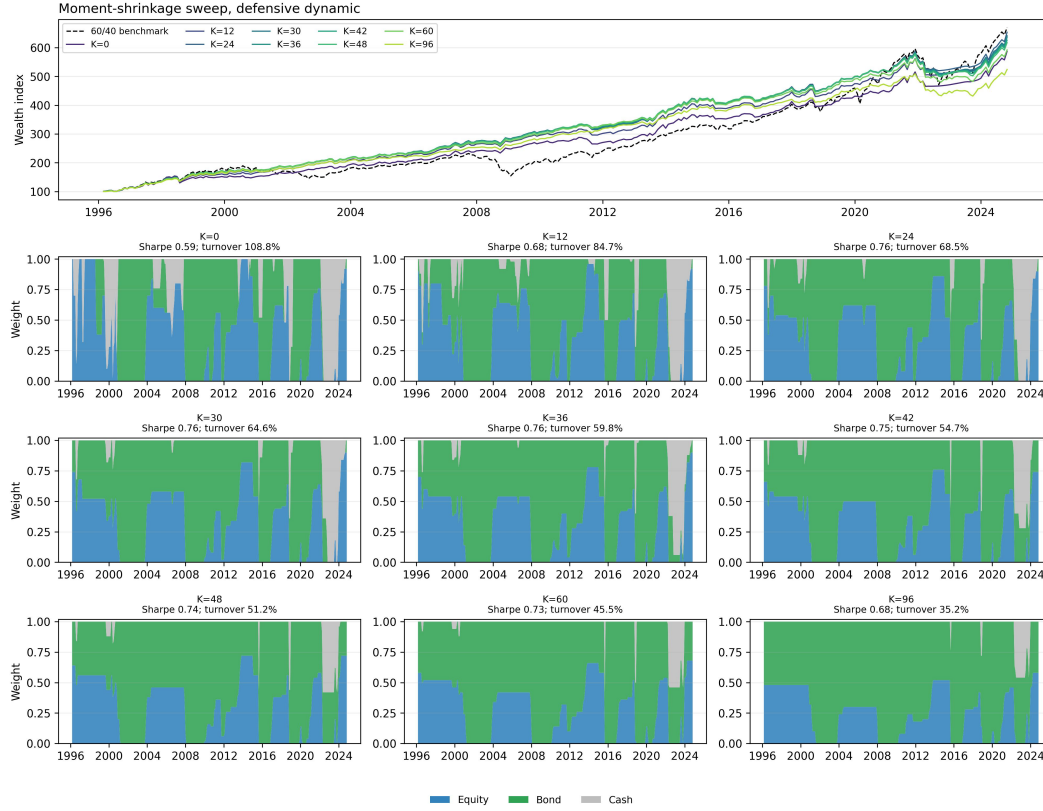


**Figure H.42:** Moment-shrinkage sweep, high-return dynamic

Note: The moment-shrinkage constant  $K$  varies across panels. The model, predictor set, risk aversion, turnover penalty, transaction cost, and smoothing half-life are fixed at the strategy-specific baseline values in Table H.25.

**Table H.37:** Moment-shrinkage sweep, high-return dynamic

| $K$ | Ann. ret. | Ann. vol. | Sharpe | Max DD | CVaR 1% | Final   | Ann. turn. |
|-----|-----------|-----------|--------|--------|---------|---------|------------|
| 0   | 7.3%      | 10.4%     | 0.71   | -16.3% | -9.7%   | 1419.16 | 173.4%     |
| 12  | 7.6%      | 10.5%     | 0.72   | -16.3% | -9.7%   | 1523.20 | 160.1%     |
| 24  | 7.5%      | 10.6%     | 0.71   | -17.3% | -10.3%  | 1496.05 | 159.9%     |
| 30  | 7.5%      | 10.6%     | 0.71   | -18.2% | -10.5%  | 1484.06 | 160.4%     |
| 36  | 7.6%      | 10.6%     | 0.71   | -19.2% | -10.8%  | 1513.59 | 152.1%     |
| 42  | 7.6%      | 10.7%     | 0.71   | -20.1% | -11.0%  | 1523.58 | 149.6%     |
| 48  | 7.8%      | 10.7%     | 0.73   | -20.3% | -11.1%  | 1598.17 | 141.6%     |
| 60  | 7.8%      | 10.7%     | 0.73   | -20.3% | -11.1%  | 1629.29 | 135.2%     |
| 96  | 7.5%      | 11.0%     | 0.68   | -21.5% | -11.1%  | 1476.33 | 135.1%     |

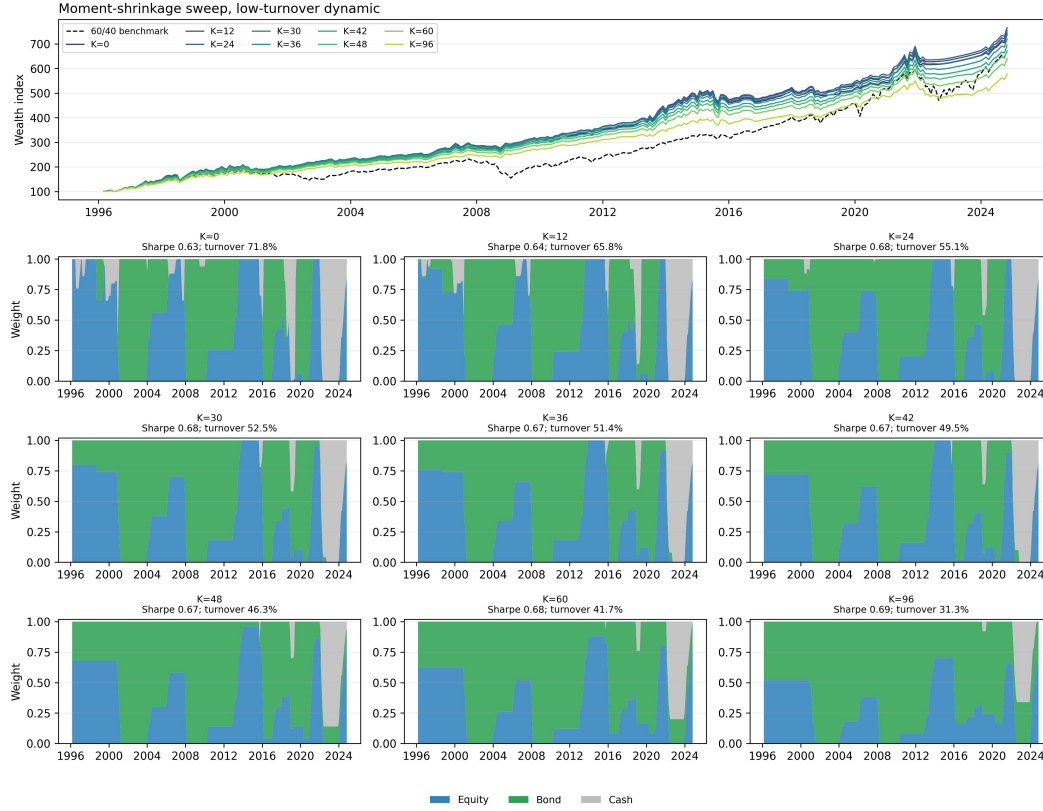


**Figure H.43:** Moment-shrinkage sweep, defensive dynamic

Note: The moment-shrinkage constant  $K$  varies across panels. The model, predictor set, risk aversion, turnover penalty, transaction cost, and smoothing half-life are fixed at the strategy-specific baseline values in Table H.25.

**Table H.38:** Moment-shrinkage sweep, defensive dynamic

| $K$ | Ann. ret. | Ann. vol. | Sharpe | Max DD | CVaR 1% | Final  | Ann. turn. |
|-----|-----------|-----------|--------|--------|---------|--------|------------|
| 0   | 4.1%      | 6.9%      | 0.59   | -16.3% | -7.9%   | 587.41 | 108.8%     |
| 12  | 4.4%      | 6.4%      | 0.68   | -12.9% | -6.7%   | 636.94 | 84.7%      |
| 24  | 4.5%      | 5.9%      | 0.76   | -12.8% | -5.4%   | 653.67 | 68.5%      |
| 30  | 4.4%      | 5.8%      | 0.76   | -14.1% | -5.4%   | 641.51 | 64.6%      |
| 36  | 4.4%      | 5.8%      | 0.76   | -14.9% | -5.5%   | 636.60 | 59.8%      |
| 42  | 4.3%      | 5.7%      | 0.75   | -16.0% | -5.5%   | 625.91 | 54.7%      |
| 48  | 4.3%      | 5.8%      | 0.74   | -16.8% | -5.6%   | 617.80 | 51.2%      |
| 60  | 4.1%      | 5.6%      | 0.73   | -17.7% | -5.5%   | 592.27 | 45.5%      |
| 96  | 3.7%      | 5.3%      | 0.68   | -18.9% | -5.1%   | 524.22 | 35.2%      |



**Figure H.44:** Moment-shrinkage sweep, low-turnover dynamic

Note: The moment-shrinkage constant  $K$  varies across panels. The model, predictor set, risk aversion, turnover penalty, transaction cost, and smoothing half-life are fixed at the strategy-specific baseline values in Table H.25.

**Table H.39:** Moment-shrinkage sweep, low-turnover dynamic

| $K$ | Ann. ret. | Ann. vol. | Sharpe | Max DD | CVaR 1% | Final  | Ann. turn. |
|-----|-----------|-----------|--------|--------|---------|--------|------------|
| 0   | 4.9%      | 7.8%      | 0.63   | -16.3% | -8.6%   | 738.97 | 71.8%      |
| 12  | 5.0%      | 7.7%      | 0.64   | -15.0% | -8.2%   | 755.63 | 65.8%      |
| 24  | 5.0%      | 7.5%      | 0.68   | -13.6% | -7.6%   | 766.50 | 55.1%      |
| 30  | 4.9%      | 7.3%      | 0.68   | -12.9% | -7.3%   | 745.09 | 52.5%      |
| 36  | 4.8%      | 7.2%      | 0.67   | -13.0% | -7.1%   | 719.99 | 51.4%      |
| 42  | 4.7%      | 7.0%      | 0.67   | -13.2% | -6.9%   | 694.83 | 49.5%      |
| 48  | 4.6%      | 6.8%      | 0.67   | -12.8% | -6.5%   | 672.44 | 46.3%      |
| 60  | 4.4%      | 6.4%      | 0.68   | -12.0% | -6.0%   | 640.71 | 41.7%      |
| 96  | 4.0%      | 5.9%      | 0.69   | -14.5% | -5.2%   | 577.82 | 31.3%      |

The sensitivity checks show how the same allocation rule changes when a single parameter is varied around each selected strategy. Higher risk aversion lowers equity exposure and reduces drawdowns in the defensive and low-turnover specifications. Probability smoothing reduces trading by slowing changes in the forecast probabilities. Higher turnover penalties reduce trading and increase cash weights, which lowers turnover but can reduce the Sharpe ratio when the penalty is too large. The moment-shrinkage sweep varies the amount of shrinkage applied to regime-conditional moments and the stock–bond correlation estimate. Larger  $K$  places more weight on unconditional pre-out-of-sample moments, while smaller  $K$

gives more weight to realised state-conditional samples.

## **H.6 Independent model-selection protocol**

This table summarises which independent-side choices are fixed design choices, which are selected on validation or grid search, and which are used only as diagnostics. The purpose is to separate the reported independent specification from robustness checks and presentation-only diagnostics.



| Component                     | Independent approach                                                               | Fixed in advance                                                             | Selected on validation/grid                                                                        | Robustness or diagnostic only                                         |
|-------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Teacher feature set           | Equity return–Sortino features and canonical bond features                         | Final reported feature blocks fixed before student prediction and allocation | No                                                                                                 | Volatility/downside equity labels used only as diagnostics            |
| Teacher jump penalty grid     | Separate equity and bond jump-penalty grids                                        | Grid fixed before final majority vote                                        | Majority-vote label selected from fitted teacher paths                                             | Path persistence and feature-separation plots reported as diagnostics |
| Distance norm                 | $L^2$ centroid and $L^1$ -medoid teacher paths                                     | Both norm families fixed as teacher candidates                               | Final label uses the combined majority vote                                                        | Separate $L^2$ and $L^1$ paths reported as diagnostics                |
| Teacher majority vote         | Final equity and bond labels combine teacher paths by majority vote                | Voting rule fixed before student prediction                                  | No refit after allocation outcomes                                                                 | Vote-scope figures report candidate-path agreement                    |
| Student predictor sets        | z-score predictors, smoothed z-score predictors, and their union                   | Transformations fixed before model fitting                                   | Predictor-set comparison reported out of sample                                                    | Feature-importance figures are diagnostic                             |
| Student model hyperparameters | Logistic regression and random forest specifications                               | Model classes fixed before evaluation                                        | Hyperparameters selected within the rolling validation design                                      | Model-frequency table reports selected specifications                 |
| Student validation metrics    | AUC, balanced accuracy, Brier score, log loss, and allocation-relevant diagnostics | Metrics fixed before reporting                                               | Best reported specifications selected from rolling outputs                                         | Full prediction diagnostics kept in appendix                          |
| Allocation rule               | Long-only equity–bond allocation with cash allowed and an $L^1$ turnover penalty   | Asset universe and allocation objective fixed before grid search             | Candidate grid varies risk aversion, turnover penalty, transaction cost, and probability smoothing | Parameter-sensitivity appendix reports one-way and joint checks       |
| Risk aversion                 | Values from the allocation grid                                                    | Grid fixed before headline reporting                                         | Representative rows use selected grid points                                                       | Risk-aversion sweeps reported as sensitivity                          |
| Transaction cost              | One-way transaction-cost grid                                                      | Grid fixed before headline reporting                                         | Headline independent rows use 10 bps                                                               | Cost and turnover-penalty sensitivity reported in appendix            |
| Turnover penalty              | $L^1$ turnover penalty in the allocation objective                                 | Grid fixed before headline reporting                                         | Representative rows use role-specific penalty values                                               | Turnover-penalty sweeps reported as sensitivity                       |
| Headline allocation rows      | High-return, defensive, and low-turnover representative rows                       | Roles fixed for presentation                                                 | Rows selected as representative points from the allocation grid                                    | Full-grid ranking rules are reported only as sensitivity diagnostics  |

**Table H.40:** Independent-side model-selection protocol. The table separates fixed design choices, validation or grid-selected choices, and robustness or diagnostic components.

## I Coupled allocation: details and sensitivity

The investable universe is the equity leg, the bond leg, and cash. Weights are long-only with a gross cap of 1.0 ( $w_{\text{cash}} \in [0, 1]$ ) and are chosen on a 0.05 grid. The COUPLED allocator uses annualised regime-conditional moments and an exponentially weighted stock–bond covariance with a 12-month half-life. This differs from the INDEPENDENT allocation grid and covariance estimator reported in Appendix H.

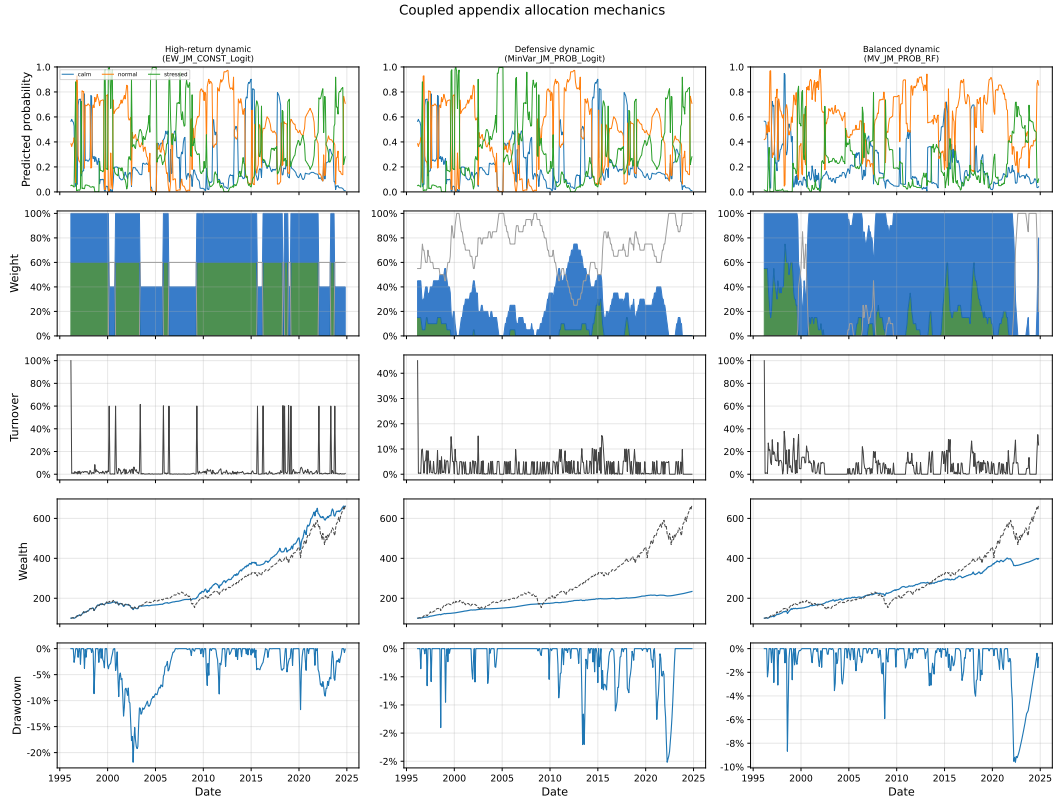
The CONST mapping uses the mean of the single predicted regime; the PROB mapping uses the probability mixture. Portfolios rebalance monthly and pay 10 bps on traded notional. Four static rules anchor the comparison: 60/40, 50/50, unconditional minimum-variance, and unconditional mean-variance. The regime-aware family comprises minimum-variance ( $\gamma=20$ ) and mean-variance ( $\gamma=5$ ) allocators under both mappings, together with a regime equal-weight overlay holding (0.6, 0.4) in calm, (0.72, 0.48) in normal, and (0, 0.4) in stressed. The unconditional mean-variance rule uses  $\gamma=3$  on an exponentially weighted mean, and the unconditional minimum-variance rule uses  $\gamma=20$ . Table I.41 reports the three headline strategies with the full metric set, and Table I.42 reports transaction-cost and turnover-penalty sensitivity for the balanced allocator.

**Table I.41:** Coupled allocation: headline results with full metrics, out of sample 1996-02 to 2024-10 (345 months). CVaR is the mean of the worst 1% of monthly excess returns; “Final” indexes 100 at the start; “Turn.” is annual turnover.

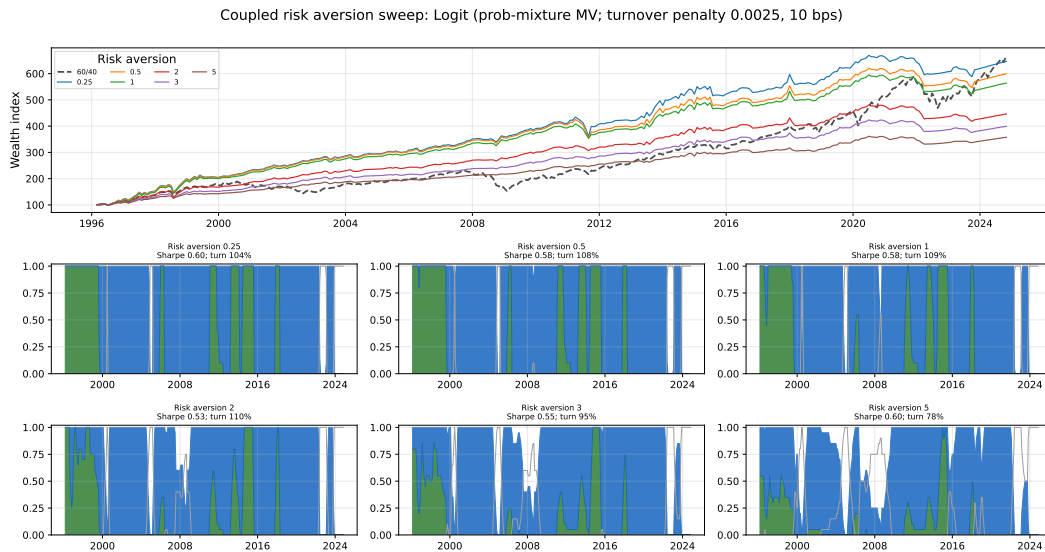
| Role        | Strategy                       | Ann. exc. | Vol.  | Sharpe | Max DD | CVaR <sub>1%</sub> | Final | Turn. |
|-------------|--------------------------------|-----------|-------|--------|--------|--------------------|-------|-------|
| Defensive   | Min-var (prob), logistic       | +0.80%    | 1.24% | 0.64   | −2.0%  | −1.03%             | 234.0 | 0.33  |
| Balanced    | Mean-var (prob), random forest | +2.74%    | 4.36% | 0.63   | −9.6%  | −4.41%             | 398.5 | 0.82  |
| High-return | Mean-var (prob), logistic      | +4.70%    | 7.57% | 0.62   | −21.9% | −6.75%             | 661.2 | 0.49  |
| Benchmark   | 60/40                          | +4.92%    | 9.61% | 0.47   | −33.9% | −8.57%             | 668.9 | 0.23  |

**Table I.42:** Coupled allocation: robustness of the probability-mixture mean-variance allocator on the logistic forecast to the one-way transaction cost and turnover penalty  $\lambda$ . Cells are out-of-sample Sharpe ratios.

| $\lambda$ | 5 bps | 10 bps | 25 bps |
|-----------|-------|--------|--------|
| 0         | 0.608 | 0.594  | 0.555  |
| 0.001     | 0.616 | 0.605  | 0.570  |
| 0.01      | 0.640 | 0.635  | 0.617  |
| 0.1       | 0.320 | 0.319  | 0.316  |



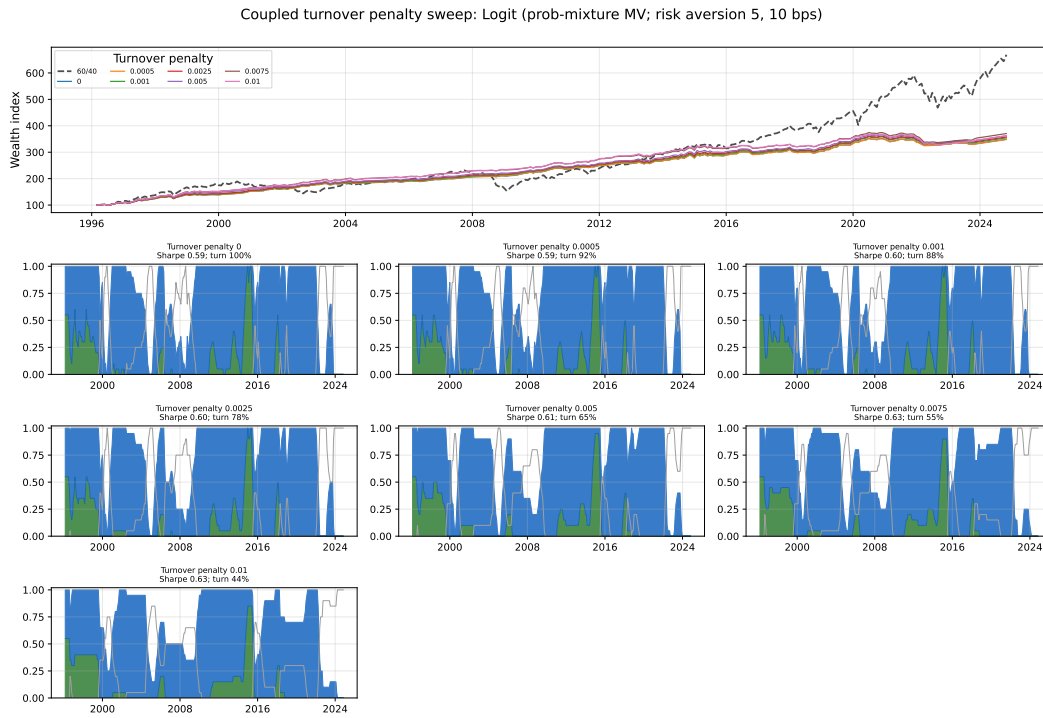
**Figure I.45:** Coupled allocation mechanics in detail for the three headline strategies. Rows, top to bottom: predicted regime probabilities; equity/bond/cash weights; monthly turnover; wealth index against the 60/40 (dashed); drawdown.



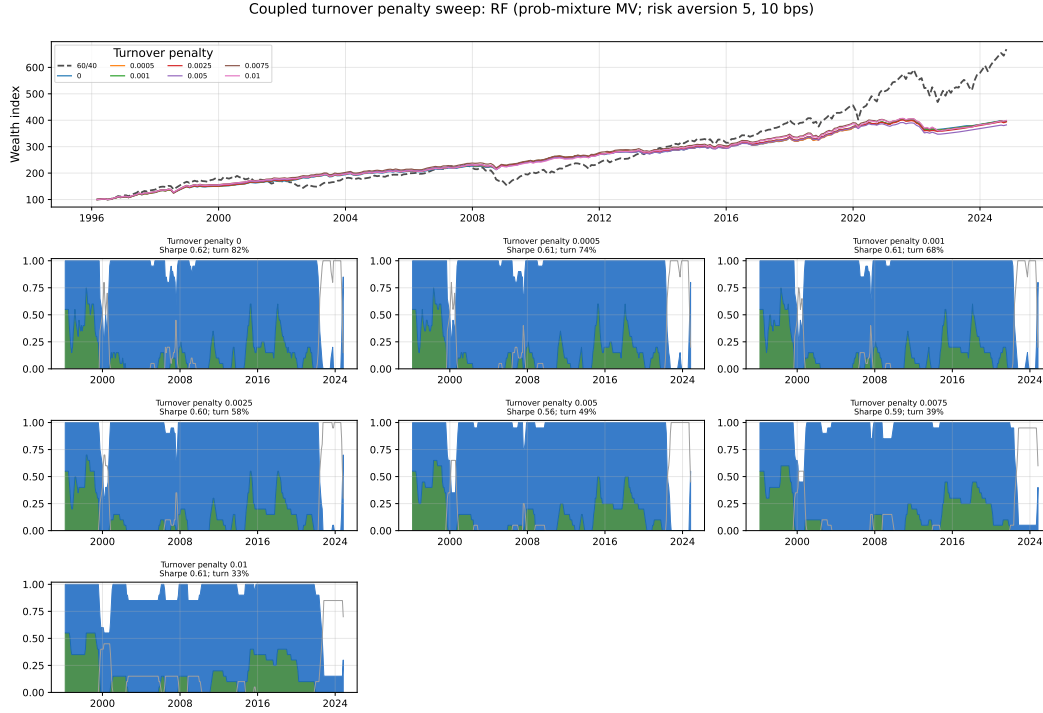
**Figure I.46:** Coupled allocation: risk-aversion sweep of the probability-mixture mean-variance allocator on the logistic forecast.



**Figure I.47:** Coupled allocation: risk-aversion sweep on the random-forest forecast.



**Figure I.48:** Coupled allocation: turnover-penalty sweep on the logistic forecast (risk aversion fixed at 5).



**Figure I.49:** Coupled allocation: turnover-penalty sweep on the random-forest forecast.

## I.1 Coupled allocation: additional benchmarks and parameter sensitivity

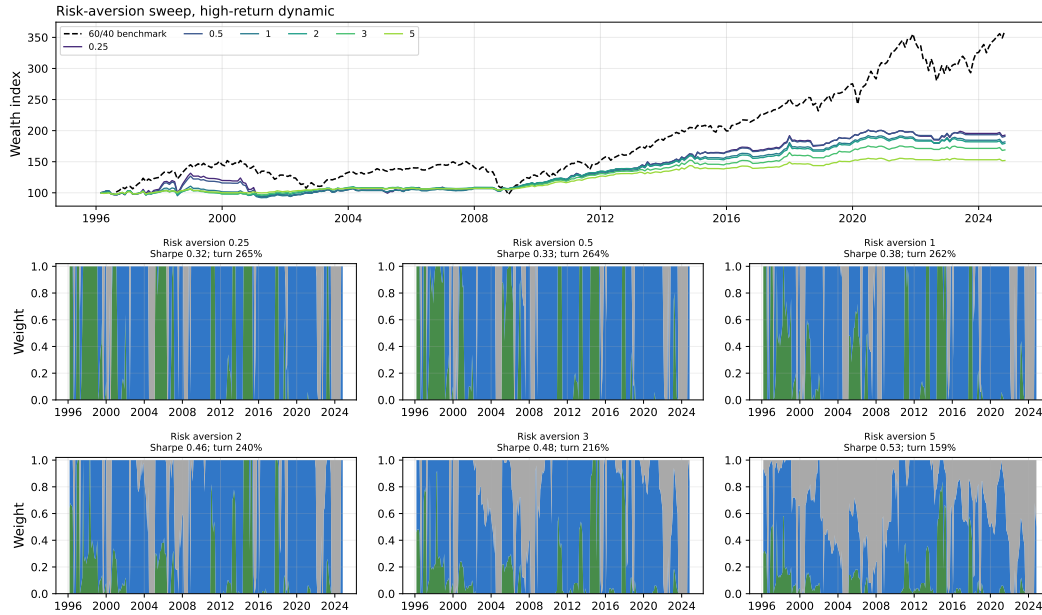
Table I.43 reports the coupled static and unconditional benchmarks over the out-of-sample span, and Table I.44 reports the sensitivity of the probability-mixture mean-variance allocator to the moment-shrinkage constant  $K$  (the headline strategies use  $K = 36$ ). Figures I.50–I.61 report one-parameter sweeps around the three selected strategies (high-return, defensive, low-turnover), matching the independent sweeps in Appendix H.

**Table I.43:** Coupled allocation: static and unconditional benchmarks, out of sample 1996-02 to 2024-10. Annualised excess returns net of 10 bps one-way costs; “Final” indexes 100 at the start; “Turn.” is annual turnover.

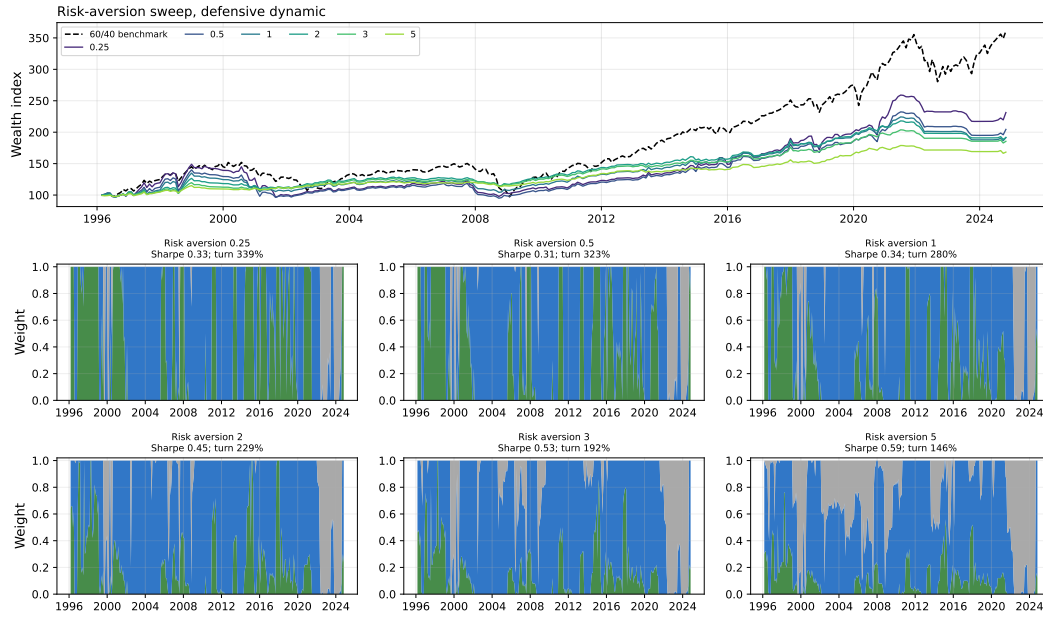
| Benchmark              | Ann. exc. | Vol.   | Sharpe | Max DD | Final | Turn. |
|------------------------|-----------|--------|--------|--------|-------|-------|
| 60/40                  | +4.94%    | 9.61%  | 0.514  | −33.8% | 672.7 | 0.03  |
| 50/50                  | +4.46%    | 8.24%  | 0.541  | −28.3% | 607.9 | 0.03  |
| Equity-only            | +6.85%    | 15.41% | 0.445  | −52.6% | 941.0 | 0.03  |
| Bond-only              | +2.07%    | 4.07%  | 0.509  | −17.2% | 330.2 | 0.03  |
| Min-variance (uncond.) | −0.00%    | 0.00%  | —      | 0.0%   | 186.6 | 0.00  |
| Vol-targeted 60/40     | +3.26%    | 6.91%  | 0.472  | −29.8% | 443.5 | 0.17  |

**Table I.44:** Coupled allocation: sensitivity of the probability-mixture mean–variance allocator to the moment-shrinkage constant  $K$  (annualised, net of 10 bps one-way costs). The headline strategies use  $K = 36$ .

| Shrinkage $K$ | Ann. return | Ann. vol. | Sharpe | Ann. turnover |
|---------------|-------------|-----------|--------|---------------|
| 0             | +2.09%      | 3.95%     | 0.529  | 88.3%         |
| 12            | +2.26%      | 4.00%     | 0.564  | 85.2%         |
| 24            | +2.40%      | 4.01%     | 0.598  | 78.3%         |
| 30            | +2.38%      | 3.91%     | 0.608  | 76.5%         |
| 36            | +2.34%      | 3.88%     | 0.604  | 76.5%         |
| 42            | +2.28%      | 3.87%     | 0.589  | 74.1%         |
| 48            | +2.25%      | 3.84%     | 0.585  | 71.3%         |
| 60            | +2.28%      | 3.71%     | 0.616  | 67.3%         |
| 96            | +2.22%      | 3.41%     | 0.652  | 53.6%         |



**Figure I.50:** Coupled allocation: risk-aversion sweep around the high-return strategy.



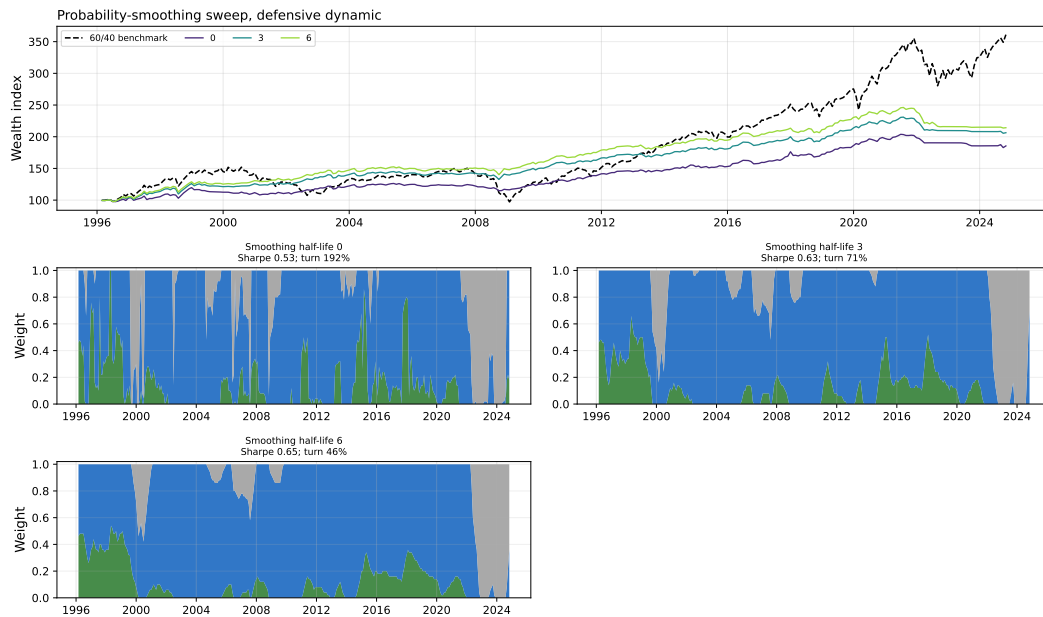
**Figure I.51:** Coupled allocation: risk-aversion sweep around the defensive strategy.



**Figure I.52:** Coupled allocation: risk-aversion sweep around the low-turnover strategy.

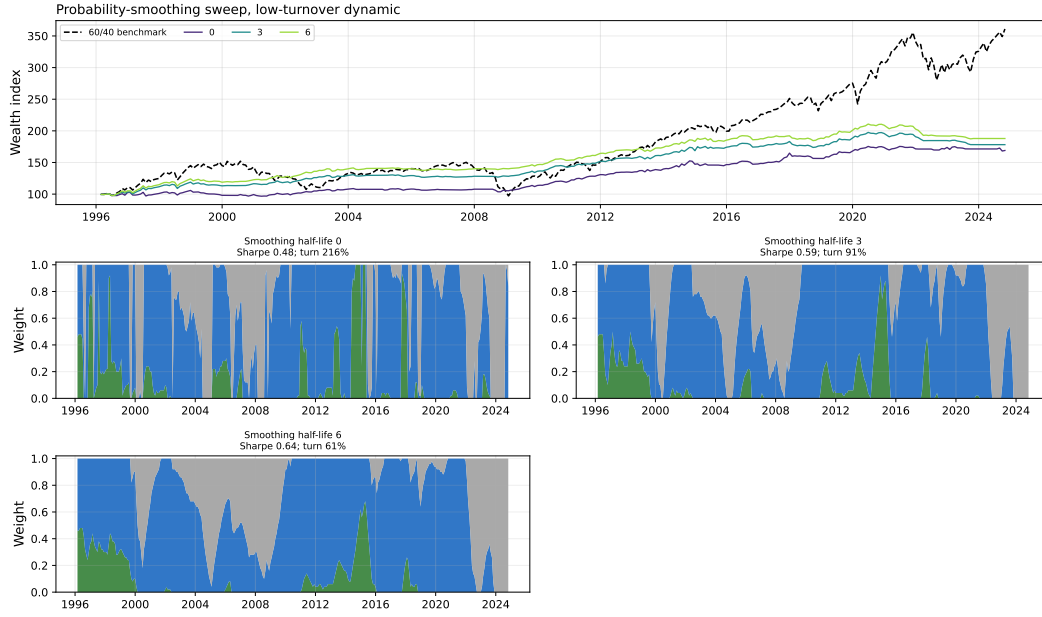


**Figure I.53:** Coupled allocation: probability-smoothing sweep around the high-return strategy.

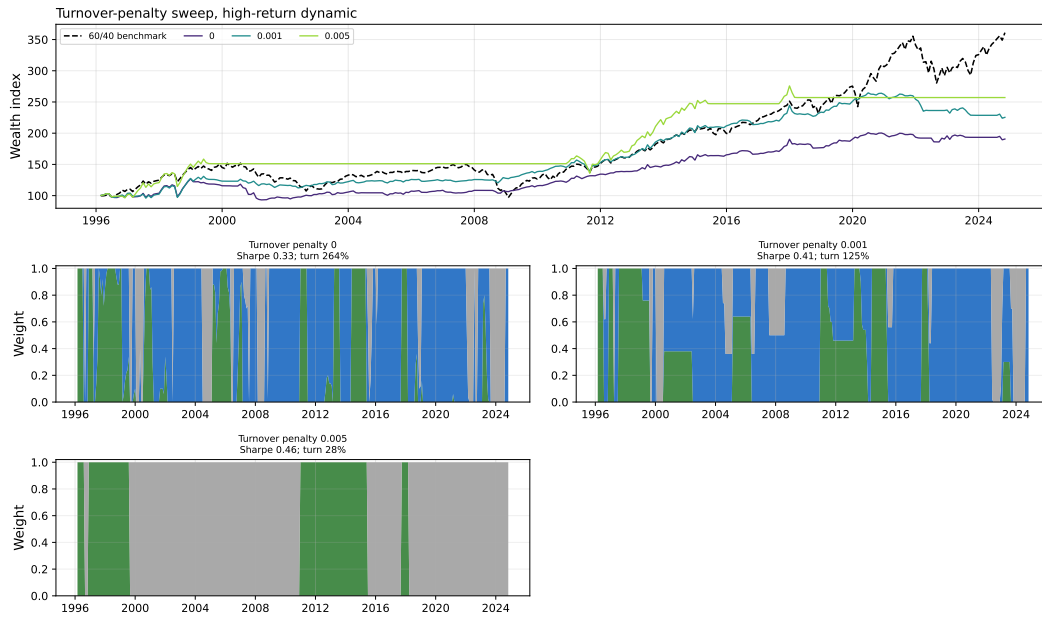


**Figure I.54:** Coupled allocation: probability-smoothing sweep around the defensive strategy.

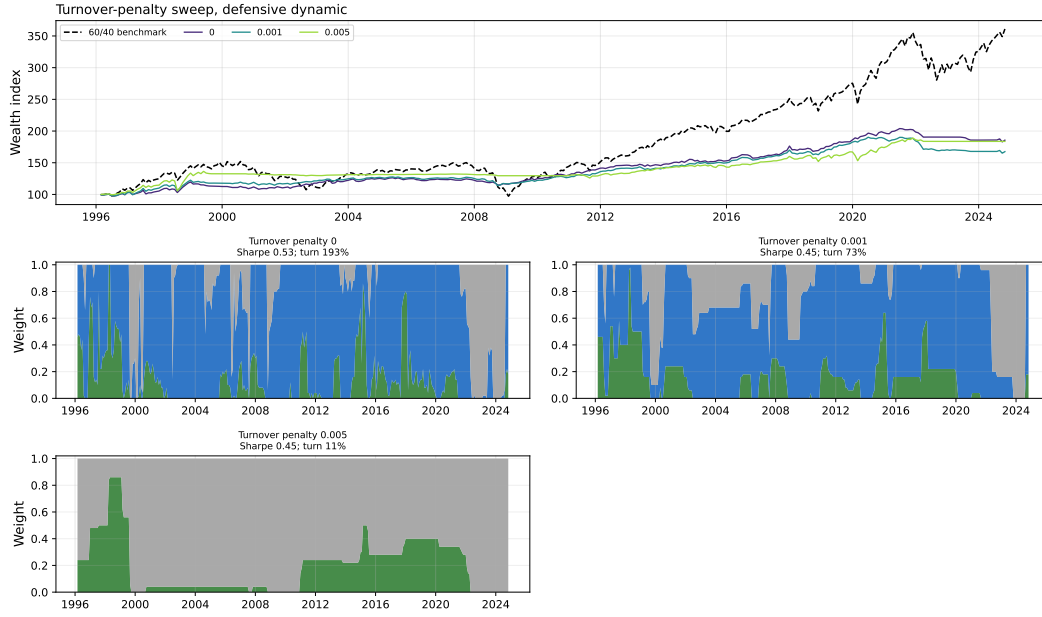




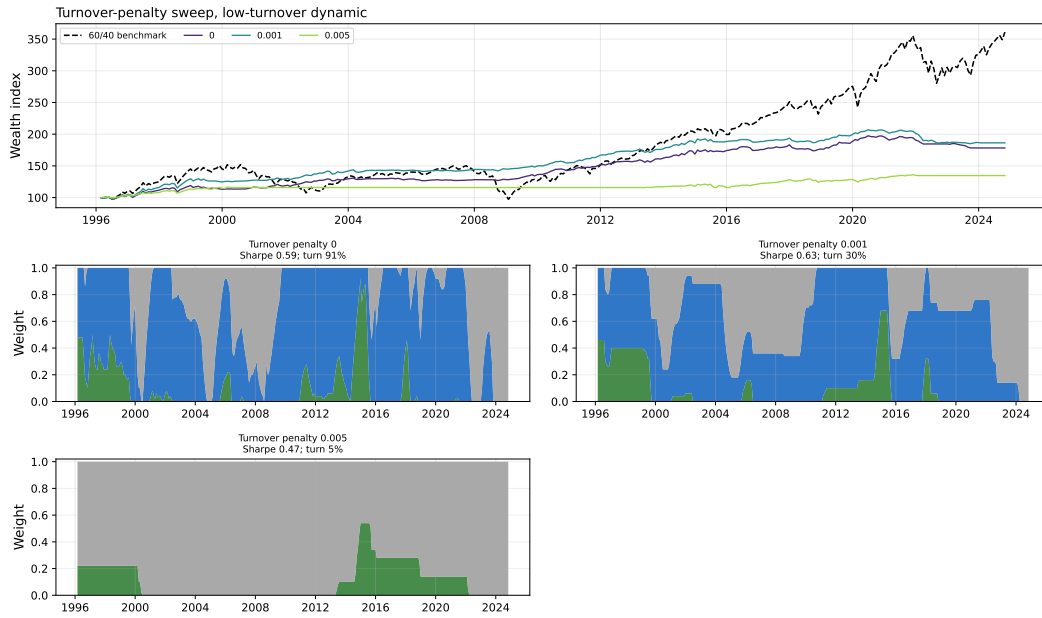
**Figure I.55:** Coupled allocation: probability-smoothing sweep around the low-turnover strategy.



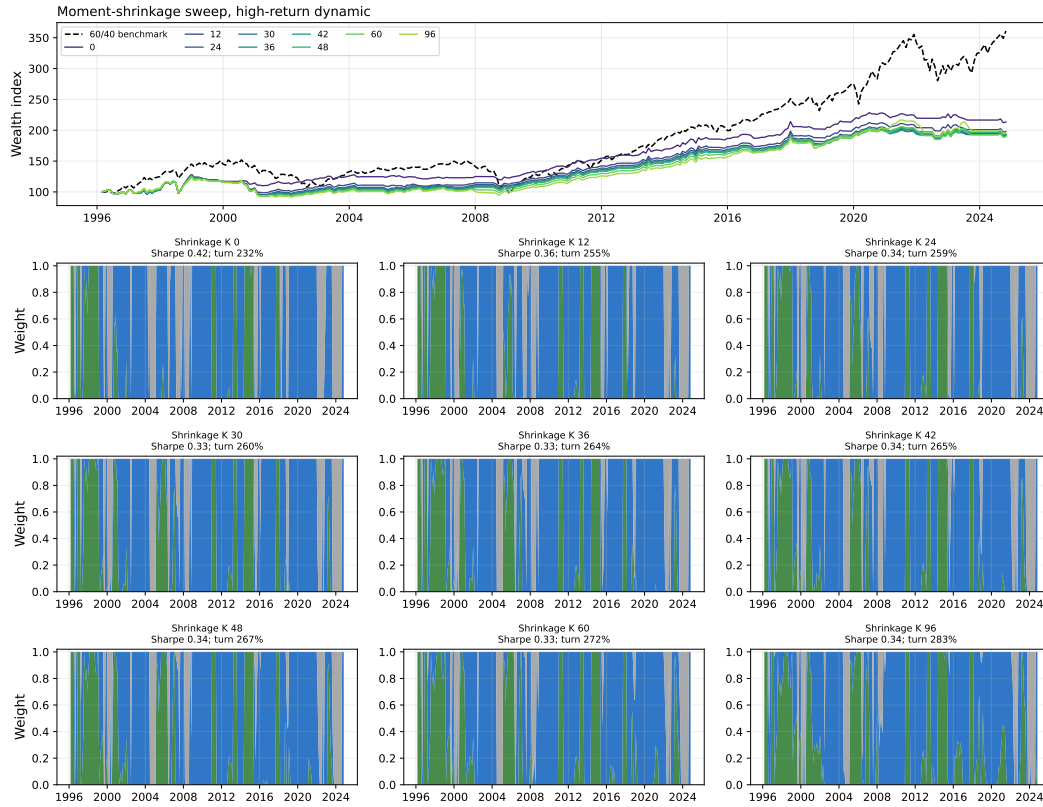
**Figure I.56:** Coupled allocation: turnover-penalty sweep around the high-return strategy. The turnover-penalty axis is the coefficient on  $\|\Delta w\|_1$  in the coupled objective (a  $\tau$ -alone penalty), distinct from the independent  $\tau \times c$  convention.



**Figure I.57:** Coupled allocation: turnover-penalty sweep around the defensive strategy. The turnover-penalty axis is the coefficient on  $\|\Delta w\|_1$  in the coupled objective (a  $\tau$ -alone penalty), distinct from the independent  $\tau \times c$  convention.



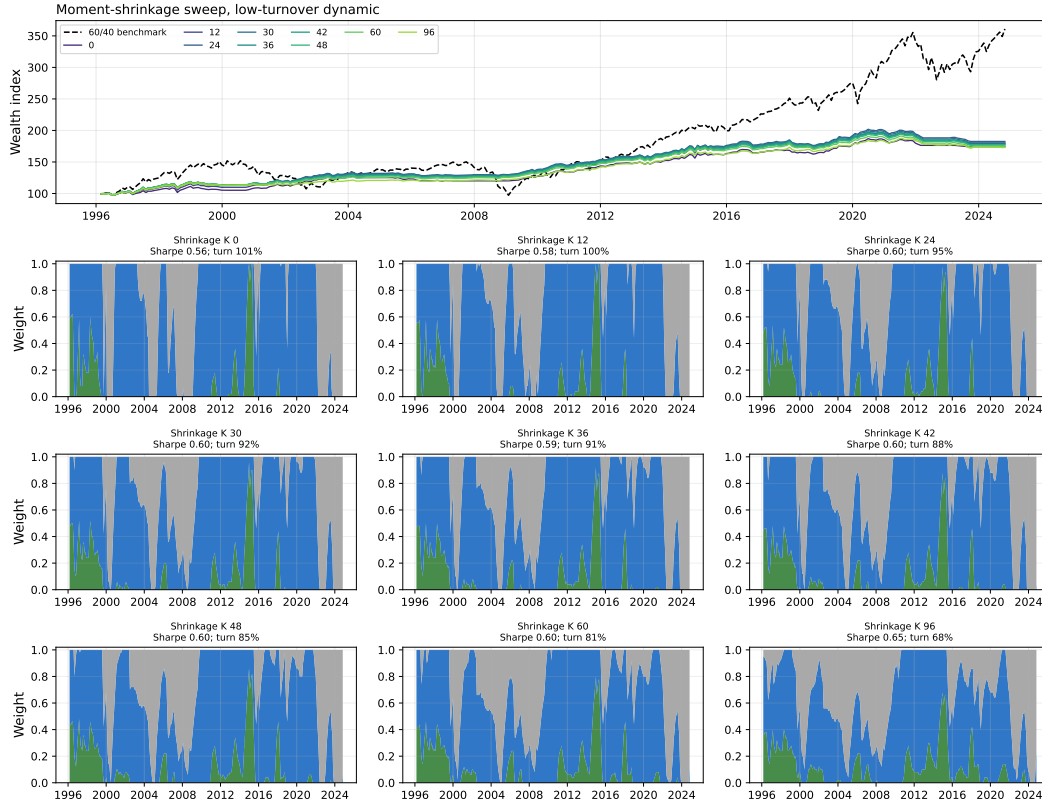
**Figure I.58:** Coupled allocation: turnover-penalty sweep around the low-turnover strategy. The turnover-penalty axis is the coefficient on  $\|\Delta w\|_1$  in the coupled objective (a  $\tau$ -alone penalty), distinct from the independent  $\tau \times c$  convention.



**Figure I.59:** Coupled allocation: moment-shrinkage sweep around the high-return strategy.



**Figure I.60:** Coupled allocation: moment-shrinkage sweep around the defensive strategy.



**Figure I.61:** Coupled allocation: moment-shrinkage sweep around the low-turnover strategy.

## I.2 Coupled model-selection protocol

This table summarises which coupled-side choices are fixed design choices, which are selected on validation or grid search, and which are used only as diagnostics. As with the independent protocol in Table H.40, the purpose is to separate the reported coupled specification from robustness checks and presentation-only diagnostics. The coupled teacher produces a single joint three-state label (calm, normal, stressed), and the student forecasts that joint label directly.

| Component                             | Coupled approach                                                                                                          | Fixed in advance                                                             | Selected on validation/grid                                                                        | Robustness or diagnostic only                                        |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Teacher feature set                   | Joint equity and bond return-based features feeding one three-state label                                                 | Final reported feature blocks fixed before student prediction and allocation | No                                                                                                 | Per-leg feature-separation summaries used only as diagnostics        |
| Teacher state structure               | Three joint regimes (calm, normal, stressed) built from the two per-leg state bits                                        | Three-state joint structure fixed before estimation                          | No; the mapping from per-leg bits to the joint label is fixed                                      | Per-leg state economics reported as diagnostics                      |
| Teacher jump penalty grid             | Shared jump-penalty grid $\lambda$ with coupling parameter $\eta$ linking the legs                                        | Grid fixed before final majority vote                                        | Majority-vote label selected from fitted teacher paths                                             | Jump-penalty diagnostics reported across $\lambda$ and $\eta$        |
| Distance norm                         | $L^2$ centroid and $L^1$ -medoid teacher paths                                                                            | Both norm families fixed as teacher candidates                               | Final label uses the combined majority vote across members                                         | Separate $L^2$ and $L^1$ paths reported as diagnostics               |
| Teacher ensemble and majority vote    | Final joint label combines the 22 ensemble members by majority vote                                                       | Voting rule fixed before student prediction                                  | No refit after allocation outcomes                                                                 | Member raster and vote-scope figures report candidate-path agreement |
| Student predictor sets                | Lagged macro-financial predictors for the next-month joint regime                                                         | Transformations fixed before model fitting                                   | Single reported predictor design evaluated out of sample                                           | Individual-predictor importance figures are diagnostic               |
| Student model hyperparameters         | Three-class logistic regression and random forest specifications                                                          | Model classes fixed before evaluation                                        | Hyperparameters selected within the rolling validation design                                      | Most-frequent-specification table reports selected settings          |
| Student validation metrics            | Macro one-vs-rest AUC, balanced accuracy, multiclass Brier score, and log loss                                            | Metrics fixed before reporting                                               | Best reported specifications selected from rolling outputs                                         | Full three-class prediction diagnostics kept in appendix             |
| Allocation rule                       | Long-only equity-bond allocation with cash allowed and an $L^1$ turnover penalty, mapping regime probabilities to moments | Asset universe and allocation objective fixed before grid search             | Candidate grid varies risk aversion, turnover penalty, transaction cost, and probability smoothing | Parameter-sensitivity appendix reports one-way checks                |
| Probability-to-moment mapping         | CONST mapping (single predicted regime) and PROB mapping (probability mixture)                                            | Both mappings fixed as candidates                                            | Headline rows use the PROB mapping                                                                 | CONST mapping retained as a diagnostic comparison                    |
| Risk aversion                         | Minimum-variance ( $\gamma=20$ ) and mean-variance ( $\gamma=5$ ) allocators from the grid                                | Grid fixed before headline reporting                                         | Representative rows use selected grid points                                                       | Risk-aversion sweeps reported as sensitivity                         |
| Transaction cost and turnover penalty | One-way transaction-cost grid with an $L^1$ turnover penalty (a $\tau$ -alone penalty)                                    | Grid fixed before headline reporting                                         | Headline coupled rows use 10 bps                                                                   | Cost and turnover-penalty sensitivity reported in appendix           |
| Headline allocation rows              | Defensive, balanced, and high-return representative rows                                                                  | Roles fixed for presentation                                                 | Rows selected as representative points from the allocation grid                                    | Full-grid ranking rules are reported only as sensitivity diagnostics |

**Table I.45:** Coupled-side model-selection protocol. The table separates fixed design choices, validation or grid-selected choices, and robustness or diagnostic components, similar to the independent protocol in Table H.40.

## References

## References

- [1] Aydinhan, A. O., Kolm, P. N., Mulvey, J. M., and Shu, Y. (2024). Identifying patterns in financial markets: extending the statistical jump model for regime identification. *Annals of Operations Research*. doi:10.1007/s10479-024-06035-z.
- [2] Bemporad, A., Breschi, V., Piga, D., and Boyd, S. P. (2018). Fitting jump models. *Automatica*, 96, 11–21. doi:10.1016/j.automatica.2018.06.022.
- [3] McCracken, M. W. and Ng, S. (2016). FRED-MD: A monthly database for macroeconomic research. *Journal of Business & Economic Statistics*, 34(4), 574–589. doi:10.1080/07350015.2015.1086655.
- [4] McCracken, M. W. and Ng, S. (2021). FRED-QD: A quarterly database for macroeconomic research. *Federal Reserve Bank of St. Louis Review*, 103(1), 1–44. doi:10.20955/r.103.1-44.
- [5] Nystrup, P., Kolm, P. N., and Lindström, E. (2021). Feature selection in jump models. *Expert Systems with Applications*, 184, 115558. doi:10.1016/j.eswa.2021.115558.
- [6] Oliveira, D. C., Sandfelder, D., Fujita, A., Dong, X., and Cucuringu, M. (2025). Tactical asset allocation with macroeconomic regime detection. *arXiv preprint arXiv:2503.11499*.
- [7] Shu, Y., Yu, C., and Mulvey, J. M. (2024). Downside risk reduction using regime-switching signals: a statistical jump model approach. *Journal of Asset Management*, 25(5), 493–507. doi:10.1057/s41260-024-00376-x.
- [8] Shu, Y., Yu, C., and Mulvey, J. M. (2024). Dynamic asset allocation with asset-specific regime forecasts. *arXiv preprint arXiv:2406.09578*.